Unforced Navier–Stokes

Seven-ray stress geometry, pressure fluxes, and fourfold nonlinear correction bounds

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Abstract

We investigate exact analytic and finite-arithmetic prerequisites for a finite-time singularity of the three-dimensional incompressible Navier–Stokes equations on the periodic torus with identically zero external force. A compact symmetric-divergence construction, pressure-neutral stress identities, and a seven-ray positivity representation give explicit compatibility conditions for a localized carrier. The one-sided coefficient cone admits an exact jointly sharp axial–azimuthal projection and pressure-compensated cylindrical flux bounds. For a shrinking fourfold-equivariant solenoidal carrier, anisotropic incompressibility selects a leading nonzero angular axial-advection channel. A compact divergence-free detector gives a conditional amplitude–concentration requirement $\delta_\tau \tau^{1/2+h} \|w\|_{L^\infty(\Omega_\tau)} \geq c > 0$ for an exact unforced correction under the stated uniform profile and local-retention assumptions. A separate conditional energy argument provides a logarithmically strengthened dyadic dissipation obstruction. We also establish a normalized tensor identity on the trigonometric unit circle and a sharp scalar seven-ray floor of $27/4$. An exact fourfold Fourier calculation exhibits 108 initially unoccupied nonlinear mediator modes and identifies their quartic initial-energy leakage, but does not give sustained transfer or control the coupling to the prescribed singular carrier. The analysis does not produce the authentic source-matched four-patch correction or a finite-time unforced singular solution.

1 The problem and the mathematical objective

We work at viscosity one on the unit three-torus $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$. For a smooth velocity $u(t,x) \in \mathbb{R}^3$ and pressure $p(t,x) \in \mathbb{R}$, the unforced incompressible equations are

$$\partial_t u + (u \cdot \nabla)u - \Delta u + \nabla p = 0, \qquad \operatorname{div} u = 0. \tag{1}$$

The specific target is to find nonzero smooth divergence-free periodic initial data and a solution smooth for $0 \leq t < 1$, of finite kinetic energy on a fundamental cell, whose speed becomes unbounded as $t \uparrow 1$. The source-free condition in (1) is essential: a verified construction with a nonzero smooth force does not establish this target. The Clay problem statement makes this distinction explicit [1]. The published OpenAI Lean formalization states Navier–Stokes breakdown with smooth forcing; its unforced finite-time result concerns Euler rather than Navier–Stokes [2].

Principal results. The exact one-sided seven-ray coefficient floor yields a sharp two-dimensional axial–azimuthal stress cone; the physical moving-frame identity retains its connection commutator. A separate fourfold selection theorem isolates the nonzero angular axial-advection channel, and a compact solenoidal test imposes a quantitative amplitude–concentration law on any locally retained unforced correction, under the hypotheses stated in Appendix AQ. The logarithmic dyadic theorem gives a distinct conditional finite-dissipation restriction. Appendix AR establishes the corrected unit-circle tensor identity and the sharp $27/4$ normalized ray floor. The associated existence problem requires a common physical

carrier, pressure, positive stress and exact nonlinear correction; none is supplied by these conditional implications.

Our approach isolates small results that are meaningful independently of the unresolved global theorem. First, a source must admit a compact symmetric-divergence correction. Second, if that correction must remain in a prescribed positive stress cone, its components need quantitative control relative to a density envelope. Third, all terms in the actual nonlinear physical residual must cancel with a single periodic pressure. Finally, a singular leading trace can survive only if the correction is provably subleading. Sections 2–7 establish exact parts of this chain.

The construction of compact symmetric anti-divergence operators under force and torque compatibility is not new; see, for example, the support-preserving operator and its explicit moment correction in [3, §3.6 and Appendix A]. The direct axisymmetric formula below is offered as an elementary source-specific coordinate realization, not as a priority claim for compact symmetric anti-divergence. The new obstacle for the present programme is the additional shrinking-support, weighted-seven-ray positivity requirement, which is not supplied by general existence of a compact tensor. Section 5 also shows that a natural *stronger* physical-component envelope is incompatible with one explicit source, even modulo pressure gradients. We subsequently exclude the original absolute coefficient envelope for the same manufactured source by a sharper dual calculation; we then determine the sharp support of the distinct one-sided cone and test the same source against it. The exact frame pullback supplies the correct dual coordinates for the genuine spatially varying source. The actual H8 source remains a separate problem.

2 A compact symmetric inverse in an annular region

Let $0 < a < b$ and $c < d$, and write $\Omega = (a, b) \times (c, d)$ in meridional cylindrical coordinates (r, z). All functions in this section are independent of the azimuthal angle θ. The orthonormal cylindrical frame is (e_r, e_θ, e_z). For a symmetric tensor with components

$$S = K\, e_r \otimes e_r + L\, e_\theta \otimes e_\theta + M(e_r \otimes e_z + e_z \otimes e_r) + C\, e_z \otimes e_z, \tag{2}$$

standard cylindrical differentiation gives

$$(\operatorname{div} S)_r = \partial_r K + \frac{K - L}{r} + \partial_z M, \tag{3}$$

$$(\operatorname{div} S)_z = \partial_r M + \frac{M}{r} + \partial_z C, \tag{4}$$

$$(\operatorname{div} S)_\theta = 0.$$

The $1/r$ terms are geometric, not lower-order terms that may be discarded near a shrinking cylindrical support.

Theorem 2.1 (Explicit compact meridional inversion)**.** *Let $f_r, f_z \in C_c^\infty(\Omega)$, and extend $f = f_r e_r + f_z e_z$ by zero to $\mathbb{R}^3$ in cylindrical coordinates. There is a symmetric $S \in C_c^\infty(\mathbb{R}^3; \mathrm{Sym}_3)$, axisymmetric and supported in the same open annular cylinder, with $\operatorname{div} S = f$, if and only if*

$$\int_a^b \int_c^d r f_z(r, z)\, \mathrm{d}z\, \mathrm{d}r = 0. \tag{5}$$

The construction uses no nonlocal spatial projection, but it asserts no positive-semidefinite or weighted-cone bound on S.

Proof. For necessity, integrate the axial component of $\operatorname{div} S = f$ against the cylindrical Jacobian r. Compact support and (4) give

$$\int_a^b \int_c^d r f_z \, \mathrm{d}z \, \mathrm{d}r = \int_a^b \int_c^d \left[\partial_r(rM) + r\partial_z C\right] \mathrm{d}z \, \mathrm{d}r = 0.$$

For sufficiency, define the axial source marginal and its radial primitive by

$$J(r) = \int_c^d f_z(r, z) \, \mathrm{d}z, \qquad H(r) = \frac{1}{r} \int_a^r sJ(s) \, \mathrm{d}s. \tag{6}$$

Since $r > 0$ on $[a, b]$, H is smooth. The vanishing moment (5) makes H identically zero outside a compact subinterval of (a, b). Choose any $\eta \in C_c^\infty((c, d))$ normalized by $\int_c^d \eta(z) \, \mathrm{d}z = 1$, and set

$$M(r, z) = H(r)\eta(z). \tag{7}$$

The identity $(rH)' = rJ$ implies

$$\int_c^d \left(f_z - \partial_r M - \frac{M}{r}\right) \mathrm{d}z = J - H' - \frac{H}{r} = 0.$$

Consequently the axial primitive

$$C(r, z) = \int_c^z \left[f_z(r, \zeta) - \partial_r M(r, \zeta) - \frac{M(r, \zeta)}{r}\right] \mathrm{d}\zeta \tag{8}$$

vanishes both below and above the axial support; it is smooth and compactly supported in Ω. Choose any $K \in C_c^\infty(\Omega)$, including $K = 0$, and define

$$L = K + r\big(\partial_r K + \partial_z M - f_r\big). \tag{9}$$

Each component of (2) is smooth and compactly supported away from the axis. Substitution in (3) gives

$$\partial_r K + \frac{K - L}{r} + \partial_z M = \partial_r K - (\partial_r K + \partial_z M - f_r) + \partial_z M = f_r,$$

and (8) gives $\partial_r M + M/r + \partial_z C = f_z$. Thus $\operatorname{div} S = f$ as claimed. □

Remark 2.2 (Scope of the construction)**.** The theorem concerns a meridional source with $f_\theta = 0$ and support strictly separated from $r = 0$. An azimuthal source introduces a different angular-momentum compatibility condition, and extending the construction across the axis requires separate regularity conditions on the cylindrical components. The tensor in the theorem may be indefinite, and its size can deteriorate when the support shrinks. None of these omissions is harmless for the weighted H8 attachment problem.

3 Pressure-neutral gauges and a sign obstruction

The mixed and hoop degrees of freedom in Theorem 2.1 are not optional decorations. A restricted diagonal ansatz exhibits a genuine obstruction to maintaining one sign.

Proposition 3.1 (Two-diagonal gauge)**.** *Let $F \in C_c^\infty(\Omega)$, put $K = rF_r$ and $C = F + rF_r$, and take $S = Ke_r \otimes e_r + Ce_z \otimes e_z$. Then*

$$\operatorname{div} S = \nabla C. \tag{10}$$

In particular its divergence is a pressure gradient, so it cannot by itself represent a prescribed source with a nonzero solenoidal part.

Proof. The radial divergence is $K_r + K/r$. Since $K = rF_r$, $C_r = F_r + K_r = K/r + K_r$. The axial divergence is C_z, which proves (10). □

Proposition 3.2 (Restricted radial sign condition)**.** *If K and C are compactly supported in $r \in (a, b)$ at a fixed z and obey $C_r = K_r + K/r$, then*

$$\int_a^b K(r, z)/r \, \mathrm{d}r = 0. \tag{11}$$

Consequently, $K \geq 0$ forces $K \equiv 0$ for this restricted two-diagonal class.

Proof. Integrate the derivative identity between a and b; the endpoint terms of K and C vanish by compact support. Since $r > 0$, continuity and (11) imply that a nonnegative K is identically zero. □

This conclusion does not exclude a sign-changing correction around a positive background tensor or a compact construction with nonzero hoop and mixed components. Indeed, Theorem 2.1 explicitly uses both of those components. The more difficult question is whether one can choose them while retaining a prescribed strict coefficient cone with uniform shrinking-support estimates.

4 The exact seven-ray coefficient margin

The finite cone appearing in the current H8 source has seven rays. With aperture $t = 1/4$ they are

$$\begin{aligned} &v_1 = (1, 0, 0), \quad v_2 = (1, t, 0), \quad v_3 = (1, -t, 0), \\ &v_4 = (1, 0, t), \quad v_5 = (1, 0, -t), \\ &v_6 = (1, t, t), \quad v_7 = (1, t, -t). \end{aligned} \tag{12}$$

Their normalized positive coefficient centre is $c_i^{(0)} = 236/1229$ for $i = 3$ and $172/1229$ for every other index. The raw source covariance and its trace-free rank-one identity are established by exact finite arithmetic and in `UnforcedPatching.SevenRayCenter.lean`.

Write $(A, B, C, D) \in \mathbb{R}^4$ for the four trace-free stress coordinates. The resulting coefficient perturbations, in ray order, are

$$\delta c = \left(-\tfrac{3}{2}D,\ 2C - 4D,\ -2C - 4D,\ 2A - 8B + 4D,\ -2A + 8B + 4D,\ 8B,\ -8B\right). \tag{13}$$

Proposition 4.1 (Sharp componentwise cone estimate)**.** *For $m = \max(|A|, |B|, |C|, |D|)$ the map (13) satisfies*

$$\max_{1 \leq i \leq 7} |\delta c_i| \leq 14m, \tag{14}$$

and the number 14 is optimal. If $\rho \geq 0$ and

$$m \leq \frac{15971}{1720600}\rho, \tag{15}$$

then for every ray $c_i^{(0)}\rho + \delta c_i \geq \rho/100$.

Proof. The absolute row sums in (13) are respectively 3/2, 6, 6, 14, 14, 8, 8; hence the operator norm from $\ell^\infty(\mathbb{R}^4)$ to $\ell^\infty(\mathbb{R}^7)$ is at most 14. At $(A, B, C, D) = (1, -1, 0, 1)$, the fourth coordinate equals 14, so the constant is sharp. Since

$$\frac{172}{1229} - \frac{1}{100} = \frac{15971}{122900},$$

inequality (15) implies $|\delta c_i| \le 14m \le (15971/122900)\rho$. As $c_i^{(0)} \ge 172/1229$, the claimed floor follows. □

The seven-ray claim is a finite linear-algebra statement. It provides the exact smallness target for a potential compact correction but gives no estimate for the particular tensor delivered by Theorem 2.1. In particular, no implication from unweighted compact solvability to (15) is available.

Proposition 4.2 (The frozen-frame physical tensor map)**.** *In the local co-rotating frame* $(e_1, e_2, e_3) = (e_r, e_\theta, e_z)$ *of the seven-ray construction, let* $G = \sum_{i=1}^{7} \delta c_i v_i \otimes v_i$*. The exact physical tensor perturbation and its trace-free pressure representative are*

$$G = \begin{pmatrix} -\frac{3}{2}D & C & A \\ C & -\frac{1}{2}D & B \\ A & B & \frac{1}{2}D \end{pmatrix}, \quad H = G + \frac{D}{2}I = \begin{pmatrix} -D & C & A \\ C & 0 & B \\ A & B & D \end{pmatrix}. \tag{16}$$

In particular, if $m = \max(|A|, |B|, |C|, |D|)$*, then* $\max_{i,j} |H_{ij}| = m$ *in that frame. Conversely, the coordinates are recovered as* $A = H_{rz}$*,* $B = H_{\theta z}$*,* $C = H_{r\theta}$*, and* $D = H_{zz}$*. The image is the four-dimensional subspace of symmetric trace-free matrices with* $H_{\theta\theta} = 0$ *(hence* $H_{rr} = -H_{zz}$*); one independent trace-free diagonal mode is absent. For a different orthonormal frame with columns* E*, the physical matrix is* EHE^{T}*; its componentwise maximum need not equal* m*.*

Proof. Substitute the seven vectors (12) and their coefficient perturbations (13) into the six independent entries of G. The diagonal entries are $-3D/2, -D/2, D/2$; the off-diagonal entries in $(r\theta, rz, \theta z)$ order are (C, A, B). Thus $\operatorname{tr} G = -3D/2$, and the scalar gauge addition $DI/2$ removes its trace. The maximum-entry identity then follows directly from the displayed matrix. Conjugation gives the last claim. □

Proposition 4.3 (The exact original seven-coefficient cone)**.** *For any* $\tau \ge 0$*, all seven constraints* $|\delta c_i| \le \tau$ *of* (13) *hold if and only if*

$$4|D| + 2|C| \le \tau, \qquad 4|D| + 2|A - 4B| \le \tau, \qquad 8|B| \le \tau. \tag{17}$$

The sufficient bound $m \le \tau/14$ *therefore need not hold for a valid coefficient perturbation: at* $(A, B, C, D) = (\tau, \tau/8, 0, 0)$*, the exact largest coefficient perturbation and* m *both equal* τ*.*

Proof. For real x, y the identity $\max(|x + y|, |x - y|) = |x| + |y|$ is exact. Apply it to the paired coefficients $(\delta c_2, \delta c_3)$, $(\delta c_4, \delta c_5)$, and $(\delta c_6, \delta c_7)$. The first pair contributes $4|D| + 2|C|$, the second $4|D| + 2|A - 4B|$, and the third $8|B|$. The remaining coefficient $|\delta c_1| = 3|D|/2$ is dominated by the first pair. Substitution verifies the equality case. □

This identifies a previously implicit assumption: the bound (15) is also a physical entrywise bound for the trace-free perturbation *in the specified cylindrical frame*. The same conclusion requires a verified change of frame when H8 rays are tilted; pointwise orthogonal conjugation alone does not control the additional derivatives of a spatially varying frame.

5 A source-matched weighted dual obstruction

The coefficient condition (15) involves four trace-free coordinates, not arbitrary physical tensor components. To test a particularly strong candidate estimate, let

$$P(r) = \begin{cases} (r-1)^2(2-r)^2, & 1 \le r \le 2, \\ 0, & \text{otherwise}, \end{cases} \qquad \eta(z) = \begin{cases} (1-z^2)^3, & |z| \le 1, \\ 0, & \text{otherwise}. \end{cases} \tag{18}$$

Set $\rho(r,z) = P(r)\eta(z) \geq 0$ and choose the meridional source $f = P(r)\eta'(z)e_z$. This is source-matched in the literal sense that $S = \rho\, e_z \otimes e_z$ is a compact symmetric inverse: $\operatorname{div} S = f$. Also $\int r f_z \,\mathrm{d}r\,\mathrm{d}z = 0$, so the force compatibility condition of Theorem 2.1 holds.

Proposition 5.1 (A pressure-independent stress bound)**.** *Let $\beta = 15971/1720600$. No smooth compact symmetric tensor H and smooth scalar pressure q satisfy both*

$$\begin{aligned} \operatorname{div} H + \nabla q &= f, \\ |H_{ij}| &\leq \beta\rho \quad (i,j \in \{r,\theta,z\}), \end{aligned} \tag{19}$$

where components refer to the orthonormal cylindrical frame*. The contradiction also persists for suitable smooth compact nonnegative approximations to the profiles in* (18)*.*

Proof. Construct an axisymmetric vector field with components

$$v_r = -rP\eta'', \qquad v_\theta = 0, \qquad v_z = (2P + rP')\eta'. \tag{20}$$

It is divergence-free because $r^{-1}\partial_r(rv_r) + \partial_z v_z = 0$. The zero-extended profiles give a compactly supported Lipschitz v: $P = P' = 0$ at $r = 1, 2$, and $\eta' = \eta'' = 0$ at $z = \pm 1$. Thus integration by parts against v is legitimate, by approximation if necessary, and the pressure term vanishes. With the angular factor 2π suppressed, the source pairing is

$$\begin{aligned} L &= \int_1^2 \int_{-1}^1 r f_z v_z \,\mathrm{d}z\,\mathrm{d}r \\ &= \left(\int_1^2 rP^2 \,\mathrm{d}r\right)\left(\int_{-1}^1 (\eta')^2 \,\mathrm{d}z\right) = \frac{1}{420}\,\frac{1024}{385} = \frac{256}{40425}. \end{aligned} \tag{21}$$

Here the middle equality follows by integrating $r^2 PP'$ by parts. The physical cylindrical gradient of v includes its hoop entry v_r/r. Its entrywise absolute sum is bounded by

$$|\nabla v|_1 \leq (4P + 2r|P'|)|\eta''| + rP|\eta'''| + (3|P'| + r|P''|)|\eta'|. \tag{22}$$

The elementary exact radial integrals are $\int_1^2 P = 1/30$, $\int_1^2 P^2 = 1/630$, $\int_1^2 rP^2 = 1/420$, and $\int_1^2 P|P'| = 1/256$. On their supports, $r \leq 2$, $|P''| \leq 2$, $|\eta''| \leq 6$ and $|\eta'''| \leq 72|z| + 120|z|^3$. Direct integration gives

$$\int_{-1}^1 \eta = \frac{32}{35}, \quad \int_{-1}^1 \eta|\eta'| = 1, \quad \int_{-1}^1 \eta|z| = \frac{1}{4}, \quad \int_{-1}^1 \eta|z|^3 = \frac{1}{20}.$$

Consequently, writing $U = \int_1^2 \int_{-1}^1 r\rho|\nabla v|_1 \,\mathrm{d}z\,\mathrm{d}r$, one obtains the rigorous upper bound

$$U \leq \left(\frac{4}{420} + \frac{2}{64}\right)\frac{192}{35} + \frac{2}{315}\,24 + \frac{3}{128} + \frac{4}{15} = \frac{313361}{470400}. \tag{23}$$

But (19) would imply $L \leq \beta U$ by the virtual-work identity $\int f \cdot v = -\int H : \nabla v$, whereas

$$L - \beta\,\frac{313361}{470400} = \frac{443015653}{2967690880000} > 0, \tag{24}$$

a contradiction. Nonnegative convolution approximations of P and η, supported away from the axis, converge in the requisite derivative integral norms. The strict gap survives for sufficiently small smoothing parameters and proves the smooth-profile assertion. □

Corollary 5.2 (Failure of the sufficient four-field envelope)**.** *For the source f and envelope ρ in* (18)*, there is no fixed-frame four-coordinate stress $H(A,B,C,D)$ of* (16) *satisfying* $\operatorname{div} H + \nabla q = f$ *and the sufficient bound* (15)*, even if the pressure is freely chosen.*

Proof. Proposition 4.2 transfers (15) to the entrywise physical cap (19). This contradicts Proposition 5.1. □

The preceding argument excludes the *stronger sufficient* four-field estimate. It cannot alone exclude the original absolute coefficient budget $\max_i |\delta c_i| \le (15971/122900)\rho$: if one simply inserts that larger constant into the conservative gradient bound (23), then

$$\frac{15971}{122900}\frac{313361}{470400} - L = \frac{51024386641}{635933760000} > 0. \tag{25}$$

The sign of this bound is inconclusive, not evidence of existence. The exact cone, however, has a smaller sharp dual support than that componentwise comparison.

Proposition 5.3 (Sharp dual of the original absolute coefficient cone)**.** *For $\tau \ge 0$ let $\mathcal{K}_\tau$ be the four-coordinate set defined by (17). For arbitrary $a, b, c, d \in \mathbb{R}$,*

$$\sup_{(A,B,C,D)\in\mathcal{K}_\tau} (aA + bB + cC + dD) = \tau\left[\max\left(\frac{|a|+|c|}{2}, \frac{|d|}{4}\right) + \frac{|4a+b|}{8}\right]. \tag{26}$$

The upper bound in (26) is proved in Lean for the actual seven coefficient constraints.

Proof. Put $x = A - 4B$. By (17), for fixed $s = |D| \in [0, \tau/4]$, the variables x, C, B are independent and satisfy

$$|x|, |C| \le \frac{\tau - 4s}{2}, \qquad |B| \le \frac{\tau}{8}.$$

The objective becomes $ax + (4a+b)B + cC + dD$. Its maximum for this s is

$$\frac{\tau - 4s}{2}(|a| + |c|) + \frac{\tau}{8}|4a + b| + s|d|.$$

This expression is affine in s, so its maximum occurs at $s = 0$ or $s = \tau/4$, yielding (26). Each bound is attained by choosing the signs of x, C, B, D to match their respective linear coefficients. Thus the value is sharp. □

Proposition 5.4 (An exact original-budget obstruction for the manufactured source)**.** *Let f, ρ be given by (18), and set $\gamma = 15971/122900$. No locally integrable symmetric tensor of the fixed-frame form $H(A, B, C, D)$ in (16), together with a locally integrable scalar pressure q, can satisfy in the distributional sense*

$$\operatorname{div} H + \nabla q = f, \qquad |\delta c_i(A, B, C, D)| \le \gamma\rho \quad (1 \le i \le 7).$$

Proof. Use the divergence-free meridional field v from (20). Its nonzero coupling coordinates to H are

$$a(r, z) = \partial_z v_r + \partial_r v_z = -rP\eta''' + (3P' + rP'')\eta',$$
$$d(r, z) = \partial_z v_z - \partial_r v_r = (3P + 2rP')\eta''.$$

Indeed, $H : \nabla v = Aa + Dd$; the B and C components vanish because v is meridional and independent of θ. By Proposition 5.3, integration by parts and $\rho \ge 0$ would force

$$L \le \gamma S, \quad S = \int_1^2 \int_{-1}^1 rP(r)\eta(z) \max\left(|a|, \frac{|a|}{2} + \frac{|d|}{4}\right) \mathrm{d}z\, \mathrm{d}r. \tag{27}$$

There is an exact rational upper bound

$$S \le \frac{583853781778722552348462424820 7}{150323855360000000000000000000000} < \frac{1}{25}. \tag{28}$$

For completeness, partition $[1,2]\times[-1,1]$ into 40^2 rational rectangles. On each rectangle express the separated polynomials a and d in a common-degree bivariate Bernstein basis. The convex-hull property bounds $|a|$ and $|d|$ by the largest absolute Bernstein coefficient. Multiply $\max(a_{ij}, a_{ij}/2 + d_{ij}/4)$ by the *exact* nonnegative cell mass $\int rP\,\mathrm{d}r \int \eta\,\mathrm{d}z$, and sum the 1600 rational terms. This yields the first inequality in (28); the second is an integer comparison. The accompanying pure-rational program reproduces each coefficient and sum. Consequently

$$L - \gamma S > \frac{256}{40425} - \frac{15971}{122900\cdot 25} = \frac{5637293}{4968232500} > 0,$$

contradicting (27). The pressure term vanishes by solenoidality of v. The zero-extended test field is compactly supported and Lipschitz, so smooth divergence-free mollifications justify the same integration-by-parts identity for locally integrable H and q. □

This obstruction concerns the *absolute correction budget* $|\delta c_i| \le \gamma\rho$ in one fixed frame. It does not rule out every nonnegative seven-ray stress: the one-sided requirement $c_i^{(0)}\rho + \delta c_i \ge \rho/100$ admits perturbations outside that sufficient absolute budget. We now compute the exact support of this larger admissible set rather than extending the obstruction by analogy.

Proposition 5.5 (Sharp support of the one-sided seven-ray cone)**.** *Let $\rho \ge 0$ and let $\mathcal{P}_\rho$ contain precisely those (A,B,C,D) for which all seven original coefficients satisfy $c_i^{(0)}\rho + \delta c_i(A,B,C,D) \ge \rho/100$. Define*

$$g = \frac{15971}{122900}, \quad h = \frac{22371}{122900}, \quad x = A - 4B.$$

Then $\mathcal{P}_\rho$ is described exactly by

$$\begin{aligned} -\frac{g\rho}{4} \le D \le \frac{(g+h)\rho}{8}, &\qquad |B| \le \frac{g\rho}{8}, \\ |x| \le \frac{g\rho}{2} + 2D, &\qquad \left|C - \frac{(h-g)\rho}{4}\right| \le \frac{(g+h)\rho}{4} - 2D. \end{aligned} \tag{29}$$

For arbitrary dual coefficients $(a,b,c,d) \in \mathbb{R}^4$ its support is

$$\begin{aligned} \sigma_\rho^+(a,b,c,d) = \rho\Big[&\frac{h-g}{4}c + \frac{g}{8}|4a+b| \\ &+ \max\left\{\frac{3g+h}{4}|c| - \frac{g}{4}d,\ \frac{3g+h}{4}|a| + \frac{g+h}{8}d\right\}\Big]. \end{aligned} \tag{30}$$

The universal upper bound in (30) is kernel-checked in Lean under the original seven coefficient floors.

Proof. The six ordinary ray centres have available negative correction $-g\rho$ and the third ray has $-h\rho$. The paired fourth and fifth coefficient inequalities yield the bound on $|x|$; the sixth and seventh yield the bound on $|B|$. The second and third give the C interval, whose nonnegative half-width forces the upper bound on D. The half-width for x forces the lower bound on D. The remaining first-ray inequality $-3D/2 \ge -g\rho$ is then redundant, since $(g+h)/8 \le 2g/3$. This proves the exact description, including when $\rho = 0$.

The objective is $ax + (4a+b)B + cC + dD$. For each fixed D, the independent intervals for x, B, C give a maximum equal to

$$\frac{h-g}{4}\rho c + \frac{g\rho}{8}|4a+b| + \left(\frac{g\rho}{2} + 2D\right)|a| + \left(\frac{(g+h)\rho}{4} - 2D\right)|c| + Dd.$$

This is affine in D. Substitution of its two endpoints gives (30). Signs of x, B and the displacement of C attain these extrema, establishing sharpness. □

Theorem 5.6 (Exact joint axial–azimuthal seven-ray cone)**.** *Fix $\rho \geq 0$, and retain the genuine one-sided seven-ray floor of Proposition 5.5. Given $A, C \in \mathbb{R}$, there exist $B, D \in \mathbb{R}$ satisfying all seven coefficient floors if and only if*

$$\begin{aligned} -g\rho \leq C &\leq \frac{g+h}{2}\rho, \\ |A| - C &\leq \frac{3g}{2}\rho, \\ |A| + C &\leq \left(g + \frac{h}{2}\right)\rho. \end{aligned} \tag{31}$$

The projection is a hexagon with vertices

$$\left(\pm\frac{g\rho}{2}, -g\rho\right), \qquad \left(\pm\frac{(5g+h)\rho}{4}, \frac{(h-g)\rho}{4}\right), \qquad \left(\pm\frac{g\rho}{2}, \frac{(g+h)\rho}{2}\right). \tag{32}$$

For $a, c \in \mathbb{R}$, its exact joint support is

$$\sigma_\rho^{AC}(a,c) = \rho\left[\frac{h-g}{4}c + \frac{g}{2}|a| + \frac{3g+h}{4}\max\{|a|,|c|\}\right]. \tag{33}$$

In particular, the separate sharp values of A and C cannot be attained simultaneously.

Proof. The fourth and fifth coefficient floors, together with the sixth and seventh, imply

$$|A - 4B| \leq \frac{g\rho}{2} + 2D, \qquad |4B| \leq \frac{g\rho}{2}.$$

Thus $|A| \leq g\rho + 2D$. The second and third floors give

$$2D \leq C + \frac{g\rho}{2}, \qquad 2D \leq \frac{h\rho}{2} - C.$$

The nonnegative half-width of the $A - 4B$ interval forces $D \geq -g\rho/4$, giving the two bounds on C and both mixed inequalities in (31).

Conversely suppose (31) holds and choose

$$D = \min\left\{\frac{g\rho + 2C}{4}, \frac{h\rho - 2C}{4}\right\}, \qquad 4B = \operatorname{clip}\left(A; -\frac{g\rho}{2}, \frac{g\rho}{2}\right),$$

where $\operatorname{clip}(s; l, u) = \max\{l, \min\{s, u\}\}$. The displayed inequalities give

$$-\frac{g\rho}{4} \leq D \leq \frac{(g+h)\rho}{8} \leq \frac{2g\rho}{3}, \qquad D \geq \frac{|A| - g\rho}{2},$$

and

$$|A - 4B| = \max\left\{|A| - \frac{g\rho}{2}, 0\right\} \leq \frac{g\rho}{2} + 2D.$$

These relations verify all seven floors, including the first-ray inequality $-3D/2 \geq -g\rho$. They also cover $\rho = 0$. The intersection of the four sloping and two horizontal half-planes has exactly the six vertices in (32). Maximizing $aA + cC$ at those vertices yields (33). □

For the unrotated cylindrical source frame the same inequalities persist upon positive-weight angular and axial averaging. Namely, at a fixed radius let

$$S_z = \int \langle K_{rz}\rangle_\theta \, dz, \qquad S_\theta = \int \langle K_{r\theta}\rangle_\theta \, dz, \qquad M_\rho = \int \langle \rho\rangle_\theta \, dz.$$

Provided the source frame agrees with the cylindrical frame, the pair (S_z, S_θ) satisfies (31) with (A, C, ρ) replaced by (S_z, S_θ, M_ρ). The axial source reconstruction must retain the $K_{zz} + p$ boundary jump, and the azimuthal reconstruction must retain its $K_{\theta z}$ endcap. For a genuinely rotating source frame the correct necessary bound instead uses the full dual support $\sigma_\rho^+(a\lambda(E^\mathsf{T} G_z E) + c\lambda(E^\mathsf{T} G_\theta E))$ integrated over the slice, where $G_z = e_r \otimes e_z$ and $G_\theta = e_r \otimes e_\theta$; fixed-frame hexagon coefficients cannot simply be transported through a tilt.

Proposition 5.7 (The manufactured dual test is inconclusive for the one-sided cone)**.** *For the fixed diagnostic source and divergence-free test field in Proposition 5.4, let a, d be the coupling expressions there, and define*

$$J = \int_1^2 \int_{-1}^1 rP(r)\eta(z)\, \sigma_1^+(-a(r,z), 0, 0, -d(r,z))\, \mathrm{d}z\, \mathrm{d}r.$$

An exact rational partition certificate proves

$$J \geq \frac{24989775099966161649388574430033054 1}{38797083829862400000000000000000000000} > \frac{256}{40425} + \frac{1}{10000}. \tag{34}$$

Consequently the necessary dual inequality $L \leq J$ is not contradicted by this test field under the one-sided cone. This asserts neither the existence of a stress nor the absence of an obstruction from a different test field.

Proof. The explicit support (30), evaluated on $(-a, 0, 0, -d)$, is the maximum of the four signed polynomial actions

$$\frac{ga}{2} + \frac{gd}{4}, \quad -\frac{ga}{2} + \frac{gd}{4}, \quad \left(\frac{g}{2} + \frac{3g+h}{4}\right) a - \frac{g+h}{8} d, \quad -\left(\frac{g}{2} + \frac{3g+h}{4}\right) a - \frac{g+h}{8} d.$$

Divide the rectangle into 40×40 rational subrectangles. On each cell, the integral of the pointwise maximum is at least the maximum of the four *exact signed polynomial integrals* against the nonnegative weight $rP\eta$. Their rational sum gives the first inequality in (34). The second is an exact integer comparison. For independent error detection, the accompanying program also bounds the same integral from above with signed bivariate Bernstein coefficients. The pressure contribution vanishes because the test field is divergence free. □

The contrast is mathematically substantive: the absolute-budget obstruction is strict, whereas the exact one-sided cone is wider and this particular necessary inequality has positive slack. A primal positive-cone construction would additionally require a compact symmetric-divergence inverse with all support and derivative bounds. In addition, $f = P\eta' e_z$ is a manufactured diagnostic source, *not* the authenticated H8 residual $g - \mathbb{P}\,\mathrm{div}(\rho R_*)$, where $\mathbb{P}$ denotes the divergence-free Leray projection; the source profiles and potentially rotating physical frame required to test that residual have not been provided in the checked project. No statement here excludes a more flexible source-matched H8 inverse or decides the global unforced problem.

Proposition 5.8 (Physical-frame pullback of the one-sided dual)**.** *Let $E(x)$ be a pointwise orthogonal 3×3 source-to-physical frame, written in Cartesian components, and set $K(x) = E(x)H(A, B, C, D)(x)E(x)^\mathsf{T}$ with H from (16). For any physical gradient $G = \nabla v$, put $M = E^\mathsf{T} G E$ and define*

$$(a_E, b_E, c_E, d_E) = (M_{rz} + M_{zr},\ M_{\theta z} + M_{z\theta},\ M_{r\theta} + M_{\theta r},\ M_{zz} - M_{rr}). \tag{35}$$

Then

$$K : G = Aa_E + Bb_E + Cc_E + Dd_E. \tag{36}$$

If every original seven-ray coefficient is at least $\rho/100$ for $\rho \geq 0$, then the pointwise necessary bound is

$$-K : G \leq \sigma_\rho^+(-a_E, -b_E, -c_E, -d_E), \tag{37}$$

where σ^+ is given exactly by (30). Both (36) and (37) are kernel-checked in Lean as pointwise finite-matrix results; the algebra does not require E to be constant.

Suppose additionally that K and q solve $\operatorname{div} K + \nabla q = f$ distributionally on a periodic cell or a region where the following integration by parts is valid. For every smooth periodic or compactly supported solenoidal v,

$$\int f \cdot v \, \mathrm{d}x \leq \int \sigma^+_{\rho(x)}(-a_E(x), -b_E(x), -c_E(x), -d_E(x)) \, \mathrm{d}x. \tag{38}$$

Proof. The Frobenius identity $(EHE^{\mathsf{T}}) : G = H : (E^{\mathsf{T}}GE)$ follows by rearranging finite sums over the three matrix indices. Inserting the nine entries of H gives (36). Apply Proposition 5.5 to the negative of this dual covector to obtain (37). Finally $\int f \cdot v = -\int K : \nabla v$ because the pressure term pairs to zero with a solenoidal v, proving (38). □

The frame transfer changes the dual coordinates even when E is constant. For example, take $G = \operatorname{diag}(1, 2, 3)$ and the orthogonal meridional rotation

$$E = \begin{pmatrix} 3/5 & 0 & 4/5 \\ 0 & 1 & 0 \\ -4/5 & 0 & 3/5 \end{pmatrix}.$$

Formula (35) gives $(a_E, b_E, c_E, d_E) = (-48/25, 0, 0, -14/25)$, whereas the frozen frame gives $(0, 0, 0, 2)$. Thus a fixed-frame numerical obstruction cannot be transferred to a varying frame without recomputing its dual integral.

For a differentiable frame, the analytic divergence additionally contains the connection terms

$$\partial_j K_{ij} = E_{ia}E_{jb}\partial_j H_{ab} + (\partial_j E_{ia})H_{ab}E_{jb} + E_{ia}H_{ab}(\partial_j E_{jb}), \tag{39}$$

with repeated indices summed. Orthogonality ensures that a source-frame scalar multiple of the identity remains a physical pressure gauge, even if E varies. Equation (39) shows why the pointwise dual theorem alone does not control the weighted derivatives of a source-matched inverse. The authenticated functions g, ρ, R_* and E, together with their support and derivative budgets, remain required to evaluate (38) on the genuine H8 residual.

Proposition 5.9 (Exact pressure-compensated annular H8 gate)**.** *Fix a time and a cylindrical annulus $a < r < b$, $c < z < d$. Let K be a smooth symmetric physical tensor, p a smooth scalar, and $f = \operatorname{div} K + \nabla p$. Denote the azimuthal average by $\langle\cdot\rangle_\theta$ and set*

$$\begin{aligned} S(r) &= \int_c^d \langle K_{rz}\rangle_\theta \, \mathrm{d}z, & F(r) &= \int_c^d \langle f_z\rangle_\theta \, \mathrm{d}z, \\ q_{\mathrm{s}}(r) &= [\langle K_{zz} + p\rangle_\theta]_{z=c}^{z=d}, & Q(r) &= \frac{1}{r}\int_a^r s\, q_{\mathrm{s}}(s) \, \mathrm{d}s. \end{aligned} \tag{40}$$

Then the exact cylindrical identity is

$$J_f(r) := \frac{1}{r}\int_a^r sF(s) \, \mathrm{d}s = S(r) - \frac{a}{r}S(a) + Q(r). \tag{41}$$

Suppose $K = EH(A, B, C, D)E^{\mathsf{T}}$, where $E(r, \theta, z)$ is orthogonal and the seven H8 ray coefficients exceed $\rho/100$ for $\rho \geq 0$. Let G be the physical matrix having $G_{rz} = 1$ and all other

entries zero, and write λ_E for the four coordinates of (35) applied to this G. Define the exact slice supports

$$\mathcal{B}_{\pm}(r) = \int_c^d \left\langle \sigma^{+}_{\rho(r,\theta,z)}(\pm\lambda_E(r,\theta,z)) \right\rangle_\theta \mathrm{d}z,$$
$$\mathcal{B}_{\mathrm{abs}}(r) = \int_c^d \left\langle \sigma_{g\rho(r,\theta,z)}(\lambda_E(r,\theta,z)) \right\rangle_\theta \mathrm{d}z, \qquad g = \frac{15971}{122900}. \tag{42}$$

The original absolute coefficient budget implies

$$\left| J_f(r) + \frac{a}{r} S(a) - Q(r) \right| \le \mathcal{B}_{\mathrm{abs}}(r), \tag{43}$$

while the larger one-sided positive-coefficient cone requires only

$$-\mathcal{B}_-(r) \le J_f(r) + \frac{a}{r} S(a) - Q(r) \le \mathcal{B}_+(r). \tag{44}$$

These bounds hold for a spatially varying orthogonal frame; they are necessary conditions and do not construct a differential inverse.

Proof. The azimuthal mean removes the $r^{-1}\partial_\theta K_{\theta z}$ term in the cylindrical divergence. The axial component therefore obeys $F = S' + S/r + q_{\mathrm{s}}$, proving $(rS)' = r(F - q_{\mathrm{s}})$ and (41) by integration. At each spatial point $K_{rz} = K : G = \lambda_E \cdot (A, B, C, D)$. The sharp one-sided support formula (30) gives the upper and lower bounds on this quantity, and (26) gives the symmetric absolute bound. Azimuthal and axial integration followed by (41) proves both inequalities. □

Proposition 5.10 (Case 006 source-native axial interval)**.** *Work in the physical cylindrical source frame of the Case 006 seven-ray centre, for which $(R_*)_{rz} = \alpha = -4443/9832$. Let T be a smooth prepared symmetric stress, let $\rho \ge 0$, and write*

$$f = \mathrm{div}(\mathsf{T} - \rho R_*) + \nabla p_0, \qquad \mathcal{T}(r) = \int_c^d \langle \mathsf{T}_{rz} \rangle_\theta \, \mathrm{d}z, \qquad \mathcal{R}(r) = \int_c^d \langle \rho \rangle_\theta \, \mathrm{d}z.$$

Set $q_0(r) = [\langle (\mathsf{T} - \rho R_*)_{zz} + p_0 \rangle_\theta]_{z=c}^{z=d}$, and let J_f be the source primitive in (41). Then

$$J_f(r) = \mathcal{T}(r) - \alpha \mathcal{R}(r) - \frac{a}{r}(\mathcal{T}(a) - \alpha \mathcal{R}(a)) + \frac{1}{r}\int_a^r s q_0(s) \, \mathrm{d}s. \tag{45}$$

Suppose that the source and the sought correction admit the corresponding compact representatives, with zero inner shear and axial pressure/tensor boundary traces. If the correction $H(A, B, C, D)$ obeys the original absolute *seven-coefficient budget $|\delta c_i| \le \kappa\rho$, $\kappa = 15971/122900$, then*

$$|\mathcal{T}(r) - \alpha \mathcal{R}(r)| \le \kappa \mathcal{R}(r).$$

In particular, whenever $\mathcal{R}(r) > 0$,

$$-\frac{143017}{245800} \le \frac{\mathcal{T}(r)}{\mathcal{R}(r)} \le -\frac{79133}{245800}. \tag{46}$$

If the angular coefficient $B = H_{\theta z}$ has already been selected and $\mathcal{B}(r) = \int_c^d \langle B \rangle_\theta \, \mathrm{d}z$, the stronger condition is $|\mathcal{T} - \alpha\mathcal{R} - 4\mathcal{B}| \le (\kappa/2)\mathcal{R}$.

This interval is not *implied by the larger one-sided positive-coefficient cone. With $g = 15971/122900$ and $h = 22371/122900$, its sharp physical rz maximum in this same frame is*

$$A_{\max} = \frac{5g + h}{4}\rho = \frac{51113}{245800}\rho > \kappa\rho.$$

Indeed $D = (g+h)\rho/8$, $B = g\rho/8$, $C = (h-g)\rho/4$ and $A = (5g+h)\rho/4$ preserve all seven coefficient floors at or above $\rho/100$. A spatially rotated source frame additionally requires the transported support in (42); neither scalar interval may simply be copied to an arbitrary rotating frame.

Proof. Equation (45) follows by applying Proposition 5.9 to $K = \mathsf{T} - \rho R_*$ and using $(R_*)_{rz} = \alpha$. Under the stated compact boundary assumptions the source primitive is the axial integral of the correction shear $H_{rz} = A$. The original coefficient norm is

$$\max\{8|B|,\ 2|C| + 4|D|,\ 2|A - 4B| + 4|D|\} \le \kappa\rho.$$

It follows that $|A| \le 4|B| + |A - 4B| \le \kappa\rho$ and $|A - 4B| \le \kappa\rho/2$. Integrating over the slice proves the two necessary bounds. Exact subtraction of the rational centre and margin gives $\alpha - \kappa = -143017/245800$ and $\alpha + \kappa = -79133/245800$. Finally the displayed one-sided witness satisfies the seven original coefficient floors by direct substitution in their rational formulas; its A strictly exceeds $\kappa\rho$. □

The boundary primitive cannot be discarded in a local test. For example, on $1 \le r \le 2$, $0 \le z \le 1$, take $K = 0$ and $p = z$. Then $F = q_{\mathrm{s}} = 1$ and $S = 0$, whereas $J_f(r) = (r^2 - 1)/(2r)$ is nonzero. The zero stress satisfies either cone with $\rho = 0$. Thus $|J_f| \le \mathcal{B}$ without the boundary correction would give a false obstruction. This example is local and does not assert compact pressure support.

For a finite exact check, let the frame be independent of z, θ on each radial slice and write $\bar\rho(r) = \int_c^d \rho(r,z)\,\mathrm{d}z$. Then the supports in (42) reduce to their respective finite-dimensional formulas at weight $\bar\rho$. For the rationally rotating frame

$$E(r) = \begin{pmatrix} \frac{1-t^2}{1+t^2} & 0 & \frac{2t}{1+t^2} \\ 0 & 1 & 0 \\ -\frac{2t}{1+t^2} & 0 & \frac{1-t^2}{1+t^2} \end{pmatrix}, \qquad t = r - 1,$$

the physical rz shear has pulled-back covector $\lambda_E = (-7/25, 0, 0, 24/25)$ at $r = 3/2$. Writing $h = 22371/122900$, exact evaluation gives

$$\mathcal{B}_+ = \frac{19h + 47g}{100}\bar\rho, \qquad \mathcal{B}_- = \frac{19g}{50}\bar\rho, \qquad \mathcal{B}_{\mathrm{abs}} = \frac{19g}{50}\bar\rho. \tag{47}$$

The asymmetry is real: replacing the original absolute budget by one-sided positivity can change a failed test into an inconclusive one. These rational point evaluations are not uniform estimates for a smooth shell, and no source authenticity follows from them.

Proposition 5.11 (Source-shaped meridional flux, including viscosity)**.** *Let $U = (U_r, U_\theta, U_z)$ be an axisymmetric divergence-free smooth velocity on the cylindrical slice. Consider finitely many poloidal stream functions*

$$\Psi_i(r,z) = K_i a_i(r) b(r) \Xi_i(z), \qquad P_{r,i} = -r^{-1}\partial_z\Psi_i, \quad P_{z,i} = r^{-1}\partial_r\Psi_i,$$

where K_i are constant and $\Xi_i = \Xi_i' = 0$ at both axial endpoints. Put $P = \sum_i P_i$ and allow any purely azimuthal toroidal field $T_\theta e_\theta$. Define the axial moments

$$I_i(r) = \int_c^d U_r(r,z)\Xi_i(z)\,\mathrm{d}z, \quad m_i = \int_c^d \Xi_i(z)\,\mathrm{d}z, \quad J_{ij} = \int_c^d \Xi_i'(z)\Xi_j(z)\,\mathrm{d}z.$$

Then the axial integral of the physical quadratic rz flux increment is

$$\begin{aligned}\mathcal{C}(r) &= \int_c^d [(U+P+T)_r(U+P+T)_z - U_rU_z]\,\mathrm{d}z = \Gamma(r)b'(r) + \Delta(r)b(r) + Q_2(r)b(r)^2,\\ \Gamma &= \frac{1}{r}\sum_i K_i a_i I_i, \qquad \Delta = \frac{1}{r}\sum_i K_i\left[a_i' I_i - a_i\left(I_i' + \frac{I_i}{r}\right)\right],\\ Q_2 &= \frac{1}{r^2}\sum_{i<j} K_iK_j(a_ja_i' - a_ia_j')J_{ij}.\end{aligned} \tag{48}$$

The contribution to the same flux from the canonical Newtonian viscous stress of P is exactly

$$\mathcal{V}_\nu(r) = -\nu\int_c^d(\partial_zP_r + \partial_rP_z)\,\mathrm{d}z = -\nu\sum_i K_im_i\frac{d}{dr}\left(\frac{(a_ib)'}{r}\right). \tag{49}$$

In particular its coefficient of b'' is $-\nu r^{-1}\sum_i K_im_ia_i$; it cannot be included in a $O(b^2)$ remainder merely by assigning a flat radial cutoff.

Proof. The mixed core–shaper product is $r^{-1}\sum_i K_i[(a_ib)'I_i - a_ib\int_c^d U_z\Xi_i'\,\mathrm{d}z]$. Axial integration by parts and $r^{-1}\partial_r(rU_r) + \partial_zU_z = 0$ give $\int U_z\Xi_i' = I_i' + I_i/r$, proving the linear terms. The self-products vanish because $\int\Xi_i'\Xi_i = 0$. For $i \neq j$ integration by parts gives $J_{ji} = -J_{ij}$; the paired cross-products reduce to $r^{-2}K_iK_j(a_ja_i' - a_ia_j')b^2J_{ij}$. All rz components of a purely toroidal velocity vanish. Finally the integral of ∂_zP_r vanishes by the endpoint conditions and $\int\partial_rP_z = \sum_i K_im_i\partial_r((a_ib)'/r)$. This proves the viscous identity and its displayed b'' coefficient. □

For an exact local diagnostic take $U_r = -2r$, $U_z = 4z$, $c = -1$, $d = 1$, $b = (r-1)^3$, $\nu = 1$, and two shapers

$$(K_1, a_1, \Xi_1) = (1, 1, (1-z^2)^2), \qquad (K_2, a_2, \Xi_2) = (2, r, z(1-z^2)^2).$$

At $r = 2$, direct polynomial integration and the reduced formula independently give

$$\Gamma = -\frac{32}{15}, \quad \Delta = \frac{32}{15}, \quad Q_2 = \frac{64}{315}, \quad \mathcal{V}_1 = -\frac{12}{5}, \quad \mathcal{C} + \mathcal{V}_1 = -\frac{2036}{315}. \tag{50}$$

The polynomial functions illustrate the exact algebra on a local slice; their zero extensions are not C^∞ compact H8 cutoffs. Neither the still-free localization stresses nor the common pressure have been inserted into this number.

Proposition 5.12 (Flat-edge second-derivative obstruction)**.** *Under the hypotheses of Proposition 5.11, put $M(r) = \sum_i K_im_ia_i(r)$. The shaper-induced kinetic and viscous stress-shear increment has the exact decomposition*

$$\begin{aligned}\mathcal{C} + \mathcal{V}_\nu &= \kappa_2b'' + \kappa_1b' + \kappa_0b + \kappa_qb^2, \\ \kappa_2 &= -\frac{\nu M}{r}, & \kappa_1 &= \Gamma - \nu\left(\frac{2M'}{r} - \frac{M}{r^2}\right),\\ \kappa_0 &= \Delta - \nu\left(\frac{M''}{r} - \frac{M'}{r^2}\right), & \kappa_q &= Q_2.\end{aligned} \tag{51}$$

Let $r_0 > 0$ and $I = [r_0, r_0+\delta]$, and suppose these four coefficients are bounded on I with $\inf_I|\kappa_2| > 0$. If $b \in C^2(I)$ satisfies $b(r_0) = b'(r_0) = 0$ and another stress-shear contribution R obeys, for finite constants $L, L' \geq 0$,

$$|R| \leq L(|b| + |b'| + b^2), \qquad |\mathcal{C} + \mathcal{V}_\nu + R| \leq L'b^2 \quad \text{on } I, \tag{52}$$

then $b \equiv 0$ throughout I.

Proof. Differentiating $M'b + Mb'$ divided by r in (49) yields $Mb''/r + (2M'/r - M/r^2)b' + (M''/r - M'/r^2)b$, which proves (51). Let $\eta = \inf_I |\kappa_2| > 0$ and $L_b = \sup_I |b| < \infty$. The two bounds in (52), boundedness of the other three coefficients and $b^2 \le L_b|b|$ yield a constant $C < \infty$ such that

$$|b''(r)| \le C(|b(r)| + |b'(r)|) \quad (r \in I).$$

By the initial conditions and the fundamental theorem of calculus, the continuous nonnegative function $w = |b| + |b'|$ satisfies $w(r) \le (1 + C)\int_{r_0}^{r} w(s)\,\mathrm{d}s$. Gronwall's elementary integral inequality gives $w \equiv 0$, and therefore $b \equiv 0$. □

For the two profiles in (50), direct integration gives $m_1 = 16/15$ and $m_2 = 0$; hence $M = 16/15$ and

$$\mathcal{C} + \mathcal{V}_1 = -\frac{16}{15r}b'' + \left(-\frac{32}{15} + \frac{16}{15r^2}\right)b' + \frac{64}{15r}b + \frac{256}{315r^2}b^2. \tag{53}$$

At $r = 2$ and $b = (r-1)^3$ this again gives $-2036/315$. Since $\kappa_2 = -16/(15r)$ is nowhere zero on any annulus with $r > 0$ (indeed $|\kappa_2| \ge 8/15$ on $1 \le r \le 2$), Proposition 5.12 prohibits a nontrivial one-sided flat cutoff from satisfying an $O(b^2)$ budget for this *restricted* flux whenever its remaining terms obey the first inequality of (52).

In the H8 application, the two slice support functions in (42) yield such a target bound only if the relevant reference-centre-subtracted shear is precisely the quantity in (52) and both supports are $O(b^2)$ on the edge collar. Neither follows from the source architecture alone. Nontrivial compactness can instead require a vanishing effective b'' coefficient, a compensating transition or overlap term of second-jet size, a pressure boundary primitive with the requisite source matching, or a density budget larger than $O(b^2)$. The lemma does not rule out these possibilities and is not a contradiction for the authentic H8 residual.

The Case 006 source specifies a compact curl localization $u_b = \nabla \times (\chi A_c)$, separable poloidal and toroidal shell shapers, and a toroidal pressure radial derivative, but does not fix the smooth profiles, the complete four-patch interactions, or a common pressure representative. Hence no value of (41) has yet been certified for the complete H8 residual. The exact code records the transition, shaper, viscosity, overlap, reference-centre and axial boundary channels separately; omitting any named channel blocks evaluation. Supplying zero for a channel requires mathematical source justification, not merely successful execution of the code.

Proposition 5.13 (Quarter-turn localization and its nonlinear overlaps)**.** *Let $Q(x, y, z) = (-y, x, z)$ and χ be invariant under Q. On a simply connected Cartesian chart let A be a smooth vector potential, and denote the active push-forward by $Q_* v(x) = Qv(Q^{-1}x)$. Write*

$$v_j = Q_*^j(\nabla \times (\chi A)), \quad 0 \le j \le 3, \qquad u_b = \frac{1}{4}\sum_{j=0}^{3} v_j.$$

Then u_b is solenoidal and equals the curl of the normalized quarter-turn average of χA. Its full kinetic stress splits exactly into

$$u_b \otimes u_b = \frac{1}{16}\sum_{j=0}^{3} v_j \otimes v_j + \frac{1}{16}\sum_{\substack{0 \le j,k \le 3 \\ j \ne k}} v_j \otimes v_k. \tag{54}$$

For the material derivative, the same diagonal and ordered-cross decomposition holds with $v_j \otimes v_k$ replaced by $(v_j \cdot \nabla)v_k$. The twelve cross products do not generally vanish under fourfold symmetry.

Proof. Rigid rotations commute with curl, and the curl of an average is the average of the curls. Each v_j is divergence free. Expanding the product of the two sums for u_b proves (54); applying the Leibniz bilinearity of directional differentiation proves the derivative formula. The normalization $1/16$ follows from two factors of $1/4$ and cannot be dropped. □

For example, on a local chart take $\chi = 1$ and $A = (-yz, xz, 0)$. Every rotated field is $v_j = (-x, -y, 2z)$, so in cylindrical notation $u_r = -r$, $u_z = 2z$, and $u_r u_z = -2rz$. The diagonal terms in (54) contribute $-rz/2$ to this shear, whereas the ordered overlaps contribute $-3rz/2$. Thus three quarters of the core shear is carried by the overlap partition even for an exactly fourfold-equivariant field. This local polynomial example does not claim a compact source profile; spatial disjointness, if asserted for different carrier supports, must be checked separately.

Proposition 5.14 (Shared-envelope second jet and core circulation)**.** *Let $r_0 < r < r_1$ be a cylindrical annulus separated from the axis, with angular coordinate θ and fixed axial endpoints $z_- < z_+$. Let $\mathcal{A} = (\mathcal{A}_r, \mathcal{A}_\theta, \mathcal{A}_z)$ be a smooth, 2π-periodic vector potential in the cylindrical orthonormal frame (possibly only C_4-equivariant), and let $U_c = \nabla \times \mathcal{A}$. Suppose the localized core and the poloidal stream functions of Proposition 5.11 share one radial envelope $b(r)$, so their total meridional correction is obtained from*

$$\nabla \times \left[b(r) \left(\mathcal{A} + \sum_i \frac{K_i a_i(r) \Xi_i(z)}{r} e_\theta \right) \right].$$

An arbitrary smooth background independent of b may be included. Denote the full angular mean by $\langle \cdot \rangle_\theta$, set

$$M(r) = \sum_i K_i m_i a_i(r), \qquad V(r) = \int_{z_-}^{z_+} \langle (U_c)_z(r, \theta, z) \rangle_\theta \, \mathrm{d}z,$$

and let $S_b(r)$ be the axial/angular integral of the increment in the physical shear $K_{rz} = u_r u_z - \nu(\partial_z u_r + \partial_r u_z)$. The coefficient of $b''(r)$ in S_b is exactly

$$\kappa_{2,\mathrm{eff}}(r) = -\nu \left[\int_{z_-}^{z_+} \langle \mathcal{A}_\theta(r, \theta, z) \rangle_\theta \, \mathrm{d}z + \frac{M(r)}{r} \right]. \tag{55}$$

All the remaining terms in S_b are expressions in b, b' with smooth bounded coefficients on compact subannuli. In particular, the quadratic overlaps among the four velocities contain no b'' term in this physical shear *channel.*

For any reference radius $r_* \in [r_0, r_1)$, put

$$H_* := r_* \int_{z_-}^{z_+} \langle \mathcal{A}_\theta(r_*, \theta, z) \rangle_\theta \, \mathrm{d}z.$$

Then (55) has the potential-free expression

$$\kappa_{2,\mathrm{eff}}(r) = -\frac{\nu}{r} \left[H_* + \int_{r_*}^{r} s V(s) \, \mathrm{d}s + M(r) \right]. \tag{56}$$

Consequently, identical *second-jet cancellation on an interval requires and is implied by a single anchor equality $H_* + \int_{r_*}^{r_a} s V(s) \, \mathrm{d}s + M(r_a) = 0$ and the differential identity*

$$r V(r) + M'(r) = 0 \tag{57}$$

throughout that interval. If the potential extends smoothly across the axis on complete disks, one may set $r_* = 0$ and $H_* = 0$.

Proof. Writing $A_\theta^{\mathrm{tot}} = \mathcal{A}_\theta + \sum_i K_i a_i \Xi_i / r$, the cylindrical curl gives

$$u_r - U_r = b\left(\frac{1}{r}\partial_\theta \mathcal{A}_z - \partial_z A_\theta^{\mathrm{tot}}\right), \qquad u_z - U_z = b\left(\frac{1}{r}\partial_r(rA_\theta^{\mathrm{tot}}) - \frac{1}{r}\partial_\theta \mathcal{A}_r\right) + b' A_\theta^{\mathrm{tot}}.$$

The quadratic product $u_r u_z$ therefore uses at most b and b'. In the viscous shear only $\partial_r(u_z - U_z)$ produces b'', with coefficient $-\nu A_\theta^{\mathrm{tot}}$. Angular/axial integration proves (55). Meanwhile

$$(U_c)_z = \frac{1}{r}\partial_r(r\mathcal{A}_\theta) - \frac{1}{r}\partial_\theta \mathcal{A}_r.$$

The full angular average annihilates the last derivative; hence $\partial_r(r\int\langle\mathcal{A}_\theta\rangle_\theta \,\mathrm{d}z) = rV(r)$. Integrating from r_* proves (56). Differentiating its bracket proves the stated characterization of identical cancellation. Smooth axis extension forces $r\langle\mathcal{A}_\theta\rangle_\theta \to 0$ as $r \downarrow 0$. □

The annular reference circulation matters. For the single profile $(K, a, \Xi) = (1, 1, (1-z^2)^2)$ on $[-1, 1]$, one has $M = 16/15$. Choosing the local, C_4-invariant azimuthal potential $\mathcal{A}_\theta = -8/(15r)$ gives $\int_{-1}^{1} \mathcal{A}_\theta \,\mathrm{d}z = -16/(15r)$ and $\kappa_{2,\mathrm{eff}} \equiv 0$. Here $(U_c)_z = 0$ on the annulus, while $H_* = -16/15$ supplies the cancellation. This is an algebraic counterexample to deleting the circulation constant, not an admissible axis-spanning H8 core. If a smooth axis-spanning potential is prescribed, its core axial flux rather than a free annular circulation must supply the needed moment.

The proposition concerns the canonical physical rz stress. The pressure-compensated source relation (41) retains its independent axial jump; neither pressure nor an H8 inverse stress is reconstructed here. If the actual transition cutoff χ differs from the shaper envelope b, the core contributions cannot be assigned to the same b'' coefficient without an additional source identity. Thus the genuine H8 test still requires the core flux V, the reference circulation, the shared-cutoff decision, and the full source-matched cone density and frame.

Proposition 5.15 (Independent core and shaper cutoffs)**.** *On an annular cylindrical patch let $\mathcal{A}$ be any smooth, 2π-periodic vector potential, let $\chi = \chi(r, \theta, z)$ be a smooth core cutoff, and let $b = b(r)$ be an* independent *radial poloidal cutoff. Put*

$$P_\theta(r, z) := \sum_i \frac{K_i a_i(r)\Xi_i(z)}{r}, \qquad u = U + \nabla \times (\chi\mathcal{A} + bP_\theta e_\theta),$$

where U is a fixed smooth background. If $S(r)$ is the axial/angular integral of the physical shear $K_{rz} = u_r u_z - \nu(\partial_z u_r + \partial_r u_z)$, its full cutoff-second-*derivative part is*

$$S_{\mathrm{jet},2}(r) = -\nu \int_{z_-}^{z_+} \left\langle \mathcal{A}_\theta(\chi_{rr} - \chi_{zz}) + \frac{\mathcal{A}_z}{r}\chi_{\theta z} - \frac{\mathcal{A}_r}{r}\chi_{r\theta} + P_\theta b'' \right\rangle_\theta \mathrm{d}z. \tag{58}$$

The difference $S - S_{\mathrm{jet},2}$ contains the cutoffs and their first derivatives, but no second derivative of either cutoff. The nonlinear cross terms, including all ordered C_4 core overlaps, belong to this remainder. In particular, a radial shaper b and a genuinely three-variable χ do not have one common second-derivative coefficient.

For any specified pressure representative p, the complete annular spatial-divergence *source still obeys*

$$q_{\mathrm{s}}(r) = [\langle u_z^2 - 2\nu\partial_z u_z + p\rangle_\theta]_{z_-}^{z_+}, \qquad J_f = S - \frac{r_a}{r}S(r_a) + \frac{1}{r}\int_{r_a}^{r} s q_{\mathrm{s}}(s)\,\mathrm{d}s.$$

For the full unforced Navier–Stokes residual $\mathcal{N}(u, p) = \partial_t u + \operatorname{div} K + \nabla p$, one must also keep

$$T_z(r) = \int_{z_-}^{z_+} \langle \partial_t u_z\rangle_\theta \,\mathrm{d}z, \qquad J_{\mathcal{N}}(r) = J_f(r) + \frac{1}{r}\int_{r_a}^{r} s T_z(s)\,\mathrm{d}s. \tag{59}$$

No smallness of the pressure primitive or the first-derivative remainder follows from (58); neither T_z nor its primitive may be set to zero without a source identity.

Proof. The cylindrical curl identities give

$$u_r - U_r = \frac{1}{r}\partial_\theta(\chi\mathcal{A}_z) - \partial_z(\chi\mathcal{A}_\theta + bP_\theta),$$
$$u_z - U_z = \frac{1}{r}\partial_r\big(r(\chi\mathcal{A}_\theta + bP_\theta)\big) - \frac{1}{r}\partial_\theta(\chi\mathcal{A}_r).$$

Each velocity contains at most one derivative of either cutoff; therefore $u_r u_z$ contains no cutoff second derivative. Differentiating the first line in z and the second in r produces exactly the four terms in (58). The rest contains only lower cutoff jets. The pressure identity is the integrated axial component of $\operatorname{div} K + \nabla p = f$ and does not identify an independent radial primitive of p with a time-only gauge. Adding the axial marginal of $\partial_t u$ and integrating the radial first-order equation proves (59). □

The source distinction is substantive: the available Case 006 structured-stress manuscript prescribes the core cutoff as a C_4-invariant function of similarity coordinates, equal to one on an inner subcore, whereas its poloidal shaper uses a separately specified shell profile $\Phi(r/\ell)\Xi(z/c)$. Those definitions neither identify the two radial envelopes nor provide a complete frozen four-patch source. Equation (58) is therefore the appropriate general decomposition; the shared-envelope Proposition 5.14 is its additional-hypothesis specialization.

Proposition 5.16 (Variable-frame dual enclosure with angular test)**.** *Consider the one-sided seven-ray coefficient cone and its trace-free representative H in (16). For every nonnegative density ρ, the pointwise coefficient floors imply*

$$|A|, |B|, |C|, |D| \le \frac{\rho}{4}, \qquad \|H\|_F \le \rho.$$

Let a cylindrical shear-test slice be partitioned into finitely many measurable cells Ω_j. On each cell suppose the physical orthogonal frame $E(x)$ admits a fixed rational orthogonal frame E_j and a certified *bound*

$$\sup_{\Omega_j}\|E(x) - E_j\|_F \le \delta_j, \qquad m_j = \int_{\Omega_j} \rho(x)\,\mathrm{d}\mu(x).$$

Here $\mathrm{d}\mu = \mathrm{d}z\,\mathrm{d}\theta/(2\pi)$ on the slice. If $\sigma^+(\lambda)$ is the exact unit-density one-sided support and $G(x) = e_r(\theta)\otimes e_z$, define

$$\widehat{B}_\pm = \sum_j \int_{\Omega_j} \rho(x)\,\sigma^+\Big(\pm\lambda(E_j^{\mathsf{T}} G(x) E_j)\Big)\,\mathrm{d}\mu(x), \qquad \varepsilon_E = 2\sum_j m_j\delta_j.$$

Then the genuine spatially varying shear supports satisfy

$$\max(0, \widehat{B}_\pm - \varepsilon_E) \le B_\pm \le \widehat{B}_\pm + \varepsilon_E. \tag{60}$$

These bounds remain valid for smooth nonanalytic frames and densities, provided the weighted cell support integrals and frame-error bounds are rigorously enclosed. Alternatively the integrals may be bounded by m_j times certified cellwise lower and upper support values. Sampling a frame and a physical test at cell centres alone does not provide such a certificate.

Proof. Applying the exact support formula of Proposition 5.5 to the eight signed coordinate covectors gives the largest respective normalized absolute extrema

$$\frac{51113}{245800}, \quad \frac{15971}{983200}, \quad \frac{19171}{122900}, \quad \frac{19171}{491600}.$$

Each is strictly less than $1/4$, so $\|H\|_F^2 = 2(A^2+B^2+C^2+D^2) \le \rho^2/2$. For orthogonal E, E_j,

$$EHE^{\mathsf{T}} - E_j H E_j^{\mathsf{T}} = (E-E_j)HE^{\mathsf{T}} + E_j H(E-E_j)^{\mathsf{T}}.$$

The Frobenius pairing with the unit-norm $G(x)$ consequently changes by at most $2\rho\|E-E_j\|_F$. Taking the pointwise supremum over the coefficient cone, retaining $G(x)$ in the frozen-frame support, and then integrating over each cell proves (60). □

An earlier cell-mass-only expression which omitted the integral in $\widehat{B}_\pm$ is invalid for physical Cartesian reference frames. Indeed, even if $E(x) = E_j = I$ and $\delta_j = 0$, its test tensor $G(\theta)$ varies. At $\theta = 0$ the unit-density support is $51113/245800$, while at $\theta = \pi/2$ it is $15971/983200$. Thus zero frame error does not imply zero angular-test error. A piecewise-constant-test implementation is valid only with an additional certified bound $\eta_j \ge \sup_{\Omega_j}\|G(x) - G_j\|_F$, in which case $\sum_j m_j\eta_j$ is added to ε_E.

The dual bound controls values of the rotating stress, not its divergence. In a Cartesian chart, $K = EHE^{\mathsf{T}}$ obeys

$$\partial_j K_{ij} = E_{ia}E_{jb}\partial_j H_{ab} + ((\partial_j E_{ia})E_{jb} + E_{ia}(\partial_j E_{jb}))H_{ab}, \tag{61}$$

with repeated indices summed. Thus even constant H can generate a nonzero source when E rotates. Cylindrical components require their geometric connection terms as well.

The angular form of this connection is now an exact Lean theorem. More generally, let $E(t)$ be a differentiable orthogonal 3×3 frame, $H(t)$ a differentiable coefficient matrix, and $\Omega(t) = E(t)^{\mathsf{T}}E'(t)$. Differentiation of $E^{\mathsf{T}}E = I$ gives $\Omega^{\mathsf{T}} = -\Omega$. The ordinary matrix product rule and $E' = E\Omega$ yield

$$\frac{d}{dt}(EHE^{\mathsf{T}}) = E(H' + \Omega H - H\Omega)E^{\mathsf{T}}. \tag{62}$$

The `FrameConnectionDerivative` module proves each scalar physical-tensor entry derivative from explicit entrywise derivative hypotheses, proves the skew and commutator identities under row orthogonality and differentiated orthogonality, and checks an actual trigonometric planar rotation. For $H = \mathrm{diag}(1,2,0)$, the $(0,1)$ entry changes at rate -1 at zero angle. This verifies the sign of (62) and refutes omission of its commutator even for a constant frame-coordinate stress.

Proposition 5.17 (Complete moving-frame source and axial trace)**.** *Let $E(x) \in SO(3)$ be a differentiable physical frame on a Cartesian chart, and let $H = H^{\mathsf{T}}$ be its differentiable stress-coordinate matrix. Write $\Gamma_i = E^{\mathsf{T}}\partial_i E$, where the index i denotes a Cartesian derivative, and set $f = \nabla\cdot(EHE^{\mathsf{T}}) + \nabla p$. Then, for each moving-frame component b,*

$$(E^{\mathsf{T}}f)_b = \sum_{i,a} E_{ia}\left(\partial_i H + \Gamma_i H - H\Gamma_i\right)_{ab} + \sum_i E_{ib}\partial_i p. \tag{63}$$

For the standard cylindrical *frame $E = (e_r, e_\theta, e_z)$ and the four-field stress in (16), the corresponding components are*

$$f_r = -\partial_r D + r^{-1}\partial_\theta C + \partial_z A - r^{-1}D + \partial_r p, \tag{64}$$

$$f_\theta = \partial_r C + 2r^{-1}C + \partial_z B + r^{-1}\partial_\theta p, \tag{65}$$

$$f_z = \partial_r A + r^{-1}A + r^{-1}\partial_\theta B + \partial_z(D+p). \tag{66}$$

If A, B, D, p are 2π-periodic in θ, have the indicated derivatives and axial traces at c, d, then

$$\int_0^{2\pi}\int_c^d r f_z \,\mathrm{d}z\,\mathrm{d}\theta = \partial_r\left(r\int_0^{2\pi}\int_c^d A\,\mathrm{d}z\,\mathrm{d}\theta\right) + r\int_0^{2\pi}[D+p]_c^d\,\mathrm{d}\theta. \tag{67}$$

In particular the pressure jump generally survives axial integration. None of these identities identifies the possibly tilted spatial frame of the unfrozen H8 construction with the standard cylindrical frame.

Proof. Orthogonality implies $\Gamma_i^{\mathsf{T}} = -\Gamma_i$ and hence $\partial_i(EHE^{\mathsf{T}}) = E(\partial_i H + \Gamma_i H - H\Gamma_i)E^{\mathsf{T}}$. Contract the Cartesian derivative index against the first tensor index and transform the remaining vector index by E^{T} to obtain (63). For the cylindrical frame,

$$E^{\mathsf{T}}\partial_\theta E = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \partial_r E = \partial_z E = 0.$$

Substitution gives (64)–(66). Integrating the last identity, periodicity cancels the angular derivative and the fundamental theorem of calculus produces the *combined* stress-pressure axial trace. □

The exact one-sided seven-ray floor also gives the sharp local necessary bound

$$|H_{rz}| = |A| \le \frac{51113}{245800}\,\rho. \tag{68}$$

This constant is attained by an exact rational ray allocation, and agrees with Proposition 5.16. Consequently, any source-matched construction in the frozen physical frame must simultaneously obey (67) and (68). For a spatially tilted frame the latter inequality remains in the *local* four-field coordinates, while (63) replaces the cylindrical expression. The symbolic verification accompanying this manuscript tests all three Cartesian divergence components, a nonconstant spatial tilt, and a nonzero axial pressure jump. No authentic H8 profile tuple has been certified by these tests, and the compact source-matched positive-stress inverse remains an open analytic problem.

Proposition 5.18 (Zero-density collar and flat-edge curl obstruction)**.** *Let $U \subset \mathbb{R}^3$ be open, and suppose $\rho \ge 0$, A, B, C, D, a source frame $E \in SO(3)$ and a pressure p are smooth. Suppose the original one-sided seven-ray floors hold pointwise and, writing $K = EHE^{\mathsf{T}}$ for $H = H(A, B, C, D)$, the actual Cartesian source identity is*

$$f = \operatorname{div} K + \nabla p. \tag{69}$$

Then on every open region $V \subset U$ where $\rho \equiv 0$,

$$A = B = C = D = 0, \qquad K = 0, \qquad \nabla \times f = 0. \tag{70}$$

More strongly, at any point $x_0 \in U$ where $\partial^\alpha \rho(x_0) = 0$ for every multi-index α,

$$\partial^\alpha(\nabla \times f)(x_0) = 0 \qquad \text{for every multi-index } \alpha. \tag{71}$$

In particular this applies at a boundary point of the support of a globally smooth compactly supported density. The conditions are necessary, not sufficient, for a positive-stress inverse.

Proof. At $\rho = 0$, the six paired ray inequalities imply first $D \ge 0$ from rays four and five, and $D \le 0$ from rays two and three (one-based indexing). They then force $C = 0$, $B = 0$ and $A = 0$. Thus $K = 0$ on V, and taking curl in (69) removes the pressure gradient. For the second assertion, the exact bounded seven-ray polytope implies $|A| + |B| + |C| + |D| \le C_*\rho$ for an absolute constant C_*. Flatness of the smooth nonnegative ρ at x_0 gives $\rho(x) = o(|x - x_0|^N)$ for every N. Each smooth stress coefficient has the same property; otherwise its first nonzero Taylor polynomial would dominate ρ along some direction. Hence all derivatives of H, and therefore of $K = EHE^{\mathsf{T}}$, vanish at x_0. The identity $\nabla \times f = \nabla \times \operatorname{div} K$ finishes the proof. □

This supplies a sharp *source-side* diagnostic for the authentic H8 collar: a nonzero source-curl value on the exterior zero-density region, or a nonzero curl jet at a smooth flat edge, would preclude a compact, seven-ray-positive correction for that same source and pressure representative. It cannot be evaluated until the full unfrozen physical residual is supplied; examples constructed with diagnostic cutoffs do not decide H8.

Proposition 5.19 (Sharp quadratic-vacuum curl parallelogram)**.** *Retain the hypotheses of Proposition 5.18, and suppose at a point x_0 that*

$$\rho(x_0) = 0, \qquad \nabla^2\rho(x_0) = \lambda\, n \otimes n, \qquad n = E(x_0)e_3, \quad \lambda \geq 0.$$

Write $\omega = E(x_0)^{\mathsf{T}}(\nabla \times f)(x_0)$. With $g = 15971/122900$ and $h = 22371/122900$, the following sharp pointwise necessary *inequalities hold:*

$$\omega_3 = 0, \qquad |\omega_1| \leq \frac{g}{8}\lambda, \qquad |\omega_2 + 4\omega_1| \leq \frac{3g+h}{4}\lambda = \frac{17571}{122900}\lambda. \tag{72}$$

The planar parallelogram is exactly attained by local smooth quadratic stress jets; no global compact-support or source-matching claim is implied. The bound is independent of the derivatives of E at x_0 and of the pressure representative.

Proof. The nonnegative smooth density has $\nabla\rho(x_0) = 0$. The bounded one-sided seven-ray polytope gives $|A| + |B| + |C| + |D| \leq C_*\rho$; hence the four coefficients and their gradients vanish at x_0. For each constant direction v, differentiate twice the seven nonnegative ray-floor slacks. Since each slack has a local minimum zero at x_0, the four numbers $D_v^2(A, B, C, D)(x_0)$ satisfy *the same* seven linear floors with density $D_v^2\rho(x_0) = \lambda(n \cdot v)^2$. This cone is bounded and reduces to the origin for zero density. In particular, the second directional jets vanish for $v \perp n$. For $v \perp n$, the coefficient jet on $n + tv$ is affine in t because $D_v^2(A, B, C, D) = 0$, but remains in the fixed bounded cone with density λ for every real t. Its mixed n, v coefficient must therefore vanish. Polarization also annihilates the cross-jets within $n^{\perp}$. Thus, for a four-tuple (a, b, c, d) satisfying the seven floors at density λ, the Hessian of each coefficient is its corresponding scalar times $n \otimes n$.

All terms involving derivatives of E disappear from the second jet of $K = EHE^{\mathsf{T}}$, because $H(x_0) = 0$ and $\nabla H(x_0) = 0$. In the positively oriented coordinates of $E(x_0)$ one obtains

$$(\nabla \times f)(x_0) = n \times \big(E(x_0)H(a, b, c, d)e_3\big), \qquad \omega = (-b, a, 0).$$

The final two ray floors give $|b| \leq g\lambda/8$. Adding the pair involving c gives $d \leq (g + h)\lambda/8$; the opposite axial pair then yields $|a - 4b| \leq g\lambda/2 + 2d \leq (3g + h)\lambda/4$. Substitution proves (72).

Conversely, any a, b in that parallelogram admit the explicit lift

$$c = \frac{h-g}{4}\lambda, \qquad d = \frac{g+h}{8}\lambda,$$

which meets all seven original floors. Taking locally $\rho(x) = \lambda s^2/2$ and $H(x) = s^2H(a, b, c, d)/2$, where $s = n \cdot (x - x_0)$, realizes the indicated second jets and all seven floors. The construction asserts sharpness only within this local quadratic model. □

Equation (72) is a stronger source diagnostic at an axial quadratic zero than separate curl component caps. The exact-arithmetic companion `h8_vacuum_curl_gate.py` checks membership, constructs the sharp ray lift, and tests a spatially rotating frame. As with the flat-edge result, no authenticated Case 006 source has yet been evaluated against these inequalities.

Corollary 5.20 (Sharp azimuthal torque-flux compatibility)**.** *Under the cylindrical-frame hypotheses of Proposition 5.17, define*

$$\mathfrak{c}(r) = \int_c^d \langle C\rangle_\theta \, \mathrm{d}z, \qquad \mathfrak{b}(r) = [\langle B\rangle_\theta]_c^d, \qquad \mathfrak{f}_\theta(r) = \int_c^d \langle f_\theta\rangle_\theta \, \mathrm{d}z,$$

where $\langle\cdot\rangle_\theta$ is the normalized full-circle angular average. Then

$$\mathfrak{f}_\theta(r) = \frac{1}{r^2}\partial_r(r^2\mathfrak{c}(r)) + \mathfrak{b}(r), \tag{73}$$

and for $a > 0$,

$$\mathfrak{c}(r) = \frac{a^2}{r^2}\mathfrak{c}(a) + \frac{1}{r^2}\int_a^r s^2(\mathfrak{f}_\theta(s) - \mathfrak{b}(s)) \, \mathrm{d}s. \tag{74}$$

If the original one-sided seven-ray positivity floor holds at every point and $\rho \geq 0$, put $\mathcal{R}(r) = \int_c^d \langle\rho\rangle_\theta \, \mathrm{d}z$. The compact positive-stress construction must satisfy

$$-\frac{15971}{122900}\,\mathcal{R}(r) \leq \mathfrak{c}(r) \leq \frac{19171}{122900}\,\mathcal{R}(r). \tag{75}$$

Both constants are attained by pointwise exact rational seven-ray allocations. The two-sided interval is asymmetric*; replacing it by a symmetric absolute-correction budget would impose a different condition.*

Proof. Average (65) over a complete angular period. The pressure term integrates to zero by periodicity, but $[\langle B\rangle_\theta]_c^d$ need not vanish. The radial identity follows by writing $\partial_r C + 2C/r = r^{-2}\partial_r(r^2 C)$. The fundamental theorem of calculus gives (74). For the one-sided ray cone, the original seven coefficient floors imply the pointwise sharp interval

$$-\frac{15971}{122900}\rho \leq C \leq \frac{19171}{122900}\rho.$$

Its endpoints follow directly by combining the second, third, fourth and fifth ray inequalities; rational endpoint allocations show attainability. Integrating the pointwise interval proves (75). □

The torque-flux test is independent of the axial meridional test: it controls a different stress coordinate and has a different radial weight (r^2 in place of r). It applies directly only when the source frame is the standard cylindrical frame. For any additional spatial tilt, the covariant Cartesian source identity (63) and the rotated cone support must be used instead. Without authenticated H8 functions, neither flux condition is a completed source-matching test.

An independently checked Cartesian rotation example and exact two-cell rational support tests accompany these formulas. Neither a bound on δ_j nor (60) controls ∇E, so the weighted symmetric-divergence inverse and its derivative budget remain open.

The form of a compact cutoff may be fixed in a *test family* without identifying it with the source's unfrozen choice. Define

$$h(s) = \begin{cases} 0, & s \leq 0, \\ \dfrac{\exp(-1/s)}{\exp(-1/s) + \exp(-1/(1-s))}, & 0 < s < 1, \\ 1, & s \geq 1, \end{cases} \qquad \chi_*(r) = h(r-1)h(4-r).$$

Then χ_* is smooth and nonanalytic, zero outside $[1,4]$, identically one on $[2,3]$, and equals $1/2$ at $r = 3/2, 7/2$. As a radial function on the surrounding chart it is C_4-invariant; the hole at

$r \le 1$ makes its extension across the axis smooth. This provides a reproducible candidate for testing derivative estimates, not a determination of the H8 similarity-coordinate cutoff.

The source's toroidal pressure prescription likewise fixes only

$$\partial_r p_T = \frac{2u_\theta T_\theta + T_\theta^2}{r}. \tag{76}$$

For a chosen inner-radius normalization $p_T(a, z) = 0$, its representative is

$$p_T(r, z) = \int_a^r \frac{2u_\theta(s, z)T_\theta(s, z) + T_\theta(s, z)^2}{s}\,\mathrm{d}s.$$

For $a = 1$, $u_\theta = r$ and $T_\theta = rz(1 - z)$ this gives

$$p_T = \frac{r^2 - 1}{2}(2z(1 - z) + z^2(1 - z)^2), \qquad [p_T]_{z=0}^{z=1} = 0.$$

By contrast $T_\theta = rz$ gives the nonzero pressure jump $[p_T]_{-1}^{1} = 2(r^2 - 1)$. The arbitrary radial integration trace $C(z)$ must be fixed by the remaining physical pressure equations; it is *not* an arbitrary gauge of the complete Navier–Stokes system.

To test all pieces in one local example, take $0 \le z \le 1$, $U_r = -r$, $U_z = 2z$, $U_\theta = r$, and the one-shaper data $K = a = 1$, $\Xi = z^2(1 - z)^2$, $b = (r - 1)^3$, $\nu = 1$, with the toroidal profile and pressure from the preceding zero-jump example. The C_4 core shear is partitioned into a diagonal $-r/4$ and ordered overlaps $-3r/4$; the shaper and viscous parts follow from Proposition 5.11. Directly integrating $K_{rz} = u_r u_z - (\partial_z u_r + \partial_r u_z)$ gives exactly

$$S(r) = \frac{-r^4 - 30r^3 - 2r + 3}{30r^2}, \qquad q_{\mathrm{s}}(r) = 4, \qquad J_f(r) = \frac{-r^4 + 30r^3 - 32r + 3}{30r^2}. \tag{77}$$

At $r = 3/2$ the exact values are $S = -63/40$ and $J_f = 91/120$. The independently computed axial source integral $F = (rS)'/r + q_{\mathrm{s}}$ reproduces J_f by integration from $a = 1$. The example uses a *polynomial local core and shaper*, not χ_*; it verifies the algebraic combination and avoidance of double-counted core terms. The smooth nonanalytic cutoff and the authentic Case 006 transition, frame, and reference-centre fields still require their own complete weighted calculation.

6 The physical residual and terminal speed

Define the viscosity-one residual

$$\mathcal{N}(u, p) = u_t + (u \cdot \nabla)u - \Delta u + \nabla p. \tag{78}$$

For smooth (u, p) and (w, q), direct expansion yields

$$\mathcal{N}(u + w, p + q) = \mathcal{N}(u, p) + \mathcal{N}(w, q) + (u \cdot \nabla)w + (w \cdot \nabla)u. \tag{79}$$

Both nonlinear cross terms are essential. Equation (79), with explicit differentiability assumptions in the upstream physical definitions, is checked in Lean by the declaration `UnforcedPatching.navierStokesResidual_add`.

An exact positive symmetric-divergence inverse does not, by itself, cancel the unforced residual. For example, on the unit three-torus let $u = (\sin 2\pi y, 0, 0)$ and $p = 0$. Then $\operatorname{div} u = 0$ and $(u \cdot \nabla)u = 0$, but

$$\mathcal{N}(u, 0) = 4\pi^2 u \neq 0.$$

The symmetric periodic tensor

$$S = 10I - 2\pi\cos(2\pi y)(e_1 \otimes e_2 + e_2 \otimes e_1)$$

is positive definite and satisfies $\operatorname{div} S = 4\pi^2 u$. This example concerns positive tensors in general, not the particular H8 seven-ray cone. It shows why a source-matched stress is only a precursor to a nonlinear correction: an actual divergence-free w and pressure q must solve

$$\partial_t w - \Delta w + (u \cdot \nabla)w + (w \cdot \nabla)u + (w \cdot \nabla)w + \nabla q = -\mathcal{N}(u, p)$$

with the necessary initial trace, periodic compatibility, regularity and terminal estimates.

Proposition 6.1 (Conditional terminal retention)**.** *Suppose that u has arbitrarily large speed at points with $t \uparrow 1$. If a correction w satisfies*

$$\|w(t,x)\| \le \alpha\|u(t,x)\| + C_0, \qquad 0 \le \alpha < 1, \quad C_0 \ge 0, \tag{80}$$

uniformly on $0 < t < 1$ and $x \in \mathbb{T}^3$, then $u + w$ also has arbitrarily large speed arbitrarily close to time one.

Proof. The reverse triangle inequality and (80) give

$$\|u + w\| \ge \|u\| - \|w\| \ge (1 - \alpha)\|u\| - C_0.$$

Given any target speed $M > 0$ and any terminal neighbourhood, choose a point therein where $\|u\| > (M + C_0)/(1 - \alpha)$. Then $\|u + w\| > M$ at that point. □

This last implication is kernel-checked by the Lean theorem

`UnforcedPatching.speedUnbounded_add_subleading`

It is deliberately conditional: no field u meeting the singularity premise, and no actual PDE correction w meeting (80), has been supplied by these lemmas.

7 What the formal development proves

The Lean project is pinned to Lean 4.34.0-rc2 and a fixed Mathlib revision. The checked physical interface states smoothness, three coordinate periods, zero forcing, nonzero initial data, finite fundamental-cell kinetic energy and endpoint speed divergence. Its existential proposition is named `UnforcedPatching.unforcedTargetStatement`; there is no proof term inhabiting that proposition.

The precise correspondence between the paper and the compiled `UnforcedPatching` declarations is as follows. In the second column, "missing" refers to mathematical constructions, not to unresolved Lean goals inside the named theorems.

Lean module	Checked scope and the missing construction
`Definitions`	Physical candidate predicates; no candidate witness.
`Residual`, `Correction`	Full residual addition and conditional cancellation; no constructed correcting field.
`Periodicity`, `PressureGauge`	Sum and time-only gauge identities; no periodic singular solution.
`ConeBudget`, `ConeOperator`, `SevenRayCenter`	Exact rational centre and sharp factor 14; no source-matched weighted positive stress.

Lean module	Checked scope and the missing construction (continued)
`FourFieldPhysical`	Exact nine-entry ray-to-cylindrical tensor map modulo an isotropic gauge; entrywise four-coordinate bound in the frozen frame, not a spatially rotated PDE bridge.
`ExactCoefficientCone`, `SharpCoefficientDual`	Exact absolute-budget seven-coefficient polytope and sharp dual upper estimate; no frozen final Case 006 residual or spatial inverse.
`PositiveCoefficientCone`	Universal one-sided seven-ray positive-cone support bound; not a spatial stress inverse or a continuum integral formalization.
`AxialFluxCone`, `JointFluxCone`	The separate axial/azimuthal caps and the coupled hexagon necessary bounds are finite algebra; new branch declarations require a fresh head-commit compilation and axiom receipt. The constructive hexagon lift is proved in the text and rational tests, not yet a continuum inverse.
`RotatingFrameDual`	Exact source-to-physical matrix Frobenius pullback and pointwise rotated one-sided dual bound; no derivative-of-frame or PDE source matching theorem.
`FrameConnectionDerivative`	Genuine entrywise product-rule derivative of EHE^{T}, skew angular connection and its commutator, including an exact moving-trigonometric-frame sign test; authentic H8 bounds and PDE source matching remain open.
`CompactGauge`	One-variable gauge derivative; no compact tensor support theorem.
`MeridionalInverse`	Actual axial and radial interval-integral FTC identities, geometric hoop cancellation, and pointwise radial/axial cylindrical inverse; no compact support theorem.
`Case006SourceBand`	Sharp finite-dimensional source-frame axial interval with an explicit one-sided-cone counterexample; no authentic spatial source identity.
`PressureCompensatedFlux`	Continuum radial FTC for the pressure-compensated axial source, conditional on the shear differential identity; axial pressure/tensor trace retained.
`CartesianAxialBridge`	Actual Cartesian axial divergence for axisymmetric mixed stress, including geometric K_{rz}/r and nonzero swirl; not a general tensor inverse.
`NonaxisymmetricAxialBridge`	Cartesian axial divergence with both radial and angular directional derivatives for arbitrary locally differentiable mixed stress; axial pressure derivative retained.
`AngularMean`	Exact periodic angular FTC removing the angular derivative after averaging, under explicit differentiability and integrability assumptions.
`PhysicalCircleAngular`	Physical circle chain rule and exact $-y\partial_x + x\partial_y$ correspondence, automatic angular endpoint equality, and integrated cancellation for Frechet-differentiable stress with integrable angular derivative.
`Retention`	Subleading corrections preserve terminal speed; neither required field constructed.

In particular, the residual-addition, correction-closure, seven-ray-floor, and terminal-retention declarations are all implications with explicit hypotheses. Their conjunction does not yield an inhabitant of `unforcedTargetStatement`. The printed axioms for the selected smoke-tested declarations are the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`.

In particular, `meridional_correction_divergence` checks both components of (3)–(4) for the hoop stress (9) and the actual interval-integral axial primitive (8), under stated local continuity and integrability assumptions. The separate declarations `radialMoment_relation` and `mixedCarrier_radial_balance` verify the antiderivative identity in (6) and its product carrier (7). The further theorem `sourceMatched_axialStress` checks the pointwise cylindrical source match $S_{zz} = \rho$, $f_z = \partial_z \rho$ used in Proposition 5.1. These results compile without proof admissions; selected axiom reports list only Lean's standard logical axioms. The further `CartesianAxialBridge` and `NonaxisymmetricAxialBridge` theorems now establish the actual Cartesian *axial* divergence component, including the nonzero angular directional derivative. The `PhysicalCircleAngular` theorem derives that derivative by the Frechet chain rule along a genuine Cartesian circle, supplies the periodic endpoint equality, and invokes `AngularMean` to remove its contribution only *after* integration. The `FrameConnectionDerivative` theorem separately derives the full product-rule differentiation of rotated stresses, including the $\Omega H - H\Omega$ connection term; it does not assume that E is spatially constant. The `PressureCompensatedFlux` theorem proves the separate radial FTC with the full axial boundary trace retained. Derivation of the integrated source–shear differential hypothesis from authenticated H8 profiles, all necessary interchange-of-differentiation conditions, authentic H8 frame profiles and uniform connection estimates, the complete Cartesian tensor divergence, global support, moment compatibility and weighted positive inverse remain open. Thus the full restricted Theorem 2.1 is *partially*, not fully, formalized.

An independent exact-arithmetic test evaluates a nontrivial annular polynomial instance of the radial compatibility identity. For the uncompensated purely radial version, the curl obstruction at $(r, z) = (5/4, 1/2)$ equals $-5103/32768$ exactly; the test explicitly rejects the pressure-neutral interpretation in this case. The polynomial is used only for interior differentiation, not as an infinitely smooth zero-extended cutoff.

A second independent exact-rational test reproduces (21)–(24) from polynomial integrals and checks the complete cylindrical divergence of the dual test field. An additional exact reconstruction and Lean theorem prove the fixed-frame four-field map (16), its inverse coordinate recovery and its entrywise consequence; the test (25) records the failure of this *conservative gradient method* at the original coefficient budget. The sharper polytope method proves Proposition 5.4: a 40×40 Bernstein partition yields the exact rational upper bound (28). No executable diagnostic substitutes for frozen final Case 006 profiles and rotated-frame PDE estimates. The recovered 122-page revision permits an additional standard-library rational check of Proposition 5.10: it verifies both interval endpoints, an attaining absolute-budget tensor, a one-sided-positive witness exceeding that budget, all seven coefficient floors, and a nonzero pressure boundary primitive. These are finite algebra checks, not an authentic-source inequality evaluation. The equivalence (17) is independently checked on exact rational samples and kernel-checked by `original_seven_coefficient_cone_exact`. The additional kernel theorem `original_cone_dual_upper` establishes the finite-dimensional upper bound in (26); exact rational maximizing witnesses independently verify sharpness. The Bernstein integral itself is a rational Python certificate, not a Lean theorem.

The separate one-sided cone is described by (29); its universal support upper bound is proved by the compiled Lean declaration `one_sided_positive_cone_dual_upper`. Independent exact rational feasible witnesses attain this value. A new 40×40 rational partition proves the lower bound (34), with a separate signed Bernstein upper enclosure. The integral certificate and its comparison with the source pairing are not formalized in Lean. The strict positivity of $J - L$ means only that this *particular* dual test supplies no one-sided positive-cone nonexistence certificate.

The further compiled declarations

`rotating_fourField_frobenius`

establish (36) and (37) for every pointwise 3×3 frame matrix. Independent rational tests verify the rotated covector, equality of matrix contractions, an attainable one-sided dual support, and preservation of isotropic pressure gauges under orthogonal (but not arbitrary) frames. The spatial differential identity (39) and integrated inequality (38) are proved on paper, not yet in Lean; neither has been evaluated against the still-unfrozen complete Case 006 prepared stress.

The source code, machine-readable G0–G8 ledger, smoke tests and reproduction commands accompany the public integration branch [7]. A prior local pinned build passed sixteen smoke modules, and the main physical Lean workflow later received a GitHub-hosted success receipt. A separate weighted source-flux workflow halted during Lean toolchain installation before executing any mathematical check. None of these receipts covers the new rational source-band test until the updated branch is independently checked. Compilation does not replace semantic theorem auditing or independent scrutiny of the paper proof.

8 Fourfold Fourier transfer and the nonlinear correction

A finite Fourier test isolates one mechanism relevant to a non-perturbative velocity correction, without assuming a source-free solution exists. On the unit torus consider the real, solenoidal 20-mode polynomial obtained from the wavevectors $p=(1,0,0)$, $q=(0,2,1)$, $r=p+q$, the polarizations $a=(0,1,0)$, $b=(0,1,-2)$, and their four axial quarter-turns. The initial coefficients are $\widehat{h}(p)=iAa$, $\widehat{h}(q)=Bb$, and $\widehat{h}(r)=\sigma Cb$ with $\sigma=\pm1$; complex conjugation gives the negative frequencies. Direct exact-arithmetic evaluation of the entire projected convolution, rather than a selected triad, finds 108 initially absent Fourier modes with nonzero generic quadratic source. Of these, 92 require interactions between distinct unrotated sectors, while 72 have squared wavenumber above the initial maximum $|r|^2=6$.

The off-support source energy is the positive quartic polynomial $\mathcal{L}(A,B,C)$ in (148). For a local isolated unforced solution starting at h,

$$\sum_{k\in\mathcal{N}}|\widehat{u}(k,t_0+s)|^2=4\pi^2s^2\mathcal{L}(A,B,C)+o(s^2)\quad(s\downarrow0).\tag{81}$$

The nonlinear interaction conserves total instantaneous kinetic energy; its flux among the occupied $|k|^2=1,5,6$ families is $0,-320\pi\sigma ABC,+320\pi\sigma ABC$, respectively. Viscosity makes the total energy strictly decrease for a nonconstant seed. Thus (81) proves immediate support expansion, not sustained forward spectral transport. The receiver at $|r|^2=6$ grows instantaneously against viscosity only for $\sigma=+1$ and $AB>6\pi\nu C$ when $A,B,C>0$.

For a prescribed divergence-free carrier U and a correction $u=U+w$, the genuine projected residual $\mathcal{R}_\nu(U)=U_t-\nu\Delta U+\mathbb{P}\nabla\cdot(U\otimes U)$ cannot be discarded. At the instant when $w=h$, the exact carrier-inclusive estimate (154) bounds the 108-mode derivative from below by the positive part of $2\pi\sqrt{\mathcal{L}}-\|\Pi_{\mathcal{N}}\mathcal{K}_U(h)\|_2$, where $\mathcal{K}_U(h)=-\mathcal{R}_\nu(U)-\mathbb{P}\nabla\cdot(U\otimes h+h\otimes U)$. The prescribed singular carrier's corresponding spectral coupling norm has not been computed from its complete Fourier profile. The 108 first-generation modes do not themselves close the isolated Fourier dynamics: Proposition AT.2 gives the explicit second-generation mode $k_*=(3,2,1)$ with $\partial_t^2\widehat{u}(k_*,t_0)=16\pi^2\sigma A^2Cb\neq0$ when $AC\neq0$. No sustained energy flux, weighted correction, or non-cancellation of the carrier's singular velocity trace is inferred. The exact finite calculation, its independent spatial-space check, and the carrier-coupling identity are documented in Appendix AT.

A complementary concentration estimate quantifies the population of frequencies required by a hypothetical corrected solution. If its total squared energy is bounded above by $C\tau^{1/5}$

while the retained speed is at least $c\tau^{-(1/2+h)}$ along terminal times, then Proposition AU.1 gives the necessary effective Fourier-participation lower bound

$$\mathcal{N}_{\mathrm{eff}}(T-\tau) \geq (c^2/C)\tau^{-(6/5+2h)}. \tag{82}$$

This standard Fourier inequality rules out any uniformly bounded effective number of Fourier modes under these assumptions. It does not demonstrate that the nonlinear Navier–Stokes evolution produces the required mode proliferation. If the full velocity were also restricted to a Fourier frequency radius $K(\tau) = O(\tau^{-\beta})$, Proposition AU.2 would require $\beta \geq 7/15$ for the illustrative $h = 1/10$ energy and speed exponents. The prescribed narrower exterior halo is not the full velocity because the shrinking carrier contains higher frequencies.

9 Remaining analytic requirements

The compact meridional inverse of Theorem 2.1 is a properly restricted solvability result. It does not resolve the weighted attachment problem because the desired source may have an azimuthal component, a noncompact Leray-projected tail, nontrivial pressure coupling, or incompatible moments. Even when compact inversion is possible, the output may leave the required seven-ray positive cone, especially as the support shrinks. The strict dual obstruction in Proposition 5.1, together with Corollary 5.2, excludes the sufficient four-coordinate estimate; Proposition 5.4 also excludes the original absolute coefficient correction budget for that same manufactured source. In contrast, Proposition 5.7 shows that the tested dual field has no obstruction for the larger one-sided cone; this leaves other dual tests and all primal construction questions open. The next mathematical question is whether the *authentic* source, perhaps using a tilted frame or a more flexible coefficient allocation, admits a weighted inverse satisfying the one-sided positive H8 ray conditions (or the absolute inequalities (17)) and all spatial derivative bounds. The sharp cone description enables a direct weighted dual optimization against the authentic source through (38), without replacing those inequalities by the factor-14 sufficient estimate. No numerical conclusion can be drawn from this inequality until the actual g, ρ, R_*, E are supplied and the frame-derivative terms in (39) are controlled.

The joint constraints (31) supply a sharper source-facing test in an aligned cylindrical frame. In a moving physical frame they must be replaced by the spatially integrated support of the complete rotated covector; the corrected Proposition 5.16 makes the needed angular test variation explicit. Failure of a fully certified necessary inequality would obstruct the proposed source-matched positive inverse. Passing these necessary tests would not construct that inverse and would not discharge the physical nonlinear correction.

A complete unforced singularity construction must still provide a single authenticated local profile, source-matched smooth shell data, a scale-sharp weighted symmetric inverse respecting the coefficient margin, a periodic nonlinear correction with a proved right-inverse/contraction scheme, the full physical zero-residual identity, and terminal retention for that actual correction. Each claim requires a further source-to-Lean semantic audit and a compiled theorem using the genuine PDE definitions.

Thus the present mathematical result is an exact compact stress construction and several independently checked necessary or conditional ingredients. The existence or nonexistence of a finite-time singular solution to (1) with smooth nonzero periodic initial data is *not decided by this manuscript.*

Appendix: H8 exterior pressure, authentic translation projection, and nonlinear modulation

Scope and release status. This integrated update reconciles the three H8 research emails of 29 October and corrects an adjoint-normalization omission. The diagnostic exterior-source formula has not been replayed from its unavailable original Appendix D; it is not asserted for the authentic singular carrier. The complete unforced finite-time blow-up theorem remains open.

Exterior-pressure record and analyticity

For the frozen diagnostic axisymmetric source the reported finite-order result is

$$|\partial_z^b p_{\mathrm{ext}}(t,0,z)| \le C_b (T-t)^{-2A-bD}, \quad A = \tfrac{1}{2}+h,\ D = \tfrac{1}{2}-h.$$

The associated diagnostic asymptotic was reported as $q^{2A} p_{\mathrm{ext}} \to -\mathfrak{I}_\beta$ with $\mathfrak{I}_\beta = 64^{-1}\int_{1/2}^1 \beta(s)^2(1-\beta(s))\,ds$, but the relevant cutoff data were not included in that email. This is *not* a newly verified constant. The same diagnostic source fails its inner-axis H8 matching gate.

Under a separated exterior support and L^1 source size $\mathcal{E}_{\mathrm{ext}}$, standard harmonic Green-kernel estimates give the conditional factorial estimate

$$|\partial^\alpha p_{\mathrm{ext}}(t,x)| \le C^{|\alpha|+1}(|\alpha|+2)!\,\mathcal{E}_{\mathrm{ext}}(t)\delta(t)^{-3-|\alpha|}.$$

In the authentic coordinates $\tau = q(1-\eta^2)$, $z = q^{1/2-h}\eta$, a source distance $\delta \asymp \sqrt{\tau}$ generally only supplies an η-radius of order q^h. The exact pullback derivative has the time term

$$\partial_\eta \widehat{f}\big|_q = q^{1/2-h}\widehat{\partial_z f} + 2q\eta\widehat{\partial_t f}.$$

Consequently an exterior fixed-radius analytic bound and terminal-uniform correction cannot be inferred from spatial derivatives alone.

Exact periodic translation projection

Let U be a smooth solenoidal periodic prepared velocity and $\mathbb{P}$ the Leray projection. Define

$$G = U_t - \Delta U + \mathbb{P}\nabla\cdot(U\otimes U), \quad L_U v = \Delta v - \mathbb{P}\nabla\cdot(U\otimes v + v\otimes U), \quad W = \partial_z U.$$

Translations commute with $\mathbb{P}$, hence $W_t = L_U W + \partial_z G$ exactly. If $N(t) = \langle W, W\rangle_{L^2} > 0$, define

$$P_t v = \frac{\langle v, W\rangle}{N} W, \quad Q_t = I - P_t.$$

The two exact residual channels are

$$P_t(-G) = -\frac{\langle G, W\rangle}{N} W, \qquad Q_t(-G) = -G + \frac{\langle G, W\rangle}{N} W.$$

They are formulae for any specified U, not a numerical evaluation of the still-unfrozen singular H8 carrier.

For the unforcing correction $w_t = L_U w - G - \mathcal{N}(w)$, $\mathcal{N}(w) = \mathbb{P}\nabla\cdot(w\otimes w)$, put $a = \langle w, W\rangle/N$ and $v = Q_t w$. Direct differentiation proves

$$a' = -a\frac{\langle \partial_z G, W\rangle}{N} + \frac{\langle -G - \mathcal{N}(w), W\rangle + \langle v, L_U^* W + W_t\rangle}{N},$$

$$v_t = Q_t L_U v - P_t' v - a Q_t \partial_z G - Q_t G - Q_t \mathcal{N}(w).$$

In particular the moving projection and the mode-defect source cannot be dropped. Integration by parts yields the sufficient amplitude bound

$$\left| a' + a \frac{\langle \partial_z G, W \rangle}{N} \right| \le \frac{|\langle G, W \rangle| + \|L_U^* W + W_t\|_2 \|v\|_2 + \|\nabla W\|_\infty \|w\|_2^2}{N}.$$

A translation mode of size $O((T-t)^{-1})$ is subleading to the proposed carrier growth $O((T-t)^{-A})$ if $a = o((T-t)^D)$; no such estimate is established here.

Adjoint normalization repair

The Appendix F equation $\ell_t' = -L_U^* \ell_t$ combined with $\ell_t(W) = 1$ is generally inconsistent when $\ell_t(\partial_z G) \neq 0$, because

$$\frac{d}{dt} \ell_t(W) = \ell_t(\partial_z G).$$

A conditionally consistent normalized adjoint instead obeys

$$\ell_t' = -L_U^* \ell_t - \ell_t(\partial_z G) \ell_t, \quad \ell_t(W) = 1,$$

so $a_\ell = \ell_t(w)$ satisfies

$$a_\ell' = -\ell_t(\partial_z G) a_\ell - \ell_t(G + \mathcal{N}(w)).$$

Existence and weighted norm control of ℓ_t remain open.

Exact regular regression and outstanding authentic source

On a 2π-torus the shear $U = e^{-t} \sin(z - \theta(t)) e_x$ has $W = e^{-t} \cos(z - \theta(t)) e_x$ and $G = -\theta' W$. For the exact unforced shear $u = e^{-t} \sin z \, e_x$, $w = u - U$ has $a = \sin\theta$ and $v = e^{-t}(\cos\theta - 1) \sin(z - \theta) e_x$. The displayed amplitude formula gives $a' = \theta' \cos\theta$; omitting the off-diagonal term would incorrectly give θ'.

The currently accessible carrier-independent source defines only $u_b = \nabla \times (\chi A_c)$ with $\nabla \times A_c = U_c$ on a core patch and leaves the actual globally fixed cutoff/frame/shapers and pressure tuple unverified. Consequently neither the claimed stable $\kappa_s < A$, nor $\sigma > 0$, nor the nonlinear estimate $a = o((T-t)^D)$ has been proved for an authentic singular H8 family. Moreover, projecting the linear source to zero in the translation channel is insufficient: the quadratic forcing $P_t \mathcal{N}(w)$ and P_t' coupling can regenerate an unstable amplitude. Thus the stable-source orthogonality condition alone does not establish terminal retention. The proof-gate ledger must retain its OPEN status.

Appendix H. H8 source-gauge authentication and projected terminal modulation

Status. This appendix proves source-gauge and moving-projection identities and supplies exact model counterexamples. The source-matched singular H8 carrier and the nonlinear unforced blow-up theorem are still open. A full source file and three-page rendered version accompany the cumulative 69-page research PDF.

Gauge-sensitive source freeze. Let (U_c, p_c) solve the unforced Navier–Stokes equations on a local chart containing the support of a smooth cutoff χ. For a local potential $\nabla \times A_c = U_c$ put

$$U = \nabla \times (\chi A_c) = \chi U_c + S, \quad S = \nabla\chi \times A_c, \quad G = U_t - \Delta U + \mathbb{P}\,\mathrm{div}(U \otimes U).$$

Replacing A_c by $A_c + \nabla\psi$ changes the prepared field by the generally nonzero amount $\nabla\chi \times \nabla\psi$. Therefore U_c and χ alone do not determine U or its translated mode. The potential gauge must be frozen as part of the physical H8 source.

Complete projected cutoff residual. Writing $\mathcal{B} = \partial_t - \Delta$ and using the genuine local core pressure p_c, the exact projected source is

$$\begin{aligned} G = \mathbb{P}\{&\chi_t U_c - 2\nabla\chi \cdot \nabla U_c - (\Delta\chi)U_c + (\chi^2 - \chi)(U_c \cdot \nabla)U_c \\ &+ \chi(U_c \cdot \nabla\chi)U_c + p_c\nabla\chi + \mathcal{B}S + \chi(U_c \cdot \nabla)S + (S \cdot \nabla)(\chi U_c) + (S \cdot \nabla)S\}. \end{aligned}$$

In particular the $p_c\nabla\chi$ and S terms must not be dropped merely because a gradient is annihilated by $\mathbb{P}$.

Translation channel and exact counterexample. Set $W = \partial_z U$ and $N = \|W\|_2^2 > 0$. Periodicity and $\mathrm{div}\, W = 0$ give the sharp scalar simplification

$$\boxed{\frac{\langle G, W\rangle_2}{N} = \frac{\langle U_t, W\rangle_2 + \langle (U \cdot \nabla)U, W\rangle_2}{N}.}$$

The viscous term is zero by z-integration by parts. To show the scalar channel is not determined by core data alone, choose on a cylindrical chart $U_c = 0$, $A_c = e_z = \nabla z$ locally and $\chi = a(r)b(z - s(t))$ with compact smooth a, b. Then $U = -a'(r)b(z - s(t))e_\theta$, $U_t = -s'(t)W$, and $(U \cdot \nabla)U$ is purely radial while W is azimuthal. Consequently

$$\boxed{\langle G, W\rangle / N = -s'(t).}$$

This is a regular diagnostic example, not the authentic H8 singular carrier. It proves that an arbitrarily improved source exponent cannot be inferred from the abstract cutoff ansatz.

Moving-projection speed. For $P_t v = \langle v, W\rangle W/N$ and $Q_t = I - P_t$,

$$\boxed{P_t' = \frac{(Q_t W_t) \otimes W + W \otimes (Q_t W_t)}{N}, \quad \|P_t'\|_{L^2 \to L^2} = \frac{\|Q_t W_t\|_2}{\|W\|_2}.}$$

For a compactly supported self-similar toroidal diagnostic field $U = \tau^{-A}\rho\Phi(\rho, \zeta)e_\theta$ with $\rho = r/\sqrt{\tau}$, $\zeta = z/\tau^D$, $A + D = 1$, and $F = \rho\partial_\zeta\Phi$, one has $W = \tau^{-1}Fe_\theta$ and

$$\|P_t'\| = c_F/\tau, \qquad c_F^2 = \frac{\|\mathcal{D}F\|_2^2}{\|F\|_2^2} - \frac{|\langle \mathcal{D}F, F\rangle_2|^2}{\|F\|_2^4}, \quad \mathcal{D}F = F + \tfrac{1}{2}\rho\partial_\rho F + D\zeta\partial_\zeta F.$$

A nonzero smooth compactly supported F has $c_F > 0$, since no nonzero compactly supported eigenfunction of this dilation operator exists. The noncompact Gaussian benchmark $\Phi = -e^{-\rho^2-\zeta^2}/2$ yields $c_F^2 = 1/2 + 3D^2/2$ exactly. A complementary propagator bound must control the order-$1/\tau$ moving-projector coupling, not only the frozen operator $Q_t L_U Q_t$.

Terminal amplitude compatibility. The corrected exact amplitude equation from Appendix G has the form $a' + b(t)a = f(t)$ with

$$b = \frac{\langle \partial_z G, W\rangle}{N}, \quad f = \frac{\langle -G - \mathcal{N}(w), W\rangle + \langle v, L_U^* W + W_t\rangle}{N}, \quad \mathcal{N}(w) = \mathbb{P}\,\mathrm{div}(w \otimes w).$$

Even for the ideal case $b = 0$, an estimate $|f(t)| \le M(T-t)^{D+\sigma-1}$ with $\sigma > 0$ does not ensure $a = o((T-t)^D)$ from arbitrary initial amplitude: generally $a(T) \neq 0$. The terminal cancellation

$$\boxed{a(t_0) + \int_{t_0}^{T} f(s)\,ds = 0}$$

is additionally needed in this ideal case; it then yields $|a(t)| \le M(T-t)^{D+\sigma}/(D+\sigma) = o((T-t)^D)$. For general b, a weighted integrating-factor estimate and corresponding terminal matching are required.

Unclosed authentic gate. One must specify U_c, p_c, χ, A_c, shell shapers, positive seven-ray stress and a single periodic pressure, then evaluate G, N, $\langle G, W\rangle/N$, $Q_t G$, and P_t' on that one source. The target $\kappa_s < A$, an improved source exponent $\sigma > 0$, nonlinear terminal tuning, and a final kernel-checked Lean theorem remain unproved.

Appendix I. Integrated H8 source and weighted terminal modulation

Claim status. This appendix reconciles the H8 source-gauge, exterior-pressure and translation-mode research of Appendices E–H and adds an exact conditional terminal-matching lemma. The complete unforced Navier–Stokes finite-time blow-up theorem remains open.

One authenticated periodic source. For the local core (U_c, p_c) and potential $\nabla \times A_c = U_c$, put

$$U = \nabla \times (\chi A_c) = \chi U_c + \nabla\chi \times A_c, \qquad G = U_t - \Delta U + \mathbb{P}\nabla \cdot (U \otimes U).$$

A potential change $A_c \mapsto A_c + \nabla\psi$ changes U by $\nabla\chi \times \nabla\psi$. The exact source tuple must therefore include the potential gauge, cutoff, shapers, H8 frame, density, positive-ray coefficients and one periodic pressure.

Projected residual and nonlinear modulation. Write $W = \partial_z U$, $N = \|W\|_2^2 > 0$, $P_t v = \langle v, W\rangle W/N$, $Q_t = I - P_t$ and $L_U v = \Delta v - \mathbb{P}\nabla \cdot (U \otimes v + v \otimes U)$. Then

$$W_t = L_U W + \partial_z G, \qquad P_t G = \frac{\langle G, W\rangle}{N} W, \qquad Q_t G = G - P_t G.$$

The exact scalar and moving-projection identities are

$$\frac{\langle G, W\rangle}{N} = \frac{\langle U_t, W\rangle + \langle (U\cdot\nabla)U, W\rangle}{N}, \quad \|P_t'\|_{L^2\to L^2} = \frac{\|Q_t W_t\|_2}{\|W\|_2}.$$

For $w_t = L_U w - G - \mathcal{N}(w)$ with $\mathcal{N}(w) = \mathbb{P}\nabla\cdot(w\otimes w)$, set $a = \langle w, W\rangle/N$ and $v = Q_t w$. The complete amplitude and complementary equations are

$$\begin{aligned} a' + ba = f, \quad b &= \frac{\langle \partial_z G, W\rangle}{N}, \\ f &= \frac{\langle -G - \mathcal{N}(w), W\rangle + \langle v, L_U^* W + W_t\rangle}{N}, \\ v_t &= Q_t L_U v - P_t' v - a Q_t \partial_z G - Q_t G - Q_t \mathcal{N}(w). \end{aligned}$$

The moving projection and quadratic regeneration cannot be suppressed.

Theorem 9.1 (Weighted terminal matching under a controlled mode coefficient)**.** *Let $D > 0$, $\sigma > 0$, $\kappa_b \geq 0$ with $\kappa_b < D + \sigma$, and let $a' + b(t)a = f(t)$ on $[t_0, T)$. Suppose*

$$|b(t)| \leq \kappa_b/(T-t), \qquad |f(t)| \leq M(T-t)^{D+\sigma-1}.$$

Write $B(t) = \int_{t_0}^t b(s)\,ds$. If the terminal compatibility condition

$$a(t_0) + \int_{t_0}^T e^{B(s)} f(s)\,ds = 0$$

holds, then

$$|a(t)| \leq \frac{M}{D+\sigma-\kappa_b}(T-t)^{D+\sigma} = o((T-t)^D).$$

Proof. The integrating factor gives $a(t) = -\int_t^T e^{B(s)-B(t)} f(s)\,ds$. For $s \geq t$, $e^{B(s)-B(t)} \leq ((T-t)/(T-s))^{\kappa_b}$. Integration gives the asserted constant and exponent. The strict inequality $\kappa_b < D + \sigma$ also guarantees convergence of the improper matching integral. □

Limits and next certificate. This theorem is an implication, not an existence proof: the forcing f depends on the unknown nonlinear correction. The authentic H8 tuple still lacks a single authenticated physical cutoff/potential/stress/pressure specification. One must first calculate N, $\langle G, W\rangle/N$, $Q_t G$ and P_t' for that source, then prove the stable complementary propagator and the bounds on b, f (and terminal matching) in one compatible shrinking-radius norm. The earlier diagnostic exterior-pressure constant is not a certificate for that authentic source. A final Lean existence witness is not available; the proof-gate ledger remains **OPEN**.

Appendix J. Translation-mode energy and H8 source-specific stability gates

Claim boundary. All periodic identities in this appendix are exact under smoothness, solenoidality and nonzero translation-mode norm. The explicit Gaussian is an independent whole-space diagnostic. Neither the singular H8 source nor the global unforced Navier–Stokes blow-up theorem is constructed.

Exact coefficient formula. For a smooth solenoidal prepared velocity U, write

$$G = U_t - \Delta U + \mathbb{P}\nabla\cdot(U\otimes U), \quad W = \partial_z U, \quad N = \|W\|_2^2 > 0, \quad b = \langle \partial_z G, W\rangle / N.$$

Commuting ∂_z with the torus Leray projection, integrating by parts, and using $\nabla\cdot U = \nabla\cdot W = 0$ proves

$$\boxed{b(t) = \frac{1}{2}\frac{d}{dt}\log N(t) + \frac{\|\nabla W\|_2^2}{N(t)} + \frac{\int_{\mathbb{T}^3} W\cdot[(W\cdot\nabla)U]\,dx}{N(t)}.}$$

The last term is the translation-mode strain form, which may have either sign. No estimate of b for the authentic H8 carrier follows without a frozen U.

Exact complementary energy cancellation. Put $P_t v = \langle v, W\rangle W/N$, $Q_t = I - P_t$, $L_U v = \Delta v - \mathbb{P}\nabla\cdot(U\otimes v + v\otimes U)$, $\mathcal{N}(w) = \mathbb{P}\nabla\cdot(w\otimes w)$, and $w = aW + v$, $v = Q_t v$, with $w_t = L_U w - G - \mathcal{N}(w)$. The full complementary equation derived in Appendix I remains valid:

$$v_t = Q_t L_U v - P_t' v - a Q_t \partial_z G - Q_t G - Q_t \mathcal{N}(aW + v).$$

Since $Q_t P_t' Q_t = 0$, we have $\langle P_t' v, v\rangle = 0$ exactly. Furthermore, the $v\otimes v$ term and one mixed quadratic term vanish in the L^2 pairing by incompressibility. The complete identity is

$$\boxed{\begin{aligned}\frac{1}{2}\frac{d}{dt}\|v\|_2^2 + \|\nabla v\|_2^2 = &-\int_{\mathbb{T}^3} v\cdot[(v\cdot\nabla)(U + aW)]\,dx \\ &- \langle G + a\partial_z G + a^2(W\cdot\nabla)W, v\rangle.\end{aligned}}$$

This cancels the direct P_t' energy cost, *not* the off-diagonal amplitude coupling or higher-regularity commutators.

Concrete conditional stability gate. Let $\tau = T - t$, $A = \frac{1}{2} + h$. Define $\mathcal{H}_a = Q_t[G + a\partial_z G + a^2(W\cdot\nabla)W]$. If, along one actual correction, every complementary v satisfies

$$-\int v\cdot[(v\cdot\nabla)(U + aW)]\,dx \le \frac{\kappa_s}{\tau}\|v\|_2^2, \quad \|\mathcal{H}_a\|_2 \le M\tau^{-A-1+\sigma}, \quad \kappa_s < A,\ \sigma > 0,$$

then $y = \|v\|_2$ satisfies $y' \le (\kappa_s/\tau)y + M\tau^{-A-1+\sigma}$ and

$$y(t) \le (\tau_0/\tau)^{\kappa_s} y(t_0) + M\tau^{-\kappa_s}\int_\tau^{\tau_0} r^{\kappa_s - A - 1 + \sigma}\,dr = o(\tau^{-A}).$$

This is only a conditional L^2 conclusion. A compatible analytic or Sobolev embedding estimate for the same shrinking geometry is still required for terminal pointwise retention, jointly with the amplitude matching theorem in Appendix I.

Adversarial toroidal Gaussian coefficient. On $\mathbb{R}^3$, separately from any physical H8 source, take $U = \tau^{-A}\rho\Phi(\rho,\zeta)e_\theta$, $\rho = r/\sqrt{\tau}$, $\zeta = z/\tau^D$, $D = \frac{1}{2} - h$, $\Phi = -\frac{1}{2}e^{-\rho^2-\zeta^2}$. Then $W = \partial_z U = \tau^{-1}\rho\zeta e^{-\rho^2-\zeta^2}e_\theta$ and $N = C\tau^{-A}$. The stretching form vanishes; the exact radial-plus-hoop and axial Rayleigh quotients are respectively 4 and 3. Thus

$$\boxed{b(t) = \frac{A/2+4}{\tau} + 3\tau^{-1+2h}.}$$

The small coefficient hypothesis of Appendix I is not automatic. The diagnostic is not a periodic singular H8 solution.

Unclosed authentic-source obligations. Freeze the complete gauge-dependent carrier U_c, p_c, A_c, χ, frame, density, shell shapers, stress and periodic pressure; compute N, the three terms of the exact b identity, the restricted strain form, $\mathcal{H}_a$ and the nonlinear amplitude f. Prove the one-sided spectral bound, positive source gain, terminal matching and an L^∞-controlling estimate in a common function space. No source-specific $\kappa_s < A$ or $\sigma > 0$ certificate, and no final Lean existential theorem, has yet been obtained.

Appendix K. Sharp complementary-strain obstruction

Status. This is a source-independent theorem and an exact periodic diagnostic. The authenticated H8 singular source and the complete unforced Navier–Stokes theorem remain open.

Theorem 9.2 (One translation projection does not improve the worst strain)**.** *Let U be smooth, periodic, solenoidal; $W = \partial_z U \neq 0$; and $S = \frac{1}{2}(\nabla U + \nabla U^{\mathsf{T}})$. Then*

$$\boxed{\sup_{\substack{\nabla\cdot v=0,\ \langle v,W\rangle=0\\ v\in C^\infty,\ v\neq 0}} \frac{-\int_{\mathbb{T}^3} v^{\mathsf{T}} S v}{\|v\|_2^2} = \max_{x\in\mathbb{T}^3}[-\lambda_{\min}(S(x))] \geq \frac{\|S\|_{L^\infty(F)}}{\sqrt{6}}.}$$

The trace-free constant $1/\sqrt{6}$ is sharp.

Proof. The pointwise eigenvalue inequality gives the upper bound. For the converse choose a point x_0 and unit eigenvector e for $\lambda_{\min}(S(x_0))$, and a unit $\xi \perp e$. With a smooth localized bump φ near x_0, the periodic solenoidal packet

$$v_k = \operatorname{curl}\left(\frac{\varphi}{k}(\xi \times e)\cos(k\xi\cdot(x-x_0))\right) = \varphi e \sin(k\xi\cdot(x-x_0)) + O(k^{-1})$$

satisfies $\langle v_k, W\rangle \to 0$. Subtracting its orthogonal projection onto W gives an admissible solenoidal packet with the same limiting quotient. Letting the bump support shrink gives the desired lower bound. For trace-free symmetric S, writing ordered eigenvalues as $(-m, \lambda_2, m-\lambda_2)$ with $\lambda_2 \in [-m, m/2]$ shows $\|S\|_F^2 \leq 6m^2$. Equality occurs at $(2m, -m, -m)$. □

An exact periodic witness. On $(\mathbb{R}/2\pi\mathbb{Z})^3$ put $U = K(t)\sin(z)e_x$, $W = K(t)\cos(z)e_x$,

$$\psi = \sin(x)\cos(z) + \cos(x)\sin(2z), \quad v = (\partial_z\psi, 0, -\partial_x\psi).$$

Then $\nabla\cdot v = 0$, $\langle v, W\rangle = 0$, and exact integration gives

$$\boxed{\|v\|_2^2 = 14\pi^3, \quad -\int v\cdot(v\cdot\nabla)U = 3\pi^3 K(t), \quad \frac{-\int v\cdot(v\cdot\nabla)U}{\|v\|_2^2} = \frac{3}{14}K(t).}$$

Taking $K(t) = (T-t)^{-1-h}$ disproves a universal fixed-κ_s bound of the form $\kappa_s/(T-t)$ for all complementary perturbations of this diagnostic prepared field. It is not an unforced or authentic H8 solution.

Proposition 9.3 (Conditional H8 supercritical-strain test)**.** *Suppose the* authenticated *periodic H8 carrier has a region where, for $\tau = T - t$ sufficiently small, $|\partial_r U_z| \geq c\tau^{-1-h}$ and $|\partial_z U_r| \leq C\tau^{-1+h}$. Then*

$$\|S\|_{L^\infty(F)} \geq \frac{c}{2\sqrt{2}}\tau^{-1-h}, \qquad \Lambda_\perp(t) \geq \frac{c}{4\sqrt{3}}\tau^{-1-h}.$$

Thus no all-perturbation complementary strain estimate $\Lambda_\perp(t) \leq \kappa_s/\tau$ holds with fixed κ_s.

Proof. For small τ, $|S_{rz}| = \frac{1}{2}|\partial_r U_z + \partial_z U_r| \geq (c/4)\tau^{-1-h}$. Thus $\|S\|_F \geq \sqrt{2}|S_{rz}|$ and the theorem applies. □

Subleading translation cannot erase the leading strain by itself. If $\|\nabla W\|_\infty \leq C_W\tau^{-3/2}$ and $a = o(\tau^D)$, $D = \frac{1}{2} - h$, then $\|a\,\mathrm{sym}\,\nabla W\|_\infty = o(\tau^{-1-h})$. The uniform all-$v$ gate remains unavailable when the authenticated supercritical strain assumptions hold. A dynamically restricted estimate on the actual nonlinear complementary correction remains conceivable.

Release status. The authentic H8 source and its pressure/stress tuple are not yet frozen in one consistent implicit-coordinate specification, so this appendix does not claim numerical values of b, f, or an H8 propagator exponent. Its theorem is verified analytically; the trigonometric witness is checked by exact symbolic integration. No complete Lean theorem or unforced singularity is claimed.

Appendix L. Dynamically restricted H8 corrections and the carrier-amplitude channel

Claim boundary. Exact periodic identities and diagnostic shears are proved below; the authentic gauge- and cutoff-frozen singular H8 source is not yet available and the full unforced blow-up theorem remains open.

Proposition 9.4 (Two orthogonal carrier modes)**.** *Let U be a nonzero smooth periodic divergence-free prepared velocity, $W = \partial_z U \neq 0$, and*

$$G = U_t - \Delta U + \mathbb{P}\,\mathrm{div}(U \otimes U), \quad M = \|U\|_2^2, \quad N = \|W\|_2^2.$$

Then $\langle U, W\rangle = 0$, and the canonical rank-two source projection is

$$\boxed{P_{AT} f = \frac{\langle f, U\rangle}{M} U + \frac{\langle f, W\rangle}{N} W, \quad Q_{AT} = I - P_{AT}.}$$

Its amplitude source coefficient is exactly

$$\boxed{c_A(t) = \frac{\langle G, U\rangle}{M} = \frac{1}{2}\frac{d}{dt}\log M + \frac{\|\nabla U\|_2^2}{M}.}$$

Proof. Periodicity gives $\langle U, W\rangle = \frac{1}{2}\int \partial_z |U|^2 = 0$. In the second identity, pair the projected prepared residual with U. Incompressibility cancels $\langle (U\cdot\nabla)U, U\rangle$ and periodic integration by parts gives the viscous quotient. □

Theorem 9.5 (Sufficient two-mode terminal retention)**.** *Put $\tau = T - t$, $A = \frac{1}{2} + h$, $D = \frac{1}{2} - h$ with $A + D = 1$. Assume $\|U\|_\infty \le C\tau^{-A}$, $\|W\|_\infty \le C\tau^{-1}$, and $\max_x |U(t,x)| \ge c\tau^{-A}$. Write the exact unforcing correction as $w = \alpha U + aW + v$ with $v = Q_{AT} w$. If*

$$\boxed{\alpha = o(1), \qquad a = o(\tau^D), \qquad \|v\|_\infty = o(\tau^{-A}),}$$

then $\|U + w\|_\infty \ge (c - o(1))\tau^{-A} \to \infty$.

Proof. Use the reverse triangle inequality at a point where $|U|$ attains its maximum and the identity $D - 1 = -A$. □

This is a sufficient implication; its hypotheses have not been established for the authentic H8 nonlinear correction.

An exact cancellation that translation modulation misses. For the diagnostic prepared shear $U = K(t)\sin z\, e_x$ on $(\mathbb{R}/2\pi\mathbb{Z})^3$ with $K(t) = \tau^{-A}$,

$$G = (K' + K)\sin z\, e_x, \quad W = K\cos z\, e_x, \quad \boxed{\frac{\langle G, W\rangle}{N} = 0, \quad Q_{AT} G = 0, \quad c_A = A/\tau + 1.}$$

Yet $u(t) = K(t_0)e^{-(t-t_0)}\sin z\, e_x$ is an exact unforced smooth shear with zero pressure. The correction $w = u - U$ has translation coefficient $a = 0$ and complementary component $v = 0$, but its amplitude coefficient

$$\boxed{\alpha = \frac{K(t_0)e^{-(t-t_0)}}{K(t)} - 1 \to -1.}$$

Thus even perfect translation matching and $Q_{AT} G = 0$ do not prevent cancellation of apparent singular growth. This deliberately forced prepared shear is not an authentic H8 carrier.

Dynamically restricted linearized growth can be subcritical. Take a different diagnostic shear $U = K(t)\sin z\, e_x$, $K(t) = \tau^{-1-h}$, for which $\|\nabla U\|_\infty = K$. The homogeneous divergence-free linearized equation has the explicit solution

$$V = C(t)\sin y\cos z\, e_x + Z(t)\sin y\, e_z, \quad Z(t) = \varepsilon e^{-(t-t_0)},$$

$$C'(t) + 2C(t) = -K(t)Z(t), \quad C(t_0) = 0, \qquad C(t) = -\varepsilon e^{-2(t-t_0)}\int_{t_0}^{t} e^{s-t_0}K(s)\,ds.$$

Hence $\boxed{\|V(t)\|_\infty = O(\tau^{-h}) = o(\tau^{-A}), \quad A = \frac{1}{2} + h.}$ At every time $\langle V, U\rangle = \langle V, W\rangle = 0$. This shows a favorable *specific* linearized orbit is possible despite supercritical instantaneous strain, but does not solve the unforcing source equation.

Quadratic amplitude regeneration. The same V satisfies $(V\cdot\nabla)V = -ZC\sin^2 y\sin z\, e_x$, which is already solenoidal. Thus

$$\boxed{P_A\mathbb{P}\,\mathrm{div}(V\otimes V) = -\frac{ZC}{2K}U, \qquad P_T\mathbb{P}\,\mathrm{div}(V\otimes V) = 0.}$$

The complementary *linear* orbit therefore generates a carrier-amplitude source at quadratic order. Any H8 nonlinear closure must track this mode alongside translation and the remaining complementary correction.

Outstanding source-specific gates. Freeze (U_c, p_c, A_c, χ), its physical frame, shapers, density, positive H8 stress and periodic pressure in the correct implicit coordinates. Calculate M, N, c_A, $\langle G, W\rangle/N$, $Q_{AT}G$, the time-dependent two-mode projection couplings and quadratic amplitude regeneration. Establish $\alpha = o(1)$, $a = o(\tau^D)$ and $\|Q_{AT}w\|_\infty = o(\tau^{-A})$ in one valid terminal analytic or anisotropic Sobolev space. The existing regression checks do not certify this analytic existence bridge or a complete Lean proof.

Appendix M. Exact two-mode H8 source-authentication gate

Status. These are exact conditional identities, not a source-specific singularity proof. The full three-page Appendix M, tests and 83-page cumulative PDF are in the downloadable research release.

Physical-chart reconciliation. The exact core uses $\tau = T - t = q(1-\eta^2)$, $z = q^D\eta$, $X = r^2/(2q)$, $D = 1/2 - h$. The source-native H8 text also writes $q = 1 - t$. These descriptions are not identical off axis. The physical axial chain rule is

$$\partial_z F = \frac{q^{-D}}{1 - 2h\eta^2}\left[(1-\eta^2)F_\eta - 2\eta X F_X + 2\eta q F_q\right].$$

A canonical potential on a star-shaped physical chart is

$$A_c(x) = \int_0^1 s\, U_c(x_0 + s(x - x_0)) \times (x - x_0)\, ds, \quad \nabla \times A_c = U_c.$$

This chooses a gauge but does not fix a physical cutoff or H8 stress.

Nonlinear two-mode projection. Set $\mathcal{B}(f,g) = \mathbb{P}[(f\cdot\nabla)g]$, $G = U_t - \Delta U + \mathcal{B}(U,U)$, $L_U f = \Delta f - \mathcal{B}(U,f) - \mathcal{B}(f,U)$. Let $W = \partial_z U$, $M = \|U\|_2^2$, $N = \|W\|_2^2$. Since $\langle U, W\rangle = 0$, write $w = \alpha U + aW + v$, $v \perp U, W$, $\beta = 1 + \alpha$, with $Q = I - P_A - P_T$. Define

$$c_A = \langle G, U\rangle/M, \quad g_T = \langle G, W\rangle/N, \quad b_T = \langle \partial_z G, W\rangle/N, \quad e_T = \langle \mathcal{B}(U,U), W\rangle/N.$$

The exact unforcing equation implies

$$\begin{aligned}
\beta' + c_A\beta &= a\langle G, W\rangle/M - \langle \mathcal{B}(w,w), U\rangle/M + \langle v, L_U^* U + U_t\rangle/M,\\
a' + b_T a &= -\beta g_T - \alpha e_T - \langle \mathcal{B}(w,w), W\rangle/N + \langle v, L_U^* W + W_t\rangle/N,\\
v_t &= QL_U v - P'v - Q[\beta G + \alpha\mathcal{B}(U,U) + a\partial_z G + \mathcal{B}(w,w)].
\end{aligned}$$

Here $P = P_A + P_T$. Both off-diagonal interactions and moving-projector couplings remain in the equations.

Conditional amplitude obstruction. The exact identity is $c_A = \frac{1}{2}(\log M)' + \|\nabla U\|_2^2/M$. If $c_A \geq \kappa/(T-t)$, $\kappa > 0$, and the total right-hand-side forcing of the β equation is $o((T-t)^{-1})$, integrating factors give $\beta \to 0$ and $\alpha \to -1$. Thus a positive coefficient requires amplitude injection rather than mere smallness of the mode forcing. The actual H8 sign of c_A is not known.

Anisotropic pointwise gate. For $\tau = T - t$, $D = 1/2 - h$ and a compact-chart correction,

$$\|v\|_\infty \leq C\tau^{-(1+D)/2} \sum_{|\gamma|\leq 2} \tau^{(\gamma_1+\gamma_2)/2 + D\gamma_3}\|\partial^\gamma v\|_2.$$

The weighted sum must be $o(\tau^{1/4-3h/2})$ to ensure $v = o(\tau^{-1/2-h})$ in L^∞. A global correction additionally needs an exterior estimate.

Remaining source-specific gate. The full periodic H8 tuple is still unfrozen, so no authentic values of c_A, g_T, b_T, e_T, QG, complementary propagator, or nonlinear terminal-retention bounds can yet be certified. The unforced finite-time blow-up theorem remains OPEN.

Appendix N. Physical Rayleigh quotients and a sharp amplitude-source gate

Status: conditional exact reduction, not a fully authenticated H8 singular source. The independent four-page Appendix N and exact verification code accompany the cumulative 87-page PDF.

Put $\tau = T - t$, $A = \frac{1}{2} + h$, $D = \frac{1}{2} - h$, $d = 1 - \eta^2$, $L = 1 - 2h\eta^2$ and use the true core chart $q = \tau/d$, $r = \sqrt{2qX}$, $z = q^D\eta$. Then

$$\boxed{dV = 2\pi\tau^{1+D}Ld^{-D-2}\,dX\,d\eta.}$$

Assume, conditionally, that the entire periodic carrier is smoothly supported on $0 \le X \le X_*$, $|\eta| \le \eta_* < 1$ and has differentiable symbol expansions

$$U_r = q^{-1/2}(V_r + O_{C^2}(q^{2h})), \quad (U_\theta, U_z) = q^{-A}((V_\theta, V_z) + O_{C^2}(q^{2h})).$$

The genuine physical derivatives act by $Z_aF = L^{-1}(2a\eta F + dF_\eta - 2\eta XF_X)$ and $T_aF = L^{-1}(-aF + D\eta F_\eta + XF_X)$. Define $E_\theta = Z_{-A}V_\theta$, $E_z = Z_{-A}V_z$, $E_r = Z_{-1/2}V_r$. Then $W = \partial_z U$ has leading components $q^{-1}(E_\theta, E_z)$ and radial component $q^{-1+h}E_r$. For integrals over the fixed compact profile domain set

$$\begin{aligned}
M_0 &= 2\pi\int Ld^{2A-D-2}(V_\theta^2 + V_z^2)\,dX\,d\eta,\\
N_0 &= 2\pi\int Ld^{-D}(E_\theta^2 + E_z^2)\,dX\,d\eta,\\
R_U &= 2\pi\int Ld^{2A-D-1}[2X(V_{\theta,X}^2 + V_{z,X}^2) + V_\theta^2/(2X)]\,dX\,d\eta,\\
R_W &= 2\pi\int Ld^{1-D}[2X(E_{\theta,X}^2 + E_{z,X}^2) + E_\theta^2/(2X)]\,dX\,d\eta.
\end{aligned}$$

Let

$$\mathcal{S} = E_r[\sqrt{2X}(E_\theta V_{\theta,X} + E_z V_{z,X}) - E_\theta V_\theta/\sqrt{2X}] + E_\theta^2(E_z + V_r/\sqrt{2X}) + E_z^3, \quad S_W = 2\pi\int Ld^{1-D}\mathcal{S}\,dX\,d\eta.$$

For positive M_0, N_0, set $p = 1 + D - 2A = \frac{1}{2} - 3h$. The exact energy identities and scaling yield

$$\boxed{\begin{aligned}
&\|U\|_2^2 = \tau^p(M_0 + O(\tau^{2h})), && \|W\|_2^2 = \tau^{-A}(N_0 + O(\tau^{2h})),\\
&c_A = \tau^{-1}(R_U/M_0 - p/2) + O(\tau^{-1+2h}), && b_T = \tau^{-1}(A/2 + (R_W + S_W)/N_0) + O(\tau^{-1+2h}).
\end{aligned}}$$

The apparent τ^{-1-h} shear does not force that rate for b_T because the radial component of W carries an extra τ^h. Writing

$$\begin{aligned}
B_\theta &= V_r\sqrt{2X}V_{\theta,X} + V_zE_\theta + V_rV_\theta/\sqrt{2X},\\
B_z &= V_r\sqrt{2X}V_{z,X} + V_zE_z,\\
J &= E_\theta B_\theta + E_zB_z - E_rV_\theta^2/\sqrt{2X},\\
I_{\mathrm{adv}} &= 2\pi\int Ld^{A-D}J\,dX\,d\eta,\\
I_{\mathrm{time}} &= 2\pi\int Ld^{A-D}(E_\theta T_{-A}V_\theta + E_zT_{-A}V_z)\,dX\,d\eta,
\end{aligned}$$

the additional projected coefficients are

$$\boxed{g_T = \tau^{-A}(I_{\mathrm{time}} + I_{\mathrm{adv}})/N_0 + O(\tau^{-A+2h}), \qquad e_T = \tau^{-A}I_{\mathrm{adv}}/N_0 + O(\tau^{-A+2h}).}$$

These formulas include the leading centripetal term in J.

Proposition 9.6 (Support-based amplitude damping and raw-source lower bound)**.** *Let $j_{0,1}$ be the first positive zero of J_0, and $d_* = 1 - \eta_*^2$. For any smooth periodic field supported on this shrinking transverse chart,*

$$\frac{\|\nabla U\|_2^2}{\|U\|_2^2} \geq \frac{j_{0,1}^2 d_*}{2X_*\tau}.$$

If the differentiated mass asymptotic above holds and $X_* < j_{0,1}^2 d_*/p$ (with $p > 0$), then $c_A \geq \kappa/\tau$ for some $\kappa > 0$ at late times. Consequently

$$\boxed{\|G\|_2 \geq \|P_A G\|_2 = |c_A|\sqrt{\|U\|_2^2} \gtrsim \tau^{-3/4-3h/2}.}$$

If the full forcing of the exact $\beta' + c_A\beta = F_A$ equation is $o(1/\tau)$, then $\beta \to 0$ and carrier-amplitude retention fails. This does not exclude a leading-order amplitude injection.

Proof. At each fixed z, the support lies in a disk of radius at most $\sqrt{2X_*\tau/d_*}$. Apply the sharp Dirichlet disk Poincare inequality to each Cartesian velocity component and integrate over z. Use $c_A = \frac{1}{2}(\log\|U\|_2^2)' + \|\nabla U\|_2^2/\|U\|_2^2$. The projected-source lower bound follows by Cauchy–Schwarz and $p/2 - 1 = -3/4 - 3h/2$. The terminal amplitude statement follows from the integrating-factor equation. □

Pressure, limitations and remaining gate. For one fully fixed periodic U, the mean-zero pressure is uniquely $P_U = (-\Delta)^{-1}\partial_i\partial_j(U_iU_j)$; it need not be compactly supported even for a compact carrier, and cannot remove the Leray-projected amplitude source. The H8 manuscripts still leave the exact physical cutoff, potential gauge, stress shapers and positive-ray certificate unfrozen. The displayed coefficients are not numerically certified for the authentic H8 source. No bound for QG, the nonlinear two-mode propagator, terminal matching or global L^∞ correction is proved. The unforced singularity claim and final Lean existence gate remain OPEN.

All computational algorithms, proof-search procedures, and certificate criteria were developed by the author(s). Large Language Models (LLMs) such as ChatGPT and Claude were utilized strictly as computational assistants for debugging code and cross-checking routines. No mathematical claims rely on LLM outputs; all results are grounded exclusively in independently reproducible, rigorously verified certificates (exact, symbolic, rational, interval, exhaustive, or controlled high-precision).

Appendix O. Solenoidal return flow and sharp transverse confinement

Status. Exact axisymmetric necessary conditions and a fully specified smooth prepared-field diagnostic. These are not the authentic H8 source-matched seven-ray stress or an unforced singularity theorem. The complete four-page appendix is preserved in the cumulative PDF and the separate source packet.

Physical chart and exact gauge. Let $\tau = T - t = q(1-\eta^2)$, $z = q^D\eta$, $X = r^2/(2q)$, $A = 1/2 + h$, $D = 1/2 - h$ and

$$\mathcal{Z}_a F = \frac{2a\eta F + (1-\eta^2)F_\eta - 2\eta X F_X}{1 - 2h\eta^2}.$$

For compact, axis-regular smooth functions Φ, B flat at their outer boundaries put

$$A_r = 0, \quad A_\theta = q^{-h}\Phi/\sqrt{2X}, \quad A_z = -q^{-h}B, \quad U = \nabla \times A.$$

Then

$$U_r = q^{-1/2}V_r, \quad U_z = q^{-A}V_z, \quad U_\theta = q^{-A}V_\theta, \qquad V_r = -\frac{\mathcal{Z}_{1-A}\Phi}{\sqrt{2X}},\ V_z = \Phi_X,\ V_\theta = \sqrt{2X}B_X.$$

This is identically solenoidal and extends smoothly by zero to a periodic chart, provided $\Phi, B = O(X)$ at the axis. Conversely compact, regular leading poloidal profiles with zero slice flux obey this representation with $\Phi(X,\eta) = \int_0^X V_z(Y,\eta)dY$.

Axial return-flow law. For a compact axisymmetric solenoidal field with support in a proper axial subinterval, integrating incompressibility at fixed physical z gives

$$\boxed{\int_0^\infty U_z(t,r,z) r\, dr = 0}, \qquad \boxed{\int_0^{X_*} V_z(X,\eta)\, dX = 0}.$$

Thus a positive central axial jet necessarily has a compensating negative annular flow. If $V_z \geq m$ on $0 < X < x_0$, then

$$\int_{x_0}^{X_*} V_z^2\, dX \geq \frac{m^2 x_0^2}{X_* - x_0}$$

by Cauchy–Schwarz.

Theorem 9.7 (Sharper axisymmetric disk energy bound)**.** *Let U be smooth, compactly confined in disks $r \leq R_*(t)$, axisymmetric, solenoidal, and supported in a proper axial subinterval. With $\theta_z = \|U_z\|_2^2/\|U\|_2^2$,*

$$\boxed{\frac{\|\nabla U\|_2^2}{\|U\|_2^2} \geq \frac{j_{1,1}^2 + (j_{2,1}^2 - j_{1,1}^2)\theta_z}{R_*^2}}$$

where $j_{1,1}^2 = 14.68197064\ldots$ and $j_{2,1}^2 = 26.37461643\ldots$.

Proof. The horizontal Cartesian components of axisymmetric U have angular Fourier order one, whose Dirichlet disk eigenvalue is $j_{1,1}^2/R_*^2$. The axial component is radial and has zero disk mean by the flux law. The sharp zero-mean radial Dirichlet quotient is $j_{2,1}^2/R_*^2$: its constrained Euler–Lagrange equation is $-(rf')' = \lambda r f + \mu r$, giving $f(r) \propto J_0(kr) - J_0(kR_*)$. The zero mean condition is $2J_1(kR_*) = kR_* J_0(kR_*)$, equivalent to $J_2(kR_*) = 0$. Add the two disk-sector bounds at every physical z and then the nonnegative axial gradient energy. □

Nonlinear amplitude-injection threshold. For $M = \|U\|_2^2 = \tau^p(M_0 + O(\tau^{2h}))$ with differentiated bounds and $p = 1/2 - 3h$, the exact identity $c_A = \frac{1}{2}(\log M)' + \|\nabla U\|_2^2/M$ gives, with $d_* = 1 - \eta_*^2$,

$$\liminf \tau c_A \geq \frac{d_*}{2X_*}[j_{1,1}^2 + (j_{2,1}^2 - j_{1,1}^2)\liminf \theta_z] - p/2.$$

If $c_A \geq \kappa/\tau$ and the exact nonlinear amplitude equation $\beta' + c_A\beta = F_A$ retains $\beta \to 1$, then

$$\boxed{\liminf_{\tau\downarrow 0} \int_{T-\tau}^{T-\tau/2} F_A(s)\, ds \geq \kappa \log 2}.$$

This follows by integrating the exact amplitude equation over the terminal dyad. A positive amplitude damping requires leading injection; an $o(1/\tau)$ amplitude forcing cannot suffice.

Fully frozen periodic prepared-field diagnostic. Take $h = 1/10$, $\eta_* = 1/2$, $X_* = 1$, $\Phi = Xf(X)g(\eta)$, $B = \Phi/4$, with

$$f(X) = e^{-X/(1-X)}\ (0 \leq X < 1), \qquad g(\eta) = e^{-4\eta^2/(1-4\eta^2)}\ (|\eta| < 1/2),$$

both extended flatly by zero. This fixes the carrier gauge, its compact support and the canonical periodic mean-zero pressure, and gives $U_z(t, 0, 0) = \tau^{-A}$. Independent Gauss–Legendre integration of the Appendix N profile quadratures at 40, 64, 90 and 128 nodes per coordinate converges to

$$M_0 = 0.829691939536,\ \ N_0 = 9.33098522102,\ \ \lim \tau c_A = 27.0163023544,\ \ \lim \tau b_T = 43.4741340312,$$

$$\lim \tau^A g_T = \lim \tau^A e_T = 0.0477816887268, \qquad \lim \theta_z = 0.95969941416.$$

The signed translation-strain and translation-time quadratures vanish by exact parity. These are convergent numerical diagnostics, not interval-certified digits. The independently proved support bound gives $\liminf \tau c_A \geq 9.61377348\ldots$, and hypothetical amplitude retention would demand approximately 18.72627 integrated forcing per terminal dyad for this prototype.

Outstanding gate. This explicit prototype is neither authenticated to the true local H8 core nor proved to satisfy its source-matched positive seven-ray stress. Its Leray residual and nonlinear complement are not closed. Freeze the authentic core/potential/cutoff/shapers/stress/pressure as one tuple, enforce the radial return-flow condition and the spectral bound, and establish amplitude injection, terminal translation matching, and global L^∞ retention for the same unforced correction. The unforced Navier–Stokes singularity theorem remains OPEN.

All computational algorithms, proof-search procedures, and certificate criteria were developed by the author(s). Large Language Models (LLMs) such as ChatGPT and Claude were utilized strictly as computational assistants for debugging code and cross-checking routines. No mathematical claims rely on LLM outputs; all results are grounded exclusively in independently reproducible, rigorously verified certificates (exact, symbolic, rational, interval, exhaustive, or controlled high-precision).

Appendix P. The complementary swirl-source obstruction

Scope. Source-independent conditional theorem for reflection-symmetric compact similarity swirls, with an exact diagnostic certificate. The authentic H8 source, nonlinear correction, and full unforced blow-up theorem remain OPEN.

Theorem 9.8 (Flat compact equatorial swirl defeats two-mode raw source gain)**.** *Let U be a smooth periodic axisymmetric solenoidal prepared field supported in a compact physical similarity chart $\tau = T - t = q(1-\eta^2)$, $X = r^2/(2q)$, $z = q^D\eta$, $A = 1/2 + h$, $D = 1/2 - h$, $0 < h < 1/6$. Assume U_r is odd in physical z, U_θ, U_z even, and $U_\theta = q^{-A}V_\theta(X,\eta)$ with nonzero $V(X) = V_\theta(X,0) \not\equiv 0$ is smooth for $X > 0$ and flat at X_* flat at $X_* > 0$. Assume the higher-order azimuthal source remainder is $O(\tau^{-A-1+2h})$ uniformly in profile coordinates, and $\lambda(t) = \tau c_A(t) \to \lambda_* \in \mathbb{R}$, where $c_A = \langle G, U\rangle/\|U\|_2^2$ and $G = U_t - \Delta U + \mathbb{P}[(U\cdot\nabla)U]$. Let $W = \partial_z U$ and Q_{AT} project orthogonally to both U and W. Then*

$$\boxed{\|Q_{AT}G(t)\|_2 \geq c\,\tau^{-3/4-3h/2}}$$

for all sufficiently small τ, with some profile-dependent $c > 0$.

Proof. The global periodic pressure need not be axisymmetric, because the torus lacks continuous rotational symmetry. Nonetheless, on each physical circle the angular average of its azimuthal gradient is zero: $(2\pi)^{-1}\int_0^{2\pi} r^{-1}\partial_\theta p\, d\theta = 0$. Since U, W are axisymmetric, the angular average of the residual equals the unprojected azimuthal defect minus both mode projections. At the equator $z = 0$, $U_r = \partial_z U_\theta = W_\theta = 0$ by reflection, so azimuthal advection and the translation projection vanish. The physical vector Laplacian gives

$$\overline{(Q_{AT}G)_\theta}(t,r,0) = \tau^{-A-1}\mathcal{D}_{\lambda(t)}V(X) + O(\tau^{-A-1+2h}), \quad \mathcal{D}_\lambda V = -2XV'' + (X-2)V' + \left(A - \lambda + \frac{1}{2X}\right)V.$$

If $\mathcal{D}_{\lambda_*}V$ vanished everywhere on $(0, X_*)$, ODE uniqueness with flat outer data $V(X_*) = V'(X_*) = 0$ would force $V \equiv 0$, a contradiction. A nonzero point persists on an open similarity-coordinate neighborhood and the exact physical Jacobian is $2\pi\tau^{1+D}(1-2h\eta^2)(1-\eta^2)^{-D-2}\,dX\,d\eta$. Jensen's inequality controls the angular average by the full vector norm; taking the L^2 norm yields $\tau^{-A-1+(1+D)/2} = \tau^{-3/4-3h/2}$. □

Two-radius exact certificate for the Attack O prototype. For $h = 1/10$, $A = 3/5$, and

$$V(X) = \frac{\sqrt{2X}}{4}\frac{d}{dX}\left[X\exp\left(-\frac{X}{1-X}\right)\right], \qquad 0 < X < 1,$$

define $R(X) = \mathcal{D}_0 V(X)/V(X)$ where nonzero. Exact symbolic arithmetic gives

$$\boxed{R(1/4) = 25819/810, \quad R(1/2) = -109/10, \quad R(1/4) - R(1/2) = 17324/405.}$$

Both $V(1/4) = 5\sqrt{2}\,e^{-1/3}/72$ and $V(1/2) = -e^{-1}/4$ are nonzero. Hence no scalar amplitude coefficient cancels both radii, and $\|Q_{AT}G\|_2 \gtrsim \tau^{-9/10}$. The genuine H8 core has a nontrivial odd axial jet $U_0(0,\eta) = j_0 + 4\eta$ and need not satisfy the assumed reflection symmetry. This theorem is therefore *not* an authentic-H8 no-go or a solution to the unforced Millennium problem.

Verification and next gate. The six fresh exact tests passed, and 12 existing Attack O tests passed alongside them (19/19 selected). The independently distributed 94-page cumulative PDF preserves 91 prior pages and adds the three-page standalone Appendix P; the local rendered pages were checked. The precise next target is a fully frozen physical H8 source, its non-even profile cancellation and Leray residual, its positive seven-ray stress, nonlinear amplitude injection, translation matching and pointwise complementary retention.

Appendix Q. Two-mode terminal-dyad compatibility for compact H8 carriers

Scope. This appendix derives necessary forcing laws from the exact coupled modulation equations of Appendix M and the profile-scaling hypotheses of Appendices N–O. Its diagnostic constants concern the prepared field of Appendix O, not an authenticated singular H8 carrier. The unforced Navier–Stokes problem remains open.

Exact equations. Put $\tau = T - t$, $0 < h < 1/6$, $A = \frac{1}{2} + h$, $D = \frac{1}{2} - h$, $p = \frac{1}{2} - 3h$, and $W = \partial_z U$. For a smooth divergence-free periodic prepared carrier U, define $G = U_t - \Delta U + \mathbb{P}[(U\cdot\nabla)U]$ and $\mathcal{B}(f,g) = \mathbb{P}[(f\cdot\nabla)g]$. Write the unforcing correction in the moving orthogonal two-mode frame

$$w = \alpha U + aW + v, \qquad \beta = 1 + \alpha, \qquad \langle v, U\rangle = \langle v, W\rangle = 0.$$

Let $M = \|U\|_2^2$, $N = \|W\|_2^2$ and

$$c_A = \frac{\langle G, U\rangle}{M}, \qquad g_T = \frac{\langle G, W\rangle}{N}, \qquad b_T = \frac{\langle \partial_z G, W\rangle}{N}, \qquad e_T = \frac{\langle \mathcal{B}(U,U), W\rangle}{N}.$$

The exact equations established in Appendix M are

$$\beta' + c_A\beta = a\,\frac{N g_T}{M} + J_A, \tag{83}$$

$$a' + b_T a = -\beta g_T - \alpha e_T + J_T, \tag{84}$$

where, with $\mathcal{L}_U f = \Delta f - \mathcal{B}(U,f) - \mathcal{B}(f,U)$,

$$J_A = -\frac{\langle \mathcal{B}(w,w), U\rangle}{M} + \frac{\langle v, \mathcal{L}_U^* U + U_t\rangle}{M},$$
$$J_T = -\frac{\langle \mathcal{B}(w,w), W\rangle}{N} + \frac{\langle v, \mathcal{L}_U^* W + W_t\rangle}{N}.$$

Both injections are exact quantities; they are not assumed small.

Theorem Q.1 (Simultaneous terminal-dyad compatibility). Assume $M_0, N_0 > 0$, real constants c_0, g_0, and some $C > 0$ satisfy, as $\tau \downarrow 0$,

$$M = \tau^p(M_0 + o(1)), \quad N = \tau^{-A}(N_0 + o(1)), \quad c_A = \frac{c_0 + o(1)}{\tau}, \quad g_T = \tau^{-A}(g_0 + o(1)),$$

together with $|b_T| \le C/\tau$, $|e_T| \le C\tau^{-A}$. If a full unforcing correction satisfies the simultaneous retention laws

$$\alpha(t) = o(1), \qquad a(t) = o(\tau^D),$$

then for $I_\tau = [T - \tau, T - \tau/2]$,

$$\lim_{\tau\downarrow 0} \int_{I_\tau} J_A(s)\,ds = c_0 \log 2, \tag{85}$$

$$\lim_{\tau\downarrow 0} \tau^{-D} \int_{I_\tau} J_T(s)\,ds = g_0\,\frac{1 - 2^{-D}}{D}. \tag{86}$$

In particular, if $c_0 \neq 0$ then $J_A = o(\tau^{-1})$ is incompatible with amplitude retention; if $g_0 \neq 0$ then $J_T = o(\tau^{-A})$ is incompatible with translation retention. These statements are necessary conditions, not sufficient existence criteria.

Proof. Write $a = \tau^D \varepsilon_a(\tau)$ with $\varepsilon_a(\tau) \to 0$. The exponent identity $2A + p = 1 + D$ implies

$$a \frac{N g_T}{M} = \varepsilon_a(\tau) \frac{N_0 g_0 / M_0 + o(1)}{\tau} = o(\tau^{-1}).$$

Integration of (83) over I_τ gives $\beta(T - \tau/2) - \beta(T - \tau) = o(1)$, $\int_{I_\tau} c_A \beta \, ds \to c_0 \log 2$, and an $o(1)$ integral of the geometric coupling. This establishes (85). For (84), the endpoint difference and the integrals of $b_T a$ and αe_T are each $o(\tau^D)$. Since $A + D = 1$,

$$\int_{I_\tau} \beta g_T \, ds = g_0 \int_{\tau/2}^{\tau} u^{-A} \, du + o(\tau^D) = g_0 \frac{1 - 2^{-D}}{D} \tau^D + o(\tau^D).$$

Rearrangement proves (86). The pointwise smallness consequences follow by integration. □

Translation damping alone is insufficient. For $b_0 > D$ the forced scalar equation $a' + (b_0/\tau)a = -g_0 \tau^{-A}$ has the exact general solution

$$a(t) = -\frac{g_0}{b_0 - D} \tau^D + C_1 \tau^{b_0}.$$

Thus damping of its homogeneous mode does not imply the strict $a = o(\tau^D)$ retention condition when $g_0 \neq 0$.

Numerical prepared-field diagnostic. The non-authentic Appendix O prototype with $h = 1/10$ has $A = 3/5$, $D = 2/5$, $p = 1/5$, and reports convergent, not interval-certified quadratures

$$c_0 \simeq 27.0163023544, \qquad b_0 \simeq 43.4741340312, \qquad g_0 = e_0 \simeq 0.0477816887268,$$
$$M_0 \simeq 0.829691939536, \qquad N_0 \simeq 9.33098522102.$$

Under the hypotheses of Theorem Q.1 these would require

$$\int_{I_\tau} J_A \, ds \longrightarrow 18.7262738\ldots, \qquad \tau^{-2/5} \int_{I_\tau} J_T \, ds \longrightarrow 0.0289248503\ldots.$$

Neglecting J_T in a scalar approximation instead predicts $a/\tau^{2/5} \to -0.00110929\ldots$, which does not meet the strict translation-retention target. No assertion is made that the full nonlinear correction follows this scalar approximation.

Outstanding proof boundary. Appendix P supplies a separate conditional complementary azimuthal-source obstruction for reflection-symmetric, flatly compact swirl profiles. It does not cover the authentic non-reflection-symmetric H8 local core. A physical H8 singularity proof still requires one authenticated source, source-specific seven-ray stresses, a periodic pressure, the two independent modal injections above, complementary propagator control and pointwise terminal retention. No unforced finite-time Navier–Stokes singularity is proved here.

All computational algorithms, proof-search procedures, and certificate criteria were developed by the author(s). Large Language Models (LLMs) such as ChatGPT and Claude were utilized strictly as computational assistants for debugging code and cross-checking routines. No mathematical claims rely on LLM outputs; all results are grounded exclusively in independently reproducible, rigorously verified certificates (exact, symbolic, rational, interval, exhaustive, or controlled high-precision).

Appendix R. Global energy compatibility for compact H8 carriers

Scope. This is a conditional obstruction derived from the unforced Navier–Stokes energy law. It does not exclude a solution with a non-negligible exterior correction or a different carrier.

Theorem R.1 (Energy compatibility of global L^2 retention). Let u solve smooth unforced viscosity-one incompressible Navier–Stokes on $\mathbb{T}^3$ for $t < T$. Let $U(t)$ be a nonzero smooth periodic, divergence-free prepared carrier, $\tau = T - t$, and suppose

$$\|U\|_2^2 = M_0\tau^p(1 + o(1)), \qquad \|\nabla U\|_2^2 = R_0M_0\tau^{p-1}(1 + o(1)), \tag{87}$$

with $M_0, p, R_0 > 0$. Assume

$$\sup_{t\in(T-\delta,T)} \frac{\|U(t)\|_2\|\Delta U(t)\|_2}{\|\nabla U(t)\|_2^2} < \infty \tag{88}$$

for some $\delta > 0$, and assume the full correction is *globally* subleading in the energy space:

$$\|u(t) - U(t)\|_2 = o(\|U(t)\|_2). \tag{89}$$

Then the necessary inequality is

$$\boxed{R_0 \le p/2.} \tag{90}$$

In particular, with differentiated similarity asymptotics the amplitude coefficient $c_A = \langle U_t - \Delta U + \mathbb{P}[(U\cdot\nabla)U], U\rangle/\|U\|_2^2$ satisfies $\tau c_A \to R_0 - p/2$; its positive limit is incompatible with (89).

Proof. Put $\varepsilon = \|u - U\|_2/\|U\|_2 = o(1)$. Periodic integration by parts and Cauchy–Schwarz show

$$\|\nabla u\|_2\|\nabla U\|_2 \ge \langle \nabla u, \nabla U\rangle = \|\nabla U\|_2^2 + \langle u - U, -\Delta U\rangle \ge (1 - o(1))\|\nabla U\|_2^2,$$

where (88) controls the last error. Thus $\|\nabla u\|_2^2 \ge (1 - o(1))\|\nabla U\|_2^2$. Likewise $\|u\|_2^2 = M_0\tau^p(1+o(1))$. On the terminal dyad $I_\tau = [T-\tau, T-\tau/2]$, the exact unforced energy identity implies

$$\begin{aligned} M_0(1 - 2^{-p})\tau^p + o(\tau^p) &= \|u(T-\tau)\|_2^2 - \|u(T-\tau/2)\|_2^2 \\ &= 2\int_{I_\tau} \|\nabla u\|_2^2\, ds \\ &\ge 2(1 - o(1))R_0M_0\int_{\tau/2}^{\tau} r^{p-1}\, dr \\ &= 2(1 - o(1))\frac{R_0M_0}{p}(1 - 2^{-p})\tau^p. \end{aligned}$$

Division by the positive common factor gives $1 \ge 2R_0/p$. The formula for c_A follows from $\langle \mathbb{P}[(U\cdot\nabla)U], U\rangle = 0$ and $\langle -\Delta U, U\rangle = \|\nabla U\|_2^2$. □

Corollary R.2 (Confinement without numerical quadrature). If U vanishes outside an embedded horizontal disk of radius $R_*\sqrt{\tau}$, then the scalar disk Dirichlet inequality applied componentwise yields

$$\|\nabla U\|_2^2 \ge \frac{j_{0,1}^2}{R_*^2\tau}\|U\|_2^2,$$

where $j_{0,1}$ is the first positive zero of the Bessel function J_0. Therefore if $j_{0,1}^2/R_*^2 > p/2$, the assumptions of Theorem R.1 cannot all hold. For the smooth Appendix O prepared-field prototype with $h = 1/10$, $p = 1/5$, $X_* = 1$ and $|\eta| \le 1/2$, one has $R_*^2 = 8/3$ and hence

$$R_0 \ge (3/8)j_{0,1}^2 > 2 > 1/10 = p/2.$$

The contradiction requires no unverified profile quadratures. The smooth flat compact prototype has a bounded Laplacian-shape ratio under its differentiated similarity scaling.

Connection with Appendix Q. For $w = \alpha U + aW + v$ with $W = \partial_z U$ and $M \sim M_0\tau^p$, $N = \|W\|_2^2 \sim N_0\tau^{-A}$, $A = \frac{1}{2} + h$, $D = \frac{1}{2} - h$, the exact exponent $A + p = 2D$ implies

$$\alpha = o(1), \quad a = o(\tau^D), \quad \|v\|_2 = o(\|U\|_2) \quad \Longrightarrow \quad \|w\|_2 = o(\|U\|_2).$$

When v is chart-supported in volume $O(\tau^{1+D})$, the pointwise condition $\|v\|_\infty = o(\tau^{-A})$ implies the displayed global L^2 smallness, since $-A + (1+D)/2 = p/2$. If instead v has a non-negligible *exterior* L^2 component, this theorem does not apply: that component needs an independent energy, pressure and complementary-propagator analysis.

Open theorem frontier. The theorem rules out globally L^2-subleading unforcing for this confined positive-Rayleigh carrier class, not an unforced singularity in general. Authentic H8 source selection, the weighted one-sided seven-ray stress inverse, a possible exterior energy reservoir, and nonlinear terminal retention remain unproved. The complete unforced Navier–Stokes theorem remains OPEN.

Appendix S. Exact physical residual and final closure interface

Scope. The identities below are exact on the periodic three-torus. Their verification does not supply the missing nonlinear correction, source-matched positive stress or singular solution. The complete unforced existence theorem remains open.

Exact periodic formulation. Let U be smooth, periodic and divergence-free, let $\mathbb{P}$ be the periodic Leray projection, and set

$$\mathcal{B}(f,g)=\mathbb{P}[(f\cdot\nabla)g],\quad G=U_t-\Delta U+\mathcal{B}(U,U),\quad L_U v=\Delta v-\mathcal{B}(U,v)-\mathcal{B}(v,U).$$

For every smooth divergence-free correction w the projected unforced residual of $u=U+w$ is the *exact* identity

$$\mathcal{N}(U+w)=G+w_t-L_U w+\mathcal{B}(w,w). \tag{91}$$

No prescribed body force has been introduced.

Proposition S.1 (Exact moving two-mode reconstruction). Let $W=\partial_z U$ and suppose U,W are L^2-orthogonal with nonzero norms. Write $w=\alpha U+aW+v$ and $\langle v,U\rangle=\langle v,W\rangle=0$. Then the full projected residual is

$$\boxed{\begin{aligned}\mathcal{N}(U+w)=\alpha' U+a'W+v_t-L_U v+(1+\alpha)G\\ +\alpha\mathcal{B}(U,U)+a\,\partial_z G+\mathcal{B}(w,w).\end{aligned}} \tag{92}$$

The identity holds for nonconstant U, varying two-mode projectors and arbitrary smooth v.

Proof. Differentiation and bilinearity expand $\mathcal{N}(U+w)$ into $G+w_t-\Delta w+\mathcal{B}(U,w)+\mathcal{B}(w,U)+\mathcal{B}(w,w)$. The coefficient of α satisfies $U_t-L_U U=G+\mathcal{B}(U,U)$. Because the periodic Leray projection commutes with ∂_z,

$$W_t-L_U W=W_t-\Delta W+\mathcal{B}(U,W)+\mathcal{B}(W,U)=\partial_z G.$$

Combining these identities gives (92) without dropping the moving modes, the time derivative, viscosity or quadratic correction. □

Pressure reconstruction. For smooth periodic divergence-free u define the zero-mean pressure uniquely by

$$-\Delta p=\sum_{i,j=1}^{3}\partial_i\partial_j(u_iu_j),\qquad \int_{\mathbb{T}^3}p\,dx=0. \tag{93}$$

The mean of the right-hand side vanishes, so this equation is solvable and $\mathbb{P}[(u\cdot\nabla)u]=(u\cdot\nabla)u+\nabla p$. Consequently $\mathcal{N}(u)=0$ and (93) are equivalent to the unforced physical Navier–Stokes PDE. This reconstruction does not solve $\mathcal{N}(u)=0$; it only fixes the pressure gauge for a candidate u.

Precise conditional closure statement. A smooth periodic divergence-free $U+w$ with a smooth pressure from (93) is an unforced Navier–Stokes solution only if the *whole vector field* in (92) vanishes at every point and time. In addition, a finite-time singularity conclusion requires smooth nonzero initial data, finite torus-cell kinetic energy, regularity for all $t<T$, and a certified pointwise trace $|U(t,x_t)+w(t,x_t)|\to\infty$ for a suitable family of points. Projected amplitude/translation equations alone do not establish the complementary equation or any of these global properties.

Final independent tests and proof status. Six independent exact symbolic regressions of the physical identities above use a nonconstant divergence-free three-dimensional Fourier field, nonlinear cross terms, the pressure Poisson sign, the periodic energy cancellation, the meridional r factor, and the implicit H8 axial chain rule. Five further tests reproduce the terminal energy-dyad algebra and the source-confinement spectral gap of Appendix R. A further independent three-dimensional periodic calculation directly projects the full nonlinear residual onto both the moving amplitude and translation modes and verifies both signed equations with nonzero correction coefficients. The combined *12-test* suite passed locally. These test receipts are evidence for the individual identities, not a machine-checked witness for the open existence theorem.

The remaining mathematical gates are an authenticated singular core and single periodic source; a compact weighted positive seven-ray inverse respecting the actual rotating frame; a constructive nonlinear correction and its exterior-energy dynamics; the entire zero-residual identity including pressure; terminal pointwise retention; and an end-to-end proof in the physical PDE definitions. The repository's G0–G8 ledger remains fail-closed with target status `OPEN`.

Appendix T. Necessary exterior-to-core energy transport

Scope. This appendix sharpens the energy-compatibility boundary of Appendix R while permitting a globally non-negligible exterior correction. Its transport law is a necessary consequence of the classical smooth unforced Navier–Stokes energy equation; neither a positive H8 stress nor a singular unforced solution is constructed.

Local conservation and the moving boundary. Let u be a smooth periodic divergence-free solution at unit viscosity, and let π be its physical periodic pressure. Put $e = \frac{1}{2}|u|^2$. The exact pointwise identity is

$$\partial_t e + \nabla \cdot \big[(e+\pi)u - \nabla e\big] = -|\nabla u|^2. \tag{94}$$

Let $\Omega(t) \Subset \mathbb{T}^3$ have a smooth moving boundary, outward normal n and normal boundary velocity $V_b \cdot n$. Reynolds transport gives

$$\frac{d}{dt}\int_{\Omega(t)} e\,dx + \int_{\Omega(t)} |\nabla u|^2\,dx + \Phi(t) = 0, \quad \Phi(t) = \int_{\partial\Omega(t)} \big[(e+\pi)u - \nabla e - eV_b\big]\cdot n\,dS. \tag{95}$$

Here $\Phi > 0$ denotes *outward* transport. The integrated pressure flux is invariant under the time-dependent gauge $\pi \mapsto \pi + c(t)$ because $\int_{\partial\Omega} u \cdot n\,dS = 0$.

Theorem T.1 (Terminal-dyad inward-flux requirement). Let $\tau = T - t$, $p > 0$ and $M_0, R_0 > 0$. Assume a smooth prepared divergence-free field $U(t)$ is compactly supported strictly inside $\Omega(t)$, vanishing in a collar of its boundary, and satisfies

$$\|U\|_2^2 = M_0\tau^p(1+o(1)), \qquad \|\nabla U\|_2^2 = R_0 M_0 \tau^{p-1}(1+o(1)). \tag{96}$$

Suppose the genuine unforced solution $u = U + w$ retains the carrier *locally* in the moving core:

$$\|w(t)\|_{L^2(\Omega(t))} = o(\|U(t)\|_2). \tag{97}$$

Assume additionally the dimensionless Laplacian shape ratio $\|U\|_2\|\Delta U\|_2/\|\nabla U\|_2^2$ remains bounded. Then for $I_\tau = [T-\tau, T-\tau/2]$,

$$\boxed{\liminf_{\tau\downarrow 0} \frac{-\int_{I_\tau}\Phi(s)\,ds}{M_0\tau^p} \geq \frac{1-2^{-p}}{p}\left(R_0 - \frac{p}{2}\right).} \tag{98}$$

If in addition $\|\nabla w(t)\|_{L^2(\Omega(t))} = o(\|\nabla U(t)\|_2)$, then (98) holds with *equality in the limit*. Consequently $R_0 > p/2$ requires a positive, leading-order inward energy transfer across the moving boundary. The theorem does not require w to be small in global L^2.

Proof. Since U is supported within $\Omega(t)$, the local retention condition implies

$$E_\Omega(t) := \int_{\Omega(t)} e\,dx = \frac{1}{2}M_0\tau^p(1+o(1)).$$

The collar permits integration by parts without surface terms:

$$\langle \nabla u, \nabla U\rangle_{L^2(\Omega)} = \|\nabla U\|_2^2 + \langle w, -\Delta U\rangle_{L^2(\Omega)}.$$

By Cauchy–Schwarz and the bounded shape ratio, $\|\nabla u\|_{L^2(\Omega)}^2 \geq (1-o(1))\|\nabla U\|_2^2$. The stronger gradient-retention condition makes this asymptotic an equality. Integrating (95) over I_τ gives

$$-\int_{I_\tau}\Phi(s)\,ds = E_\Omega(T-\tau/2) - E_\Omega(T-\tau) + \int_{I_\tau}\int_{\Omega(s)} |\nabla u|^2\,dx\,ds.$$

The first term is $-\frac{1}{2}M_0(1-2^{-p})\tau^p + o(\tau^p)$; the second is at least $(1-o(1))R_0M_0(1-2^{-p})\tau^p/p$. Together they prove (98). □

Corollary T.2 (Source-independent axisymmetric diagnostic). For the compact axisymmetric solenoidal Appendix O prepared profile, $h = 1/10$, $p = 1/5$, $X_* = 1$, $|\eta| \le 1/2$. The transverse disk radius obeys $R_*^2 = 8/3$. Because the horizontal Cartesian components of an axisymmetric vector have angular order one and the axial component has zero horizontal mean, the exact disk inequalities in Appendix O imply

$$R_0 \ge \frac{3}{8} j_{1,1}^2 = 5.50573899079645\ldots.$$

Therefore any unforced solution satisfying the local-retention hypotheses of Theorem T.1 for this diagnostic must obey

$$\liminf_{\tau \downarrow 0} \frac{-\int_{I_\tau} \Phi(s)\, ds}{M_0 \tau^{1/5}} \ge \frac{1 - 2^{-1/5}}{1/5} \left(\frac{3}{8} j_{1,1}^2 - \frac{1}{10} \right) = 3.49884933663390\ldots. \tag{99}$$

The Bessel eigenvalue is classical; the numerical display is a rounding of an exact spectral expression, not a certificate for the non-authentic Appendix O source coefficients.

Proposition T.3 (An isolated compact core cannot retain). Under the hypotheses of Theorem T.1 with $R_0 > p/2$, if $w = 0$ in an open collar of $\partial\Omega(t)$ for all sufficiently late times, then no such unforced solution exists. Indeed $U = 0$ in that collar, hence $u = \nabla u = e = 0$ on the moving boundary and (95) has $\Phi = 0$, contradicting (98). More generally, any proposed boundary estimates making $\int_{I_\tau} |\Phi(s)|\, ds = o(\tau^p)$ contradict the same theorem.

Corollary T.4 (A bounded exterior collar cannot supply the flux). Assume also the H8 compact chart has lateral scale $r = O(\tau^{1/2})$, axial scale $z = O(\tau^D)$, where $D = 1/2 - h$, $0 < h < 1/6$, and $p = 1/2 - 3h$. Choose a smooth moving collar boundary with surface area $|\partial\Omega(t)| = O(\tau^{1-h})$ and normal speed $|V_b \cdot n| = O(\tau^{-A})$, $A = 1/2 + h$. If the full unforced pressure π and all traces $w, \nabla w$ are uniformly bounded on this collar as $t \uparrow T$, then, since $U = \nabla U = 0$ there,

$$|\Phi(t)| = O(\tau^{1-h-A}) = O(\tau^{1/2-2h}), \qquad \int_{I_\tau} |\Phi(s)|\, ds = O(\tau^{3/2-2h}) = o(\tau^p).$$

This contradicts Theorem T.1 whenever $R_0 > p/2$. Consequently, any exterior-coupled realization retaining such a rapidly dissipating core must violate at least one of these uniform boundary trace bounds: the correction or its gradient or the physical pressure must develop a sufficiently large interface contribution. This conditional exclusion does not identify which channel supplies it, and does not require an isolated or globally small correction.

A quantitative global-error alternative. Let $\epsilon(t) = \|u(t) - U(t)\|_2 / \|U(t)\|_2$ globally, $K = \limsup_{t \uparrow T} \|U\|_2 \|\Delta U\|_2 / \|\nabla U\|_2^2 < \infty$, and $\delta = \limsup_{t \uparrow T} \epsilon(t)$. If $0 \le \delta < \min\{1, 1/K\}$, then the exact global unforced energy law and the same projection argument give the necessary inequality

$$\boxed{(1+\delta)^2 - 2^{-p}(1-\delta)^2 \ge \frac{2R_0}{p}(1 - 2^{-p})(1 - K\delta)^2.} \tag{100}$$

To prove it, bound the early dyad energy above by $(1 + \delta + o(1))^2 M_0 \tau^p$ and the late energy below by $(1 - \delta + o(1))^2 M_0 2^{-p} \tau^p$. Meanwhile $\|\nabla u\|_2^2 \ge (1 - K\delta - o(1))^2 \|\nabla U\|_2^2$. Insert these into the energy identity and pass to the limit. At $\delta = 0$ this recovers Theorem R.1's necessary bound $R_0 \le p/2$. No numerical value for the authentic H8 ratio K is asserted.

Where the exterior correction enters. Since U and all its derivatives vanish on the selected boundary collar, the integrand of the exact flux becomes

$$\left[\left(\frac{|w|^2}{2}+\pi\right)w-\nabla\left(\frac{|w|^2}{2}\right)-\frac{|w|^2}{2}V_b\right]\cdot n.$$

Its nonlocal pressure trace is essential; neither the amplitude and translation ODEs nor the pointwise stress cone provides the required sign or size. The decisive next source-specific certificate is a lower bound on this *actual* inward pressure–convection–viscous–boundary flux on every sufficiently late dyad while independently proving the complementary PDE and pointwise terminal retention. The authentic H8 source tuple remains unfrozen. The complete unforced Navier–Stokes theorem remains OPEN.

Appendix U. Anisotropic interface flux and pressure-oscillation threshold

Scope. This appendix sharpens the necessary exterior-to-core transport gate of Appendix T. All results concern a hypothetically retained compact carrier and an actual smooth unforced solution; they neither authenticate the H8 source nor construct a singularity.

Exact anisotropic moving-boundary geometry. Put $\tau = T-t$, $0 < h < 1/6$, $D = 1/2-h$, $A = 1/2+h$, and $p = 1/2-3h$. Fix a smooth bounded reference region Ω_0 and a fixed centre x_0. Let $\Omega(t) = x_0 + S_\tau \Omega_0$, where $S_\tau = \mathrm{diag}(\tau^{1/2}, \tau^{1/2}, \tau^D)$, embedded in a single torus chart for small τ. If n_0 denotes the reference outward unit normal, then the area Jacobian is $\det(S_\tau)|S_\tau^{-T} n_0|$ and

$$|\partial\Omega(t)| \le C_{\Omega_0} \tau^{1-h}. \tag{101}$$

For the physical boundary velocity $V_b(t, S_\tau y) = \partial_t(S_\tau y)$, the oriented measure identity is exact:

$$(V_b \cdot n)\, dS = -\tau^D \left[\frac{1}{2}(y_1 n_{0,1} + y_2 n_{0,2}) + D y_3 n_{0,3}\right] dS_0. \tag{102}$$

Consequently $\int_{\partial\Omega(t)} |V_b \cdot n|\, dS \le C_{\Omega_0}\tau^D$. This is sharper than multiplying the maximum boundary speed by its total surface area.

Theorem U.1 (Necessary interface-growth threshold). Let u be smooth, periodic, divergence-free and solve the unforced viscosity-one Navier–Stokes equation on $t < T$. Let U be a smooth divergence-free carrier supported strictly within $\Omega(t)$, vanishing with its gradient in a neighbourhood of $\partial\Omega(t)$. Suppose

$$\|U\|_2^2 = M_0 \tau^p (1+o(1)), \quad \|\nabla U\|_2^2 = R_0 M_0 \tau^{p-1}(1+o(1)), \quad M_0 > 0,\ R_0 > p/2,$$

and suppose the ratio $\|U\|_2 \|\Delta U\|_2 / \|\nabla U\|_2^2$ stays bounded and $\|u - U\|_{L^2(\Omega(t))} = o(\|U\|_2)$. Define $I_\tau = [T-\tau, T-\tau/2]$ and

$$V_\tau = \sup_{s\in I_\tau,\ x\in\partial\Omega(s)} |u(s,x)|, \quad G_\tau = \sup_{s\in I_\tau,\ x\in\partial\Omega(s)} |\nabla u(s,x)|.$$

Let P_τ be the maximum over this dyad of the physical pressure *oscillation* on each boundary, namely $\max_{\partial\Omega(s)} \pi(s,\cdot) - \min_{\partial\Omega(s)} \pi(s,\cdot)$. Then for some $c > 0$ and all sufficiently small τ,

$$\boxed{\tau^{3/2+2h}\left(V_\tau^3 + P_\tau V_\tau + V_\tau G_\tau\right) + \tau^{1+2h} V_\tau^2 \ge c.} \tag{103}$$

The hypothesis concerns *local* energy retention and permits a globally non-negligible exterior correction.

Proof. The smooth unforced local energy equality reads

$$\partial_t(|u|^2/2) + \mathrm{div}[(|u|^2/2 + \pi)u - \nabla(|u|^2/2)] = -|\nabla u|^2.$$

The local-retention theorem of Appendix T gives, with Φ the outward moving-boundary energy flux,

$$-\int_{I_\tau} \Phi(s)\, ds \ge \left[\frac{1-2^{-p}}{p}(R_0 - p/2) + o(1)\right] M_0 \tau^p.$$

The divergence-free identity $\int_{\partial\Omega(s)} u \cdot n \, dS = 0$ permits subtraction of any time-dependent constant from π; thus the pressure boundary term is controlled by its *oscillation*, independent of gauge. The collar of zero U makes $u = w$ on $\partial\Omega$. Since $|\nabla(|u|^2/2)| \le |u||\nabla u|$, the exact geometry (101)–(102) yields

$$\left| \int_{I_\tau} \Phi(s)\, ds \right| \le C \left[\tau^{2-h}(V_\tau^3 + P_\tau V_\tau + V_\tau G_\tau) + \tau^{1+D} V_\tau^2 \right].$$

Divide by τ^p; use $2 - h - p = 3/2 + 2h$ and $1 + D - p = 1 + 2h$ to conclude. □

Corollary U.2 (Four competing interface channels). If for all late terminal dyads $V_\tau = O(\tau^{-\sigma})$, $P_\tau = O(\tau^{-\rho})$ and $G_\tau = O(\tau^{-\kappa})$, then at least one of

$$\boxed{3\sigma \ge 3/2 + 2h, \quad \rho + \sigma \ge 3/2 + 2h, \quad \sigma + \kappa \ge 3/2 + 2h, \quad \sigma \ge 1/2 + h} \tag{104}$$

is necessary. If the pressure and viscous terms are subcritical, then $\sigma \ge 1/2 + 2h/3$ is necessary in this growth-exponent sense.

Corollary U.3 (Gauge-invariant pressure burden at $h = 1/10$). Suppose the hypotheses of Theorem U.1 hold with $h = 1/10$ and $V_\tau \le C_V \tau^{-1/2}$, $G_\tau \le C_G \tau^{-1}$. The convection, viscosity and boundary-motion contributions in (103) are then $O(\tau^{1/5})$ and vanish. The pressure term must therefore remain nonzero; hence for some $c_\pi > 0$ on every sufficiently small dyad,

$$\boxed{P_\tau \ge c_\pi \tau^{-6/5}.} \tag{105}$$

This concerns pressure oscillation, not an arbitrary gauge representative, and is conditional on the two interface bounds.

Corollary U.4 (General critical pressure resonance). For every $0 < h < 1/6$, under Theorem U.1's assumptions, the two interface bounds $V_\tau = O(\tau^{-1/2})$ and $G_\tau = O(\tau^{-1})$ imply the *gauge-invariant* pressure-oscillation requirement

$$\boxed{P_\tau \ge c_h \tau^{-(1+2h)} = c_h \tau^{-2A}.} \tag{106}$$

Indeed all other terms in (103) are $O(\tau^{2h})$ and vanish. The exponent $2A$ equals the exponent of the square of the leading H8 carrier velocity, but this scaling coincidence does not establish a nonzero physical pressure trace or a positive pressure-work flux for an actual source-matched exterior field.

Open proof gate and verification. The coefficient needed to enter Theorem U.1 is positive for the independently specified, non-authentic Appendix O prototype by Appendix T's sharp Bessel estimate. The existence of a correction and its pressure are not established. Seven independent tests check the exact cofactor normal-motion identity, area and motion exponents, terminal flux thresholds, pressure exponent and diagnostic spectral coefficient; these do not constitute a proof of a singular Navier–Stokes solution. The authentic source cutoff, physical frame, periodic Poisson pressure, seven-ray positive inverse, complementary nonlinear equation and pointwise terminal retention all remain OPEN.

Appendix V. Periodic Poisson pressure and the signed interface-work gate

Scope. We isolate the signed requirement missing from the pressure-amplitude condition of Appendix U. The exact Fourier counterexample is a smooth diagnostic, not an authenticated H8 carrier, nonlinear correction, or finite-time singularity.

The complete Poisson pressure. For smooth periodic divergence-free vector fields F, H, write

$$\mathcal{Q}(F,H) = (-\Delta)_0^{-1}\partial_i\partial_j(F_iH_j), \qquad \pi[u] = \mathcal{Q}(u,u),$$

where $(-\Delta)_0^{-1}$ denotes the zero-mean periodic inverse. Repeated indices are summed. Symmetry of the double derivative implies $\mathcal{Q}(F,H) = \mathcal{Q}(H,F)$, hence for $u = U + w$

$$\boxed{\pi[U+w] = \pi[U] + 2\mathcal{Q}(U,w) + \pi[w].} \tag{107}$$

For a region $\Omega(t)$ with $U = 0$ on a collar of its boundary, the outward physical pressure work is exactly

$$\boxed{\mathcal{W}_\pi(t) = \int_{\partial\Omega(t)} [\pi[U] + 2\mathcal{Q}(U,w) + \pi[w]]\,(w\cdot n)\,dS.} \tag{108}$$

Since $\int_{\partial\Omega} w\cdot n\,dS = 0$, this is gauge invariant. There is no sign constraint from an oscillation bound for π alone.

Proposition V.1 (Exact signed Fourier certificate). On $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$, set $\widehat{f}(k) = (2\pi)^{-3}\int f(x)e^{-ik\cdot x}\,dx$ and $\widehat{\mu}(k) = \int_{\partial\Omega} g(x)e^{-ik\cdot x}\,dS$, with $g = u\cdot n$. For smooth periodic u,

$$\boxed{\int_{\partial\Omega} \pi[u]g\,dS = -\sum_{k\in\mathbb{Z}^3\setminus\{0\}} \frac{k_ik_j}{|k|^2}\widehat{u_iu_j}(k)\widehat{\mu}(-k).} \tag{109}$$

The sum converges absolutely by smoothness of the pressure. It is invariant under the pressure gauge because $\widehat{\mu}(0) = 0$. If U vanishes in a collar, then $\pi[U]$ is harmonic in that collar, but its boundary trace need not be constant.

Proof. The Fourier transform of $-\Delta\pi = \partial_i\partial_j(u_iu_j)$ is $\widehat{\pi}(k) = -k_ik_j|k|^{-2}\widehat{u_iu_j}(k)$ for $k \neq 0$. Expand the pressure on the boundary and integrate the series. If U vanishes in a collar, the Poisson source vanishes there. □

Proposition V.2 (Exact pressure-work sign reversal). Define on $\mathbb{T}^3$

$$f(x) = \sin x + \tfrac{1}{3}\sin 2x, \quad U = (f(x)\sin y,\ f'(x)\cos y,\ 0), \quad v = (\cos 2y, 0, 0), \quad u_\lambda = U + \lambda v.$$

Let $\Omega = (0,\pi)\times\mathbb{T}^2$, and construct the *full* zero-mean Poisson pressure $\pi_\lambda = \pi[u_\lambda]$ for each real λ. Every u_λ is smooth, divergence free and periodic; $U\cdot n = 0$ on the two boundary faces. Then

$$\boxed{\int_{\partial\Omega} \pi_\lambda u_\lambda\cdot n\,dS = \frac{224\pi^2}{195}\lambda.} \tag{110}$$

Thus choosing λ of either sign reverses the instantaneous pressure-work direction even after consistently recomputing its pressure. This is not a family of singular Navier–Stokes solutions.

Proof. The carrier pressure is exactly

$$\pi[U] = -\tfrac{1}{2}f(x)^2 + \tfrac{5}{18} + b(x)\cos 2y, \quad b(x) = -\tfrac{13}{36} - \tfrac{3}{10}\cos x + \tfrac{1}{78}\cos 3x.$$

Direct differentiation verifies $-\Delta\pi[U] = \partial_i\partial_j(U_iU_j)$ and zero mean. Since $f(0) = f(\pi) = 0$, the normal velocity on each face is $\lambda\cos 2y$. The pressure difference is

$$\pi[U](\pi, y, z) - \pi[U](0, y, z) = \tfrac{112}{195}\cos 2y.$$

The mixed pressure $2\lambda\mathcal{Q}(U, v)$ only contains the y-frequencies 1 and 3, orthogonal to $\cos 2y$; and $\pi[v] = 0$ because v_x is independent of x. Integrating $\cos^2 2y$ over y and the constant z factor gives $\lambda(112/195)\pi(2\pi) = 224\pi^2\lambda/195$.

□

The actual H8 signed certificate. With $\mathcal{W}_{\pi,U}$, $\mathcal{W}_{\pi,Uw}$ and $\mathcal{W}_{\pi,w}$ denoting the three terms in (108), and Φ_{other} the convection, viscous and boundary-motion flux from Appendix T, an actual unforced solution under the local-retention hypotheses must obey

$$-\int_{I_\tau}[\mathcal{W}_{\pi,U} + \mathcal{W}_{\pi,Uw} + \mathcal{W}_{\pi,w} + \Phi_{\text{other}}]\,dt \geq \left[\frac{1-2^{-p}}{p}\left(R_0 - \frac{p}{2}\right) + o(1)\right] M_0\tau^p. \tag{111}$$

The complete signed pressure and boundary-normal velocity must be computed or rigorously enclosed for *one common authentic H8 source* and a correction satisfying the full nonlinear unforced equation. A lower bound on pressure oscillation does not imply the signed work. Even a verified signed flux condition is necessary, not sufficient, for constructing a singular solution.

Verification and claim boundary. An independent exact symbolic regression verifies the solenoidal fields, Poisson source and zero mean, boundary traces, frequency decoupling, rational coefficient and sign reversal. The original Case 006/H8 tuple remains unfrozen: the common physical potential, cutoff, shapers, frame, pressure, stress centre and residual are not authenticated together. Consequently no authentic H8 pressure-work coefficient, positive seven-ray stress or nonlinear terminal-retained solution is proved. The full unforced Navier–Stokes theorem remains OPEN.

Appendix W. A source-specific signed pressure-work certificate

Scope. This appendix reduces the pressure part of the H8 energy-transfer problem to one signed coefficient and gives an explicit finite-Fourier error test. It does not authenticate an original H8 source or construct a nonlinear unforced solution. The periodic Poisson pressure is used in full; a local pressure ansatz is not a substitute.

Critical lateral trace. Put $\tau = T - t$, $0 < h < 1/6$, $D = 1/2 - h$, $A = 1/2 + h$, $p = 1/2 - 3h$. Fix a smooth bounded region Ω_0 and let $\Omega(t) = x_0 + S_\tau \Omega_0$, with $S_\tau = \mathrm{diag}(\tau^{1/2}, \tau^{1/2}, \tau^D)$, contained in one periodic chart. Suppose the smooth prepared solenoidal carrier U vanishes in a collar of this boundary, and write the genuine divergence-free unforced velocity as $u = U + w$. Reconstruct the pressure from $-\Delta\pi = \partial_i \partial_j (u_i u_j)$ and $\int_{\mathbb{T}^3} \pi = 0$.

Theorem W.1 (Critical signed pressure coefficient). Suppose the following boundary profiles exist uniformly on $\partial\Omega_0$ as $\tau \downarrow 0$:

$$\tau^{1/2} w_\perp(t, x_0 + S_\tau y) \longrightarrow V_\perp(y), \qquad \tau^{1/2} w_z(t, x_0 + S_\tau y) = O(1),$$

$$\tau^{2A}(\pi(t, x_0 + S_\tau y) - c(t)) \longrightarrow \Pi(y),$$

where $c(t)$ is a time-dependent pressure-gauge choice. Then

$$\boxed{\int_{\partial\Omega(t)} \pi\,(u \cdot n)\, dS = \tau^{p-1}(C_\pi + o(1)), \qquad C_\pi = \int_{\partial\Omega_0} \Pi(y) V_\perp(y) \cdot n_{0,\perp}(y)\, dS_0.} \tag{112}$$

Moreover, $\int_{\partial\Omega_0} V_\perp \cdot n_{0,\perp}\, dS_0 = 0$, so C_π is gauge independent.

Proof. The exact oriented surface measure is

$$n\, dS = \det S_\tau\, S_\tau^{-T} n_0\, dS_0 = (\tau^{1/2+D} n_{0,1}, \tau^{1/2+D} n_{0,2}, \tau n_{0,3})\, dS_0.$$

Since $U = 0$ in the collar, insert the boundary profiles. The lateral pressure-work term scales as $\tau^{1/2+D-1/2-2A} = \tau^{p-1}$, whereas the axial term is $O(\tau^{1-1/2-2A}) = O(\tau^{p-1+h})$. Solenoidality gives zero total normal flux. Divide that identity by τ^D and pass to the limit to obtain zero mean for the leading lateral normal trace. Thus the pressure gauge cancels. □

Corollary W.2 (The necessary signed coefficient). Assume also the local core-retention and dissipation hypotheses of Appendix T, with constants M_0, R_0, p, and suppose the combined convection, viscous and boundary-motion flux has the leading expansion $\Phi_{\mathrm{other}}(t) = \tau^{p-1}(C_{\mathrm{other}} + o(1))$. Then

$$\boxed{C_\pi + C_{\mathrm{other}} \le -M_0(R_0 - p/2).} \tag{113}$$

If $\Phi_{\mathrm{other}} = o(\tau^{p-1})$, the pressure coefficient alone must satisfy $C_\pi \le -M_0(R_0 - p/2)$. This is a necessary condition on the *actual* pressure and actual boundary-normal velocity, not an existence criterion.

Proposition W.3 (Explicit finite-Fourier tail certificate). For smooth periodic u, let $g = u \cdot n$ on the boundary of any smooth domain $\Omega \subset \mathbb{T}^3$ and set

$$\widehat{A}_{ij}(k) = (2\pi)^{-3} \int_{\mathbb{T}^3} u_i u_j e^{-ik\cdot x}\, dx, \quad \widehat{\mu}(k) = \int_{\partial\Omega} g(x) e^{-ik\cdot x}\, dS.$$

Let W_K be the pressure-work Fourier sum from Appendix V restricted to $0 < \|k\|_\infty \le K$, $K \ge 1$ an integer. Define the Fourier-normalized tensor Sobolev norm

$$\|u \otimes u\|_{H^3_F}^2 = \sum_{k \in \mathbb{Z}^3} (1 + |k|^2)^3 \sum_{i,j=1}^{3} |\widehat{A}_{ij}(k)|^2.$$

Then, with $W = \int_{\partial\Omega} \pi[u] g \, dS$,

$$\boxed{|W - W_K| \le \sqrt{\frac{26}{3}} \, K^{-3/2} \|g\|_{L^1(\partial\Omega)} \|u \otimes u\|_{H^3_F}.} \tag{114}$$

The bound may be integrated over a terminal dyad. Rigorous enclosures of the finite Fourier sum and right-hand side can therefore certify the sign of the full pressure work *only after* a source-specific velocity and moving interface are authenticated.

Proof. For $k \neq 0$, $\widehat{\pi}(k) = -k_i k_j |k|^{-2} \widehat{A}_{ij}(k)$. The matrix $k \otimes k/|k|^2$ has Frobenius norm one, hence $\|\pi\|_{H^3_F} \le \|u \otimes u\|_{H^3_F}$; also $|\widehat{\mu}(k)| \le \|g\|_{L^1(\partial\Omega)}$. The lattice shell $\|k\|_\infty = n$ has exactly $(2n+1)^3 - (2n-1)^3 = 24n^2 + 2$ points. Thus

$$\sum_{\|k\|_\infty > K} (1 + |k|^2)^{-3} \le 26 \sum_{n=K+1}^{\infty} n^{-4} \le \frac{26}{3K^3}.$$

Cauchy–Schwarz in the omitted Fourier series proves the result. □

Independent mode-overlap diagnostic. On the reference unit sphere, choose $V_\perp = (y_1, -y_2)$ and $\Pi = c(y_1^2 - y_2^2)$. Then $\int_{\partial\Omega_0} V_\perp \cdot n_{0,\perp} \, dS_0 = 0$ but $C_\pi = 16\pi c/15$, of either sign. This prescribed boundary profile is *not* claimed to solve the complete H8 periodic Poisson or unforced nonlinear problem.

Open source-specific proof gate. The genuine Case 006 source tuple (vector potential, three-variable cutoff, shapers, rotating frame, density, stress centre, complete periodic pressure and common residual), a compatible positive seven-ray stress, and a correction solving the full unforced equation remain unauthenticated. Neither the displayed coefficient criterion nor the elementary Fourier-tail bound closes the complete three-dimensional unforced Navier–Stokes theorem, which remains OPEN.

Appendix X. The axisymmetric pressure-mode filter at a shrinking H8 interface

Scope. We prove an exact cancellation for the leading *carrier self-pressure* in the conditional transverse axisymmetric limit. This is not an authentic H8 calculation: the original source-dependent nonaxisymmetric shapers, the full periodic pressure of the nonlinear correction, and the source-matched positive stress are not yet fixed.

The reference geometry and leading solenoidality. Fix $0 < h < 1/6$, and let $D = 1/2 - h$, $A = 1/2 + h$, $p = 1/2 - 3h$, and $\tau = T - t$. As in Appendices U and W, put $S_\tau = \mathrm{diag}(\tau^{1/2}, \tau^{1/2}, \tau^D)$ and let $\Omega(t) = x_0 + S_\tau \Omega_0$ in a torus chart. Assume Ω_0 is a smooth region of revolution whose horizontal sections are disks $\{R < a(\eta)\}$, with a regular smooth tip at either endpoint. Assume U vanishes on an open collar of $\partial\Omega(t)$, and that the solenoidal exterior correction w admits differentiated reference-coordinate profiles

$$\tau^{1/2} w_\perp(t, x_0 + S_\tau y) \longrightarrow V_\perp(y), \qquad \tau^{1/2} w_z(t, x_0 + S_\tau y) \longrightarrow V_z(y)$$

in C^1 on each closed reference cross-section, uniformly away from the tips (with integrable uniform control at the tips). The physical incompressibility identity is

$$0 = \mathrm{div}\, w = \tau^{-1}\, \mathrm{div}_{y_\perp}(\tau^{1/2} w_\perp) + \tau^{-1+h} \partial_\eta (\tau^{1/2} w_z).$$

Thus $\mathrm{div}_{y_\perp} V_\perp = 0$ on every reference disk and, by the planar divergence theorem,

$$\boxed{\int_0^{2\pi} q(\theta, \eta)\, d\theta = 0, \qquad q(\theta, \eta) = V_\perp(a(\eta), \theta, \eta) \cdot e_R(\theta).} \tag{115}$$

Lemma X.1 (Pressure generated by compact axisymmetric swirl in the transverse limit). Let $v_\theta(R, \eta)$ be an axisymmetric regular azimuthal velocity with $v_\theta = 0$ for $R \geq R_*(\eta) < a(\eta)$. Assume the leading carrier self-pressure Π_U is the regular radial solution, up to an additive function of η, of the transverse Poisson equation generated by $v_\theta e_\theta$:

$$-\Delta_{y_\perp} \Pi_U = \partial_{y_i} \partial_{y_j} [(v_\theta e_\theta)_i (v_\theta e_\theta)_j] = -\frac{1}{R} \partial_R (v_\theta^2). \tag{116}$$

Then $\partial_R \Pi_U = v_\theta^2 / R$ and $\Pi_U = C(\eta)$ in the entire exterior radial collar $R \geq R_*(\eta)$. In particular its trace on the reference boundary has no azimuthal Fourier modes except order zero.

Proof. The convective acceleration of axisymmetric pure swirl is $-(v_\theta^2/R) e_R$. Taking its transverse divergence proves the second equality in (116). Radial integration yields $\partial_R \Pi_U = v_\theta^2/R + c(\eta)/R$; regularity at $R = 0$ forces $c(\eta) = 0$. Since the swirl vanishes on the collar, the radial pressure derivative vanishes there. □

Theorem X.2 (Vanishing carrier-only leading pressure work). Assume the hypotheses of Lemma X.1 and the leading-solenoidality conditions above. With the normal cofactor of Appendix W, the leading carrier self-pressure coefficient is

$$\boxed{C_{\pi,U} = \int_{\partial\Omega_0} \Pi_U(y) V_\perp(y) \cdot n_{0,\perp}(y)\, dS_0 = 0.} \tag{117}$$

The claim is about the leading radial self-pressure Π_U, not the full pressure Π of $U + w$.

Proof. Parametrize the lateral surface of revolution as $y = (a(\eta)\cos\theta, a(\eta)\sin\theta, \eta)$. Its oriented lateral surface element is $n_{0,\perp} dS_0 = a(\eta) e_R(\theta)\, d\theta d\eta$. The pressure trace is $C(\eta)$ by Lemma X.1. Hence

$$C_{\pi,U} = \int a(\eta) C(\eta) \Big[\int_0^{2\pi} q(\theta, \eta)\, d\theta \Big] d\eta = 0$$

by (115). □

Corollary X.3 (Exact angular-mode selection). Let $\Pi(\theta, \eta)$ be any sufficiently regular leading *full* pressure on the boundary, and expand $\Pi(\theta, \eta) = \sum_{m\in\mathbb{Z}} \Pi_m(\eta) e^{im\theta}$ and $q(\theta, \eta) = \sum_{m\in\mathbb{Z}} q_m(\eta) e^{im\theta}$. Then

$$\boxed{C_\pi = 2\pi \int a(\eta) \sum_{m\neq 0} \Pi_m(\eta) q_{-m}(\eta)\, d\eta.} \tag{118}$$

The axisymmetric $m = 0$ pressure channel cancels *exactly*, because $q_0 = 0$ at each η. A successful pressure-only mechanism must supply a nonzero, signed overlap of matching $m \neq 0$ pressure and normal-velocity modes. Such overlap may arise from the original nonaxisymmetric H8 shapers, the full $2\mathcal{Q}(U, w)$ interaction, or $\pi[w]$, but it has not been demonstrated for the authentic source. The other energy fluxes remain relevant and can also supply the needed inward transfer.

An exact prepared-field control. For $R < 1$, take $v_\theta(R, \eta) = R(1 - R^2)^3 b(\eta)$ and extend it by zero for $R \geq 1$; this is a finite-differentiability diagnostic, not an H8 source. A radial self-pressure is

$$\Pi_U(R, \eta) = -\tfrac{1}{14}(1 - R^2)^7 b(\eta)^2 \quad (0 \leq R \leq 1), \qquad \Pi_U(R, \eta) = 0 \quad (R \geq 1).$$

Direct differentiation gives $\partial_R \Pi_U = v_\theta^2 / R$. It is therefore identically constant on every exterior circular interface. An angular pressure mode can nevertheless have nonzero signed pairing: for $q = \cos\theta$ and $\Pi = \cos\theta$, one has $\int_0^{2\pi} \Pi q\, d\theta = \pi$, whereas $\Pi \equiv 1$ gives zero. These are source-independent diagnostics.

Scope of the remaining physical theorem. The exact identity (118) is a useful *mode-selection gate*, not an existence proof. It presumes the stated differentiated anisotropic boundary limits; it does not derive the transverse pressure limit for the original nonaxisymmetric, rotating-frame H8 source. Before applying it, one must freeze the genuine Case 006 vector potential, common cutoff, all shapers and overlaps, rotated stress frame, density and stress centre, periodic Poisson pressure, and the exterior correction solving the full unforced nonlinear equation. The preceding nine algebraic checks validate only the model identities. The complete unforced Navier–Stokes theorem remains OPEN.

Appendix Y. Quarter-turn symmetry and certified signed pressure

Scope. This appendix gives a source-independent exact periodic-pressure test of the quarter-turn symmetry used by H8 localization and a fourfold angular-mode error bound. It does not authenticate the Case 006 H8 data or construct a nonlinear unforced solution.

Proposition Y.1 (Fourfold selection with a symmetry caveat). On the square periodic three-torus let $Q(x,y,z)=(-y,x,z)$. If a smooth divergence-free velocity satisfies $u(Qx)=Qu(x)$, its unique mean-zero Poisson pressure $\pi[u]=(-\Delta)_0^{-1}\partial_i\partial_j(u_iu_j)$ is Q-invariant. Hence its trace on a circle centred on the rotation axis has Fourier coefficients $\pi_m=0$ unless $m\in 4\mathbb{Z}$. The quarter-turn average of a vector field has this covariance. However, if only the prepared carrier U is equivariant, the conclusion is valid for $\pi[U]$ but does *not* automatically hold for $\pi[U+w]$: the exact polarization $\pi[U+w]=\pi[U]+2\mathcal{Q}(U,w)+\pi[w]$ can break the symmetry when w does. Averaging a proposed nonlinear correction does not generally preserve the correction equation.

Proof. The scalar Poisson source is invariant under Q and the periodic mean-zero inverse is unique. Quarter-turn invariance of the scalar trace forces its Fourier support to lie in $4\mathbb{Z}$. The last distinction follows from pressure polarization. □

Theorem Y.2 (Fourfold-symmetric pressure work with opposite signs). Let

$$\psi_\alpha(x,y)=\cos x+\cos y+\alpha\sin x\sin y(\cos x-\cos y),\qquad u_\alpha=(\partial_y\psi_\alpha,-\partial_x\psi_\alpha,0)$$

on $(\mathbb{R}/2\pi\mathbb{Z})^3$. Each u_α is smooth, divergence-free and C_4-equivariant. Reconstruct its *actual* mean-zero periodic Poisson pressure π_α and consider the embedded unit cylindrical region $\Omega=\{x^2+y^2<1\}\times\mathbb{T}_z$. Define work *per unit axial length*,

$$\mathcal{W}(\alpha)=\frac{1}{2\pi}\int_{\partial\Omega}\pi_\alpha u_\alpha\cdot n\,dS=\int_0^{2\pi}\pi_\alpha(\cos\theta,\sin\theta)\,u_\alpha\cdot e_R\,d\theta.$$

For $\alpha=3/10$ there is the exact finite Bessel formula

$$\boxed{\mathcal{W}(3/10)=2\pi\sum_{n\in\mathcal{N}}c_n\frac{J_1(\sqrt{n})}{\sqrt{n}},\qquad \mathcal{N}=\{1,5,9,13,17,29,37,41\},}\tag{119}$$

with

n	1	5	9	13	17	29	37	41
c_n	$-2427/32000$	$21/100$	$-10791/64000$	$3/640$	$1047/32000$	$-81/16000$	$27/8000$	$-81/64000$

and the *rigorously rationally enclosed* signs

$$\boxed{-0.0079871<\mathcal{W}(3/10)<-0.0079870,\qquad \mathcal{W}(-3/10)=-\mathcal{W}(3/10)>0.}\tag{120}$$

The full three-dimensional work equals $2\pi\mathcal{W}(\alpha)$. Neither snapshot is claimed to solve the time-dependent unforced Navier–Stokes equation.

Proof. Since $\sin x\sin y(\cos x-\cos y)=(\sin 2x\sin y-\sin x\sin 2y)/2$, the stream function is invariant under $(x,y)\mapsto(-y,x)$. Writing $\psi_\alpha=\sum_a s_a e^{ia\cdot(x,y)}$ gives rational finite Fourier

coefficients s_a and velocity coefficients $i(a_y, -a_x)s_a$. For $k \neq 0$, the complete periodic Poisson pressure has the finite Fourier coefficient

$$\widehat{\pi}_\alpha(k) = \frac{1}{|k|^2} \sum_{a+b=k} (k_x a_y - k_y a_x)(k_x b_y - k_y b_x) s_a s_b, \qquad \widehat{\pi}_\alpha(0) = 0.$$

On the unit circle use $\int_0^{2\pi} e^{iq\cdot e_R} e_R \, d\theta = 2\pi i J_1(|q|)\, q/|q|$. Exact rational convolution and grouping by $n = |q|^2$ produce the eight coefficients displayed. The expansion

$$\frac{J_1(\sqrt{n})}{\sqrt{n}} = \frac{1}{2} \sum_{j=0}^{\infty} \frac{(-n/4)^j}{j!(j+1)!}$$

has total truncation error after $j = 20$ at most

$$E_{20} = \frac{1}{2} \sum_{n \in \mathcal{N}} |c_n| \frac{(n/4)^{21}}{21!\, 22!} < 3 \times 10^{-23};$$

the alternating tails are decreasing there because $n \leq 41$. Using rational bounds $314159/100000 < \pi < 314160/100000$ establishes the strict interval (120) with no floating-point sign assumptions. Finally $\psi_{-\alpha}(x, y) = \psi_\alpha(y, x)$: coordinate exchange reverses the oriented stream velocity while retaining the corresponding Poisson pressure, and consequently reverses $\mathcal{W}$. □

Corollary Y.2a (Realization in short-time unforced dynamics). Each smooth solenoidal datum $u_{\pm 3/10}$ is admissible initial data for classical short-time smooth unforced Navier–Stokes evolution on the periodic three-torus. Its physical initial pressure is uniquely given up to a constant by the same Poisson equation as in Theorem Y.2. Thus there are genuine smooth unforced evolutions, each preserving C_4 symmetry by uniqueness, whose instantaneous pressure work across the fixed unit circular cylinder at the initial time has opposite signs. This is a direct consequence of standard smooth local well-posedness, *not* a construction of the terminal shrinking H8 state or a finite-time singularity.

Proposition Y.3 (Quantitative $m = 4$ dominance gate). For a circular reference interface $R = a(\eta)$ under Appendix X's leading lateral solenoidality, let $q(\theta, \eta) = V_\perp \cdot e_R$ have $q_0(\eta) = 0$. If the *full* leading pressure trace Π is C_4-invariant, then

$$C_\pi = 4\pi \int a(\eta)\, \mathrm{Re}(\Pi_4(\eta)\overline{q_4(\eta)})\, d\eta + \mathcal{E}_{\geq 8}$$

and

$$\boxed{|\mathcal{E}_{\geq 8}| \leq \frac{1}{64} \int a(\eta) \|\partial_\theta^2 \Pi(\cdot, \eta)\|_{L^2_\theta} \|q(\cdot, \eta)\|_{L^2_\theta}\, d\eta.} \tag{121}$$

The bound also applies to carrier self-pressure when only U is C_4-equivariant. It cannot be applied to *full* pressure unless the complete velocity and nonlinear correction retain the symmetry.

Proof. Only angular frequencies $m \in 4\mathbb{Z}$ occur in Π, and $q_0 = 0$. After the $m = \pm 4$ modes, $|m| \geq 8$, so $|m|^{-2} \leq 1/64$; Parseval and Cauchy–Schwarz give (121). □

Conditional sign test and remaining frontier. Let C_4 denote the explicit $m = \pm 4$ term and B_8 the bound on the right of (121). If a single authenticated H8 solution supplies $C_{\mathrm{other}} \le C^{+}_{\mathrm{other}}$ and

$$C_4 + B_8 + C^{+}_{\mathrm{other}} \le -M_0(R_0 - p/2),$$

then the *necessary inward-energy coefficient gate* of Appendix W is certified. Conversely, a lower bound $C_4 - B_8 + C^{-}_{\mathrm{other}} > -M_0(R_0 - p/2)$ excludes that source/solution under the stated hypotheses. Neither sign test proves that a nonlinear correction exists. The original H8 vector potential, cutoff, shapers and overlaps, frame, density, stress centre, complete pressure, exterior correction and terminal retention remain unauthenticated together. The full unforced Navier–Stokes finite-time blow-up theorem remains **OPEN**. The Fourier selection and Bessel techniques are classical, not new general theorems.

Appendix Z. A fourth-multipole pressure certificate for compact fourfold carriers

Scope. These are exact two-dimensional free-space pressure results for a compact transverse carrier. They supply a moment calculation for a potential leading H8 transverse limit; they do not establish that the original three-dimensional periodic pressure has that limit. They do not authenticate a source-matched positive stress or a nonlinear unforced correction.

Theorem Z.1 (Exact exterior fourth pressure mode). Let $U = (U_1, U_2)$ be sufficiently regular, compactly supported in the disk $R \le R_*$, divergence-free on $\mathbb{R}^2$ and covariant under rotation through $\pi/2$. Let Π decay at infinity and solve $-\Delta_\perp \Pi = \partial_i \partial_j (U_i U_j)$. Define $\zeta = x + iy$, $U_+ = U_1 + iU_2$ and $\Pi(R,\theta) = \sum_{m\in\mathbb{Z}} \Pi_m(R) e^{im\theta}$. For $R > R_*$,

$$\boxed{\Pi_4(R) = \frac{3}{4\pi R^4}\overline{\mathcal{Q}_4(U)}, \qquad \mathcal{Q}_4(U) = \int_{\mathbb{R}^2} \zeta^2 U_+^2 \, dx\, dy.} \tag{122}$$

Only angular frequencies divisible by four occur outside the support.

Proof. Write $f = \partial_i\partial_j(U_iU_j)$ and $M_m = \int \zeta^m f$. The fundamental solution $-(2\pi)^{-1}\log|x-y|$ gives the convergent exterior expansion

$$\Pi(R,\theta) = \frac{1}{2\pi}\sum_{m\ge 1}\frac{1}{mR^m}\operatorname{Re}(e^{-im\theta}M_m).$$

The zero total mass and dipole of a double divergence eliminate the logarithmic and first moments. The quarter-turn symmetry eliminates every $m \notin 4\mathbb{Z}$. Two integrations by parts give $M_4 = \int \partial_i\partial_j(\zeta^4)U_iU_j = 12\int \zeta^2(U_1+iU_2)^2 = 12\mathcal{Q}_4(U)$. The coefficient of $e^{i4\theta}$ is $\overline{M_4}/(16\pi R^4)$. □

Corollary Z.2 (All four-patch cross terms matter). For $U = \frac{1}{4}\sum_{j=0}^{3} v_j$,

$$\boxed{\mathcal{Q}_4(U) = \frac{1}{16}\sum_{j,k=0}^{3}\int \zeta^2 (v_{j,1}+iv_{j,2})(v_{k,1}+iv_{k,2})\,dx\,dy.} \tag{123}$$

There are twelve ordered off-diagonal terms as well as four diagonal terms. Their possible cancellations require checking the actual source and cannot be assumed from the fourfold average.

Proposition Z.3 (An exact compact C_4 pressure witness). Let $F(R) = (1-R^2)^6$ for $0 \le R < 1$ and zero beyond, and set $\psi_\varepsilon = F(R)(1+\varepsilon R^4\cos 4\theta)$ and $U_\varepsilon = (\partial_y\psi_\varepsilon, -\partial_x\psi_\varepsilon)$. This is compactly supported, C^4, divergence-free and C_4-equivariant. The moment and exterior pressure mode are

$$\boxed{\mathcal{Q}_4(U_\varepsilon) = \frac{8\pi\varepsilon}{1001}, \qquad (\Pi_{U_\varepsilon})_4(R) = \frac{6\varepsilon}{1001R^4} \quad (R>1).} \tag{124}$$

Compactly supported C^∞ approximations retain the strict nonzero moment by continuity in H^1.

Proof. Put $G = R^4F$, $U_R = R^{-1}\partial_\theta\psi$, $U_\theta = -\partial_R\psi$. Angular orthogonality in the exact moment gives

$$\mathcal{Q}_4 = -2\pi\varepsilon\int_0^1 R^3F'G'\,dR - 8\pi\varepsilon\int_0^1 R^2F'G\,dR = \frac{8\pi\varepsilon}{1001}.$$

The ε^2 terms carry only angular orders zero and eight. Apply Theorem Z.1. □

Proposition Z.4 (Full planar-pressure sign control with a disjoint exterior correction). Fix $a > 1$. Let $\chi(R)$ be a smooth annular cutoff, equal to one near $R = a$ and supported outside $R = 1$. Set $\varphi = \chi(R)R^4 \sin 4\theta/(4a^3)$ and $V = \nabla^\perp \varphi$. Then V is compact, solenoidal and C_4-equivariant, is disjointly supported from U_ε, and has $V \cdot e_R = \cos 4\theta$ on $R = a$. For $u_\delta = U_\varepsilon + \delta V$ and its *complete* planar free-space Poisson pressure,

$$\boxed{\int_{R=a} \Pi[u_\delta] u_\delta \cdot e_R \, dS = \delta \frac{12\pi\varepsilon}{1001a^3} + \delta^3 \int_{R=a} \Pi[V](V \cdot e_R) \, dS.} \tag{125}$$

Therefore either signed pressure-work direction occurs for sufficiently small δ of either sign, provided $\varepsilon \neq 0$.

Proof. Disjoint supports give $U_\varepsilon \otimes V = V \otimes U_\varepsilon = 0$, so the exact pressure polarization is $\Pi[u_\delta] = \Pi[U_\varepsilon] + \delta^2 \Pi[V]$. On $R = a$, the only velocity normal trace is $\delta \cos 4\theta$ and the carrier's fourth mode is given by (124). Integrating the product yields the stated linear term; the remaining term is cubic. □

Application boundary for the original H8 problem. The moment (123) can be evaluated on a single frozen genuine H8 four-patch velocity *only after* authenticating every common cutoff, shaper, frame and overlap, and proving that the full periodic three-dimensional Poisson pressure admits the transverse free-space limit used in Theorem Z.1 with controlled periodic, axial and exterior errors. The positive seven-ray tensor, exact nonlinear unforced correction and terminal pointwise retention are independent unresolved gates. Formula (125) involves static planar diagnostics; it is not an unforced singular solution. The complete three-dimensional unforced Navier–Stokes theorem remains OPEN. The Green-function multipole method is classical; no general novelty claim is made.

Appendix AA. Anisotropic carrier pressure and the near-field necessity

Scope. The following results use the physical H8 component hierarchy and a compact smooth similarity profile. They prove the carrier-only periodic-to-transverse pressure limit under explicitly stated hypotheses and bound the fourth angular pressure mode from distant kinetic stress. The authentic complete Case 006 profile, positive seven-ray stress, nonlinear unforcing correction and terminal retention remain unconstructed.

Theorem AA.1 (Leading swirl is necessarily axisymmetric). Set $\tau = T - t$, $0 < h < 1/6$, $A = 1/2 + h$, $D = 1/2 - h$, $p = 1/2 - 3h$ and $\epsilon = \tau^h$. In a fixed-centre anisotropic chart $x = x_0 + \mathrm{diag}(\tau^{1/2}, \tau^{1/2}, \tau^D)y$ with $y = (R\cos\theta, R\sin\theta, \eta)$, suppose a smooth divergence-free prepared field has the H8 component hierarchy

$$U_r = \tau^{-1/2} V_r^\epsilon, \quad U_\theta = \tau^{-A} V_\theta^\epsilon, \quad U_z = \tau^{-A} V_z^\epsilon.$$

Then the exact cylindrical divergence equation is

$$\boxed{\frac{1}{R}\partial_\theta V_\theta^\epsilon = -\epsilon\left[\frac{1}{R}\partial_R(RV_r^\epsilon) + \partial_\eta V_z^\epsilon\right].} \tag{126}$$

If the three profiles converge in C^1_{loc}, or strongly in H^s with $s > 5/2$, the leading swirl V_θ^0 is independent of θ. An order-one leading $m = 4$ swirl requires an equally leading compensating derivative and cannot coexist with bounded $V_r^\epsilon, V_z^\epsilon$ under these scales.

Proof. The radial, azimuthal and axial pieces of the physical divergence scale as τ^{-1}, τ^{-1-h} and τ^{-1}, respectively. Multiply their sum by τ^{1+h} and pass to the differentiated limit. □

Theorem AA.2 (Full carrier periodic pressure has a planar limit). Assume, additionally, that $V^\epsilon \to V^0$ strongly in $H^s(\mathbb{R}^3)$ with $s > 5/2$, are uniformly compactly supported in one fixed smooth reference cell, and are regular across the rotation axis. Put

$$v^\epsilon = \epsilon V_r^\epsilon e_R + V_\theta^\epsilon e_\theta + V_z^\epsilon e_z, \quad v^0 = V_\theta^0(R,\eta)e_\theta + V_z^0(y)e_z.$$

Let π_U be the *actual* mean-zero periodic pressure determined by $-\Delta\pi_U = \partial_i\partial_j(U_iU_j)$, not a prescribed local pressure. Let Π^0 be the decaying transverse free-space pressure of $V_\theta^0 e_\theta$:

$$-\Delta_\perp \Pi^0 = \partial_a\partial_b[(V_\theta^0 e_\theta)_a (V_\theta^0 e_\theta)_b] = -R^{-1}\partial_R (V_\theta^0)^2. \tag{127}$$

Then uniformly in the fixed reference cell,

$$\boxed{\tau^{2A}\pi_U(t, x_0 + S_\tau y) \longrightarrow \Pi^0(R,\eta) = -\int_R^\infty \frac{(V_\theta^0(s,\eta))^2}{s}\, ds,} \tag{128}$$

where $S_\tau = \mathrm{diag}(\tau^{1/2}, \tau^{1/2}, \tau^D)$. In particular, if the swirl vanishes on the exterior radial collar of the chosen boundary, then the limiting *carrier-only* pressure is zero on the collar, and all of its leading angular modes, including $m = 4$, vanish there.

Proof. Rescaling the whole-space pressure of U produces $\tau^{-2A}\Pi^\epsilon$, with Fourier multiplier

$$\widehat{\Pi}^\epsilon(k,\xi) = -\frac{k_a k_b \widehat{T}_{ab}^\epsilon + 2\epsilon k_a \xi \widehat{T}_{az}^\epsilon + \epsilon^2 \xi^2 \widehat{T}_{zz}^\epsilon}{|k|^2 + \epsilon^2\xi^2}, \quad T_{ij}^\epsilon = v_i^\epsilon v_j^\epsilon.$$

For each nonzero frequency this multiplier has norm at most one as a contraction on tensor entries. The products converge strongly in H^s, and for $k \neq 0$ its symbol converges to $-k_a k_b/|k|^2$ on horizontal entries and to zero on other entries. The set $k = 0$ has measure zero in $\mathbb{R}^3$, so dominated convergence yields $\Pi^\epsilon \to \Pi^0$ strongly in $H^s(\mathbb{R}^3)$, hence uniformly. This is a strong limit, not a claimed uniform multiplier-norm bound at low horizontal frequency.

The periodic and whole-space pressure kernels differ by a smooth uniformly bounded kernel when both observation and source are in the shrinking torus chart. Therefore their pressures differ there by at most $C\|U \otimes U\|_1 = O(\tau^p)$. After multiplying by τ^{2A} the error is $O(\tau^{p+2A}) = O(\tau^{3/2-h})$. Theorem AA.1 gives axisymmetry of V_θ^0. The unique regular decaying transverse solution of (127) has radial derivative $(V_\theta^0)^2/R$, yielding (128). □

Corollary AA.3 (Carrier self-pressure is subcritical on a zero-carrier collar). Assume the interface and exterior normal-velocity trace bounds in Appendix W, including $w_\perp = O(\tau^{-1/2})$ and $w_z = O(\tau^{-1/2})$, and the collar hypothesis of Theorem AA.2. Then

$$\boxed{\int_{\partial\Omega(t)} \pi_U (u \cdot n)\, dS = o(\tau^{p-1}).} \tag{129}$$

Consequently this route requires leading contributions from the *mixed* pressure $2\mathcal{Q}(U, w)$, the *correction* pressure $\pi[w]$, an alternative velocity scaling, or nonpressure boundary flux. It does not infer that the complete pressure is subcritical.

Proposition AA.4 (Quantitative remoteness bound for fourfold pressure). Suppose a C_4-covariant periodic quadratic stress T^{far} satisfies $\sum_{ij} \|T_{ij}^{\mathrm{far}}\|_{L^1} \leq E_*$ and is supported at physical distance at least $d(\tau)$ from all relevant circles on the shrinking interface. Assume $\tau^{1/2} \ll d(\tau) < d_0$. Let $\pi_{\mathrm{far}} = (-\Delta)_0^{-1} \partial_i \partial_j T_{ij}^{\mathrm{far}}$. For a circle of radius $a(\eta)\tau^{1/2}$,

$$\boxed{\sup_{m \in 4\mathbb{Z}\setminus\{0\}} |(\pi_{\mathrm{far}})_m| \leq C E_* \tau^2 d(\tau)^{-7}, \quad \tau^{2A} \sup_{m \neq 0,\, m \in 4\mathbb{Z}} |(\pi_{\mathrm{far}})_m| \leq C E_* \tau^{3+2h} d(\tau)^{-7}.} \tag{130}$$

If $d(\tau)/\tau^{(3+2h)/7} \to \infty$, this remote stress cannot generate a critical fourfold pressure coefficient. At $h = 1/10$ the threshold exponent is $16/35$. The statement concerns pressure-generating *stress*, not the entire correction's spatial support.

Proof. The fourth spatial derivatives of the periodic pressure kernel away from its pole satisfy $|\nabla^4 K_{ij}(x)| \leq C|x|^{-7}$. Thus $\|\nabla_\perp^4 \pi_{\mathrm{far}}\|_\infty \leq C E_* d^{-7}$ near the interface. On a circle of radius r, subtract the transverse Taylor polynomial through total degree three; the remainder is $O(E_* r^4 d^{-7})$. Fourfold covariance forces that Taylor polynomial to have only angular frequency zero, so every nonzero allowed harmonic is bounded by the remainder. Insert $r = O(\tau^{1/2})$, then multiply by τ^{2A}. □

Source-native seed and the remaining nonlinear gate. For the available local vector-potential seed $A = (-yz, xz, 0) = zre_\theta$ and axisymmetric $\chi(r, z)$,

$$U_r = -r(\chi + z\partial_z \chi), \quad U_\theta = 0, \quad U_z = z(2\chi + r\partial_r \chi).$$

Its radial and axial divergences cancel exactly in three dimensions, but its transverse divergence need not vanish; therefore it must *not* be treated as a planar solenoidal carrier when invoking Appendix Z. Its axisymmetric fourth angular moment vanishes, but it is not the fully source-matched H8 field. A complete proof still requires an authenticated original Case 006 source, correct off-axis physical chain rule, all four-patch and shaper products, periodic *mixed and*

exterior Poisson pressure, the signed remaining flux, compatible one-sided seven-ray positive stress, a true nonlinear zero-residual correction and pointwise terminal retention. The three-dimensional unforced singularity theorem is OPEN.

Appendix AB. The near-field requirement for the nonlinear exterior pressure

Scope. A regular correction at the shrinking H8 scale, together with a finite-energy exterior velocity whose quadratic stress remains sufficiently distant, cannot retain the rapidly dissipating prepared core in an exact unforced solution. This is a conditional exclusion, not an authentication of the original Case 006 source or a solution of the global regularity problem.

Setup. Let $\tau = T - t$, $0 < h < 1/6$, $A = 1/2 + h$, $D = 1/2 - h$, $p = 1/2 - 3h$, and $\Omega(t) = x_0 + S_\tau \Omega_0$, where $S_\tau = \mathrm{diag}(\tau^{1/2}, \tau^{1/2}, \tau^D)$ and Ω_0 is fixed and smooth. Assume a compact solenoidal U, vanishing in a collar of the interface, obeys

$$\|U\|_2^2 = M_0 \tau^p (1 + o(1)), \qquad \|\nabla U\|_2^2 = R_0 M_0 \tau^{p-1}(1 + o(1)), \quad M_0 > 0,\ R_0 > p/2, \tag{131}$$

with Appendix T's bounded dimensionless Laplacian ratio. Assume also the strong compact reference-profile hypotheses and zero-swirl collar of Appendix AA. Its full *periodic* self-pressure then satisfies $\sup_{\partial\Omega(t)} |\pi[U]| = o(\tau^{-2A})$.

Theorem AB.1 (Necessary correction-pressure gradient). Suppose a smooth unforced periodic solution $u = U + w$ retains the carrier locally, with $\|w\|_{L^\infty(\Omega(t))} \le C\tau^{-1/2}$ and $\|\nabla w\|_{L^\infty(\partial\Omega(t))} \le C\tau^{-1}$. Set $\pi_c = \pi[u] - \pi[U] = 2\mathcal{Q}(U, w) + \pi[w]$, with the full mean-zero periodic Poisson bilinear form. Then some $c > 0$ satisfies, for every sufficiently late terminal dyad $I_\tau = [T - \tau, T - \tau/2]$,

$$\boxed{\sup_{s \in I_\tau} \|\nabla \pi_c(s)\|_{L^\infty(\Omega(s))} \ge c\tau^{-(3/2+2h)}.} \tag{132}$$

Proof. Appendix T forces $-\int_{I_\tau} \Phi \ge c_0 \tau^p$. The convection, viscosity and moving-boundary fluxes, bounded by the interface traces and Appendix U's cofactor formula, have dyadic integral $O(\tau^{p+2h}) = o(\tau^p)$. The carrier self-pressure work is also $o(\tau^p)$ from Appendix AA. Therefore the correction-pressure work has absolute dyadic integral at least $c_1 \tau^p$.

Since $\operatorname{div} w = 0$,

$$\int_{\partial\Omega(t)} \pi_c w \cdot n \, dS = \int_{\Omega(t)} w \cdot \nabla \pi_c \, dx. \tag{133}$$

The moving volume is $|\Omega(t)| = \tau^{3/2-h}|\Omega_0|$ and hence $\|w\|_{L^1(\Omega(t))} \le C\tau^{1-h}$. Integrating (133) over a dyad of length $O(\tau)$ gives $c_1 \tau^p \le C\tau^{2-h} \sup_{I_\tau} \|\nabla \pi_c\|_\infty$, equivalent to (132). □

Theorem AB.2 (Exclusion of a regular near field and distant finite-energy halo). Under the preceding carrier hypotheses, suppose a hypothetical smooth unforced solution admits $u = U + w_N + w_F$, with w_N, w_F individually solenoidal, and the following additional structure:

1. $w_N(t, x_0 + S_\tau y) = \tau^{-1/2} W^\epsilon(y)$, $\epsilon = \tau^h$, where $\sup_\epsilon \|W^\epsilon\|_{H^s(\mathbb{R}^3)} < \infty$, $s > 5/2$, and the reference supports lie in one fixed compact set;
2. $U + w_N$ and w_F have disjoint supports, $w_F = 0$ in a neighbourhood of $\overline{\Omega(t)}$, and $\mathrm{dist}(\mathrm{supp}(w_F \otimes w_F), \overline{\Omega(t)}) \ge d(\tau)$, with $0 < d(\tau) < d_0$ and $\|w_F\|_2^2 \le E_*$;
3. $d(\tau)/\tau^{3/8+h/2} \to \infty$.

Then there is *no* such exact unforced solution retaining the core with $R_0 > p/2$.

Proof. The local correction energy is $\|w_N\|^2_{L^2(\Omega)} = O(\tau^{1/2-h}) = O(\tau^{p+2h})$, so its relative L^2 norm is $O(\tau^h) \to 0$. The disjoint supports make the *full* quadratic pressure split exactly:

$$\pi[u] = \pi[U + w_N] + \pi[w_F].$$

Under the anisotropic reference rescaling, $U + w_N = \tau^{-A}(v^\epsilon + \epsilon W^\epsilon)$. The uniform H^s bounds and the norm-one Fourier pressure multiplier used in Appendix AA show that its rescaled periodic pressure has the same limit as the carrier, namely zero on the selected boundary collar. Therefore $\sup_{\partial\Omega} |\pi[U + w_N]| = o(\tau^{-2A})$; its pressure work is $o(\tau^{p-1})$, since $u = w_N = O(\tau^{-1/2})$ on the boundary and $|\partial\Omega| = O(\tau^{1-h})$.

The derivative of the three-dimensional periodic double-Riesz pressure kernel away from its pole is bounded by Cd^{-4}. Because $\|w_F \otimes w_F\|_1 \le E_*$ and the stress is separated,

$$\|\nabla\pi[w_F]\|_{L^\infty(\Omega(t))} \le CE_* d(\tau)^{-4}. \tag{134}$$

The exact divergence theorem, followed by compact divergence-free support of U, gives

$$\int_{\partial\Omega(t)} \pi[w_F] u \cdot n \, dS = \int_{\Omega(t)} (U + w_N) \cdot \nabla\pi[w_F] \, dx = \int_{\Omega(t)} w_N \cdot \nabla\pi[w_F] \, dx.$$

Its modulus is at most $CE_*\tau^{1-h} d(\tau)^{-4}$, which is $o(\tau^{p-1})$, because

$$\frac{\tau^{1-h} d(\tau)^{-4}}{\tau^{p-1}} = \tau^{3/2+2h} d(\tau)^{-4} \longrightarrow 0.$$

The remaining convection, viscosity and moving-boundary fluxes have dyadic integral $O(\tau^{p+2h})$. The total outward dyadic flux is consequently $o(\tau^p)$, contrary to Appendix T's positive inward-flux requirement. □

Corollary AB.3 (Exact proximity threshold). At $h = 1/10$, the separation exponent is

$$\boxed{\frac{3}{8} + \frac{h}{2} = \frac{17}{40} = 0.425.} \tag{135}$$

Within Theorem AB.2's two-component class, any surviving correction must fail at least one of its regularity, separation or size assumptions. For an energy-bounded distant halo, approaching the critical distance is necessary but not sufficient. Unlike the fourfold pressure-tail bound of Appendix AA, this theorem does not require symmetry of the exterior correction.

Boundary of the result. The no-go theorem excludes a regular self-similar near correction combined with a separated finite-energy exterior halo, but does not establish existence of a singular near-field correction or of an admissible positive seven-ray stress. The genuine Case 006 H8 carrier, its complete mixed and exterior pressure, the signed remaining flux, exact unforced nonlinear correction and terminal retention remain unauthenticated. No unforced three-dimensional Navier–Stokes singularity or regularity theorem is proved.

Appendix AC. Local retention forces quadratic-stress concentration

Scope. This appendix derives a source-independent necessary quadratic-stress gradient from pressure-driven inward energy transport and local retention. No separation or disjoint-support assumption is made on the nonlinear exterior correction. The statement is conditional on the nonpressure and carrier-only pressure fluxes being subcritical and does not construct an unforced singularity.

Theorem AC.1 (Pressure-to-stress concentration). Let $0 < h < 1/6$, $\tau = T - t$, $p = \frac{1}{2} - 3h > 0$ and $I_\tau = [T-\tau, T-\tau/2]$. Assume the smooth periodic divergence-free unforced velocity $u = U + w$ and moving regions $\Omega(t)$ satisfy all core and local-retention hypotheses of Appendix T, with

$$\|U(t)\|_2^2 = M_0\tau^p(1+o(1)), \qquad \|\nabla U(t)\|_2^2 = R_0 M_0 \tau^{p-1}(1+o(1)), \quad M_0 > 0, \quad R_0 > p/2.$$

Assume, in addition, the integrated sum of nonpressure boundary flux and *carrier-only* pressure flux is $o(\tau^p)$:

$$\int_{I_\tau} \big[\Phi_{\text{other}}(t) + \int_{\partial\Omega(t)} \pi[U](u\cdot n)\, dS\big] dt = o(\tau^p). \tag{136}$$

The premise can follow from the additional interface bounds of Appendix U and the periodic carrier-pressure collar limit of Appendix AA, but it is not implied by local L^2 retention alone.

Define $\delta_\tau = \tau^{-p/2} \sup_{t\in I_\tau} \|w(t)\|_{L^2(\Omega(t))}$, so $\delta_\tau \to 0$, and let

$$\pi_c = \pi[U+w] - \pi[U] = (-\Delta)_0^{-1}\partial_i\partial_j S_{ij}, \qquad S = U\otimes w + w\otimes U + w\otimes w.$$

Here S is the *velocity quadratic stress*, not the positive seven-ray H8 engineering stress. For some $c_0 > 0$ and all sufficiently small τ,

$$\boxed{\sup_{t\in I_\tau} \|\nabla\pi_c(t)\|_{L^2(\Omega(t))} \;\geq\; c_0\, \delta_\tau^{-1}\tau^{p/2-1}.} \tag{137}$$

Moreover, with the globally Fourier-defined L^2 tensor norm,

$$\boxed{\sup_{t\in I_\tau} \|\nabla S(t)\|_{L^2(\mathbb{T}^3)} \;\geq\; c_0\, \delta_\tau^{-1}\tau^{p/2-1}, \qquad \tau^{1-p/2}\sup_{t\in I_\tau}\|\nabla S(t)\|_2 \longrightarrow \infty.} \tag{138}$$

If $\delta_\tau = 0$ on any sufficiently late dyad, the assumed inward pressure transfer is impossible.

Proof. Appendix T yields the dyadic inward energy estimate

$$-\int_{I_\tau} \Phi(t)\, dt \geq (K + o(1))\tau^p, \qquad K = \frac{1-2^{-p}}{p}(R_0 - p/2)M_0 > 0.$$

Subtracting (136) gives $-\int_{I_\tau}\int_{\partial\Omega} \pi_c u\cdot n\, dS dt \geq (K+o(1))\tau^p$. Since U is solenoidal and vanishes on a collar of the boundary, $\int_{\Omega(t)} U\cdot\nabla\pi_c = 0$. The exact divergence theorem therefore gives

$$\int_{\partial\Omega(t)} \pi_c u\cdot n\, dS = \int_{\Omega(t)} w\cdot\nabla\pi_c\, dx.$$

By Cauchy–Schwarz, the time interval length $|I_\tau| = \tau/2$, and the definition of δ_τ,

$$(K+o(1))\tau^p \leq \frac{\tau}{2}\,\delta_\tau\tau^{p/2}\sup_{t\in I_\tau}\|\nabla\pi_c(t)\|_{L^2(\Omega(t))}.$$

For small enough τ, take $c_0 = K$, yielding (137).

For $k \neq 0$, the normalized periodic Poisson multiplier obeys

$$\widehat{\partial_\ell \pi_c}(k) = -ik_\ell \frac{k_i k_j}{|k|^2} \widehat{S}_{ij}(k).$$

The matrix $(k_i k_j/|k|^2)_{ij}$ has Frobenius norm one. Parseval and Cauchy–Schwarz give the exact universal inequality

$$\|\nabla \pi_c\|_{L^2(\mathbb{T}^3)} \leq \|\nabla S\|_{L^2(\mathbb{T}^3)}.$$

Its left side dominates the local $L^2(\Omega(t))$ norm. Combining with (137) proves (138). □

Corollary AC.2 (No-go for natural-scale quadratic stress). Under the hypotheses of Theorem AC.1, if for all late dyads

$$\sup_{t \in I_\tau} \|\nabla(U \otimes w + w \otimes U + w \otimes w)(t)\|_{L^2(\mathbb{T}^3)} \leq C\tau^{p/2-1},$$

then the locally retained unforced solution cannot exist. No spatial separation, fourfold symmetry, stress positivity, or restriction on globally non-negligible exterior energy is assumed. The critical growth exponent is $p/2 - 1 = -3/4 - 3h/2$, equal to $-9/10$ when $h = 1/10$.

Corollary AC.3 (Pointwise pressure concentration). For $\Omega(t) = x_0 + \mathrm{diag}(\tau^{1/2}, \tau^{1/2}, \tau^{1/2-h})\Omega_0$, the moving volume equals $|\Omega_0|\tau^{3/2-h}$. Therefore

$$\boxed{\sup_{t \in I_\tau} \|\nabla \pi_c(t)\|_{L^\infty(\Omega(t))} \geq \frac{c_0}{|\Omega_0|^{1/2}} \, \delta_\tau^{-1} \tau^{-3/2-h}.} \tag{139}$$

For the regular near-correction scaling $\delta_\tau = O(\tau^h)$, this recovers or strengthens the $\tau^{-3/2-2h}$ gradient threshold of Appendix AB. The new inequality needs no a priori L^∞ bound on w; instead it explicitly retains the local L^2 deficit.

Final claim boundary. Theorem AC.1 places a quantitative necessary condition on a *hypothetical* pressure-driven exact solution; its hypotheses do not follow from the unforced equation without the remaining source-specific checks. A correction with sufficiently concentrated quadratic stress is not excluded by this result and may still fail the positivity, source-matching, nonlinear zero-residual, and terminal-retention gates. The original Case 006 H8 source is not authenticated as a complete common tuple. No finite-time unforced Navier–Stokes singularity is constructed or proved.

Appendix AD. An energy-dissipation barrier to strong local H8 retention

Scope. The result strengthens the pressure concentration constraint of Appendix AC using the finite total dissipation of genuine smooth unforced flows. It excludes one class of correction/retention rates under explicit global pointwise bounds, but does not construct a singularity.

Theorem AD.1 (Integrated pressure-driven dissipation lower bound). Put $\tau = T-t$, $0 < h < 1/6$, $A = \frac{1}{2} + h$, $p = \frac{1}{2} - 3h > 0$ and $I_\tau = [T-\tau, T-\tau/2]$. Assume a smooth finite-energy unforced periodic Navier–Stokes solution $u = U + w$ satisfies the retained-core hypotheses of Appendix AC, including the dyad-integrated requirement that the carrier-only pressure and all nonpressure fluxes sum to $o(\tau^p)$. Write

$$\delta_\tau = \tau^{-p/2} \sup_{I_\tau} \|w(t)\|_{L^2(\Omega(t))} \to 0, \qquad S = U \otimes w + w \otimes U + w \otimes w.$$

Assume uniform global bounds on every sufficiently late dyad

$$\|w\|_{L^\infty(\mathbb{T}^3)} \le C\tau^{-1/2}, \quad \|U\|_{L^\infty(\mathbb{T}^3)} \le C\tau^{-A}, \quad \|\nabla U\|_{L^2(\mathbb{T}^3)} \le C\tau^{(p-1)/2},$$

understood with comparable instantaneous terminal time on the dyad. Then for some $c > 0$,

$$\boxed{\int_{I_\tau} \|\nabla w(t)\|_2^2 \, dt \ge c\, \delta_\tau^{-2} \tau^{1/2-h}.} \tag{140}$$

In particular no such finite-energy smooth unforced solution exists if

$$\boxed{\delta_\tau = O(\tau^{1/4-h/2}).} \tag{141}$$

At $h = 1/10$ this excludes the rate $\delta_\tau = O(\tau^{1/5})$. No spatial halo separation or fourfold symmetry is needed.

Proof. The signed correction-pressure work requirement from Appendices T and AC, the exact boundary-to-volume identity, the local retention ratio and the periodic Fourier Poisson multiplier give, for some $K > 0$,

$$K\tau^p \le \left| \int_{I_\tau} \int_{\Omega(t)} w \cdot \nabla \pi_c \, dx dt \right| \le \delta_\tau \tau^{p/2} \int_{I_\tau} \|\nabla \pi_c\|_2 dt \le \delta_\tau \tau^{p/2} \int_{I_\tau} \|\nabla S\|_2 dt.$$

The product rule and global velocity bounds yield

$$\|\nabla S\|_2 \le 2\|w\|_\infty \|\nabla U\|_2 + (2\|U\|_\infty + 2\|w\|_\infty)\|\nabla w\|_2 \le C_1 \tau^{p/2-1} + C_2 \tau^{-A} \|\nabla w\|_2,$$

where $(p-1)/2 - 1/2 = p/2 - 1$. Integrating over I_τ and using $\delta_\tau \to 0$ to absorb the first term gives

$$\int_{I_\tau} \|\nabla w\|_2 dt \ge c_1 \delta_\tau^{-1} \tau^{p/2+A} = c_1 \delta_\tau^{-1} \tau^{3/4-h/2}.$$

Cauchy–Schwarz in time, with $|I_\tau| = \tau/2$, proves (140) since $p + 2A - 1 = 1/2 - h$. Under (141), each late dyad has $\int_{I_\tau} \|\nabla w\|_2^2 dt \ge c_2 > 0$, whereas $\int_{I_\tau} \|\nabla U\|_2^2 dt = O(\tau^p) \to 0$. The reverse triangle inequality in $L^2(I_\tau; L^2_x)$ gives $\int_{I_\tau} \|\nabla u\|_2^2 dt \ge c_3 > 0$ on every sufficiently late dyad. The disjoint sequence $\tau_n = 2^{-n}\tau_0$ would force infinite total dissipation, contrary to the smooth unforced global energy equality for finite initial kinetic energy. □

Claim boundary. This does not rule out weaker local retention, a correction without the global L^∞ bound, substantial nonpressure inward flux, or a different physical scaling. For $0 < h < 1/6$ the natural regular near-correction rate $O(\tau^h)$ is weaker than the excluded $O(\tau^{1/4-h/2})$ rate. No complete original Case 006 source, positive seven-ray stress, zero-residual nonlinear correction or terminal blow-up theorem is established.

Appendix AE. A logarithmic dyadic retention barrier

Scope. We strengthen Appendix AD's conditional energy-dissipation theorem by summing over disjoint terminal intervals. This argument does not authenticate the original H8 source or construct a singular solution.

Theorem AE.1 (Logarithmic strengthening). Under every hypothesis of Appendix AD, including the true unforced equation, local retention of a carrier with gradient energy $O(\tau^{p-1})$, pressure-dominated inward flux, bounded shape ratio, and global bounds $\|w\|_\infty \le C\tau^{-1/2}$, $\|U\|_\infty \le C\tau^{-1/2-h}$, set $\tau_n = 2^{-n}\tau_0$ and $\delta_{\tau_n} = \tau_n^{-p/2}\sup_{I_{\tau_n}} \|w\|_{L^2(\Omega(t))}$. Then

$$\boxed{\sum_{n\ge 0} \frac{\tau_n^{1/2-h}}{\delta_{\tau_n}^2} < \infty.}$$

Consequently $\delta_\tau = O(\tau^{1/4-h/2}\sqrt{\log(1/\tau)})$ on all late dyads is impossible. At $h = 1/10$, the excluded rate is $O(\tau^{1/5}\sqrt{\log(1/\tau)})$.

Proof. Appendix AD gives $\int_{I_{\tau_n}} \|\nabla w\|_2^2 dt \ge c\tau_n^{1/2-h}/\delta_{\tau_n}^2$ with a uniform constant. The dyads are pairwise disjoint up to endpoints. Since $w = u - U$, the integral of $|\nabla w|^2$ up to T is finite: it is bounded by twice the global unforced dissipation of u plus twice the carrier dissipation, which is integrable because $\|\nabla U\|_2^2 = O(\tau^{p-1})$ and $p > 0$. Summing proves series convergence. For the stated logarithmic upper bound on δ_{τ_n}, the n-th series term is at least $c/\log(1/\tau_n) \asymp c/n$, contradicting divergence of the harmonic series. □

Claim boundary. This excludes a specific retention rate under additional global velocity and pressure-flux assumptions. It does not exclude all near-field corrections, solve the spatial H8 positive stress inverse, or establish a finite-time singularity for unforced Navier–Stokes.

Appendix AF. Source-native affine localization and periodic pressure

Scope. This exact calculation uses the available Case 006 local vector-potential seed. The affine full-space pressure is an established elementary control. The theorem detects a pressure mismatch when the seed is cut off anisotropically and periodized. It neither identifies the final Case 006 nonaxisymmetric four-patch source nor proves unforced singularity.

Theorem AF.1 (An explicit localized-source residual). Let $0 < h < 1/6$, $\tau = T - t$, $A = 1/2 + h$, $D = 1/2 - h$, $p = 1/2 - 3h$ and $\epsilon = \tau^h$. The seed $\mathcal{A}_0 = (-yz, xz, 0)$ satisfies

$$\nabla \times \mathcal{A}_0 = (-x, -y, 2z).$$

The affine field $u_{\mathrm{aff}} = \tau^{-1}(-x, -y, 2z)$ solves the unforced three-dimensional Navier–Stokes equation locally with $p_{\mathrm{aff}} = -3z^2/\tau^2$ but has infinite whole-space kinetic energy.

Choose any smooth axisymmetric $\chi(R, \eta)$ compactly supported in a fixed reference cell, smooth at $R = 0$, and equal to one on a nonempty plateau $R < r_0$, $|\eta| < \eta_0$. For sufficiently small τ define one periodic compact velocity

$$U_\tau = \nabla \times \{\tau^{-1}\chi(r/\sqrt{\tau}, z/\tau^D)(-yz, xz, 0)\}.$$

Its components are exactly

$$U_r = \tau^{-1/2}F_r, \quad U_\theta = 0, \quad U_z = \tau^{-A}F_z, \quad F_r = -R(\chi + \eta\chi_\eta), \quad F_z = \eta(2\chi + R\chi_R).$$

Let π_τ be the actual mean-zero *periodic* quadratic pressure, $-\Delta\pi_\tau = \partial_i\partial_j(U_{\tau,i}U_{\tau,j})$. With $\Pi_\tau(R, \eta) = \tau^{2A}\pi_\tau(\sqrt{\tau}R, \tau^D\eta)$, uniformly on fixed compact subsets of the reference cell,

$$\boxed{\|\partial_\eta\Pi_\tau\|_\infty \le C\tau^{2h}(1 + |\log\tau|).} \tag{142}$$

On its plateau the actual physical unforced residual $\mathcal{R}_\tau = \partial_t U_\tau - \Delta U_\tau + (U_\tau \cdot \nabla)U_\tau + \nabla\pi_\tau$ obeys

$$\boxed{\tau^{3/2+h}(\mathcal{R}_\tau)_z(\sqrt{\tau}R, \tau^D\eta) = 6\eta + O(\tau^{2h}(1 + |\log\tau|)).} \tag{143}$$

Its normalized periodic pressure satisfies the *exact* plateau Poisson equation

$$-(\Delta_\perp + \epsilon^2\partial_\eta^2)\Pi_\tau = 6\epsilon^2, \quad \epsilon^{-2}\Delta_\perp\Pi_\tau = -6 + O(\epsilon^2(1 + |\log\epsilon|)).$$

Thus the local affine Poisson source is balanced largely by *transverse* pressure curvature; its axial momentum cancellation is not preserved by compact periodization.

Proof. The affine assertion is immediate from $M = \mathrm{diag}(-1, -1, 2)$, $\mathrm{tr}\, M = 0$ and $M + M^2 = \mathrm{diag}(0, 0, 6)$. The cutoff curl is solenoidal, and on the plateau is exactly Mx/τ.

In the anisotropic reference Fourier variables $(k, \xi) \in \mathbb{R}^2 \times \mathbb{R}$, write the compact velocity as $\tau^{-A}(\epsilon F_\perp, F_z)$. Its rescaled whole-space Poisson pressure has the exact multiplier

$$\widehat{\Pi}^\epsilon(k, \xi) = -\epsilon^2\frac{k_a k_b\widehat{F_aF_b} + 2k_a\xi\,\widehat{F_aF_z} + \xi^2\widehat{F_z^2}}{|k|^2 + \epsilon^2\xi^2}.$$

All numerator factors are Schwartz. Split the two-dimensional transverse Fourier integral at $|k| = \epsilon|\xi|$. Inside the disk the integrable multiplier contributes $O(\epsilon^2)$, after a Schwartz-weighted integration in ξ. Outside, the only logarithmic integral is $\epsilon^2\xi^2\int_{\epsilon|\xi|}^1 dk/k$. Therefore every fixed scaled derivative, including ∂_η and ∂_η^2, is $O(\epsilon^2(1 + |\log\epsilon|))$.

The periodic and whole-space pressure kernels differ smoothly within one shrinking chart; their first physical derivatives differ by $O(\|U_\tau\|_2^2) = O(\tau^p)$. Hence the rescaled η derivative of this difference is $O(\tau^{p+2A+D}) = O(\tau^{2-2h})$. On the plateau $\partial_t U_\tau + (U_\tau \cdot \nabla)U_\tau - \Delta U_\tau = (0, 0, 6z/\tau^2)$. Finally $2A + D = 3/2 + h$ gives (143). Rescaling the physical pressure Poisson equation with its exact plateau source $6/\tau^2$ gives the transverse-curvature identity. □

Corollary AF.2 (Signed correction-pressure reconstruction gate). If an actual unforced periodic solution has $u = U_\tau + w$ and the complete axial nonpressure correction term

$$Q_z = \big[w_t - \Delta w + (U_\tau \cdot \nabla)w + (w \cdot \nabla)U_\tau + (w \cdot \nabla)w\big]_z$$

satisfies $\tau^{3/2+h} Q_z = o(1)$ uniformly on a compact plateau, then for $\pi_c = \pi[U_\tau + w] - \pi_\tau$,

$$\partial_\eta(\tau^{2A}\pi_c) = -6\eta + o(1).$$

In particular, for $0 \le a < b < \eta_0$,

$$\boxed{\tau^{2A}\big[\pi_c(\sqrt{\tau}R, \tau^D b) - \pi_c(\sqrt{\tau}R, \tau^D a)\big] = -3(b^2 - a^2) + o(1).} \tag{144}$$

This signed *axial* pressure jump is not the signed *radial interface work* required in Appendix W. The correction may also satisfy the momentum equation by a leading nonpressure term, in which case the premise here does not apply.

Exact energy scale and limitations. The compact carrier has $\|U_\tau\|_2^2 = M_0\tau^p(1 + O(\tau^{2h}))$, $M_0 = \int F_z^2 > 0$, and $\|\nabla U_\tau\|_2^2 = M_0 R_0 \tau^{p-1}(1 + O(\tau^{2h}))$, where $R_0 = \int |\nabla_\perp F_z|^2 / \int F_z^2 \ge j_{0,1}^2/R_1^2$ for horizontal support $R < R_1$. This is an exact singular *prepared field*, not a solution of the compact unforced problem.

The elementary affine field and pressure nonlocality are prior mathematical knowledge. This source-seed result has 14 independent local symbolic, Gaussian-spectral and Fourier tests, but none authenticate the full nonaxisymmetric Case 006 patch tuple, compatible positive seven-ray tensor, exact nonlinear correction or retained terminal blow-up. Full unforced singularity status: OPEN.

Appendix AG. GitHub mathematical verification and independent replay

Purpose. This appendix separates finite algebra verified in the research code, source-available but not yet hosted-certified Lean modules, and unproved physical Navier–Stokes construction gates. The theorem inventory and independent Python checker record exactly which finite arithmetic and Fourier identities are reproducible. The constructive PDE conditions are stated separately.

Exact stress centre and coefficient cones. The pinned seven-ray source at aperture $t = 1/4$ has positive reference coefficients $c_3^{(0)} = 236/1229$, $c_i^{(0)} = 172/1229$ for $i \neq 3$. The exact minimum coefficient margin and the sharp componentwise factor are

$$g = \frac{172}{1229} - \frac{1}{100} = \frac{15971}{122900}, \qquad \|\delta c(A,B,C,D)\|_\infty \leq 14 \max(|A|,|B|,|C|,|D|).$$

The original absolute-perturbation cone and the larger one-sided positivity cone are different mathematical sets. In the latter, with $g_* = 22371/122900$, the sharp source-frame meridional cap is

$$|A| \leq \frac{5g + g_*}{4}\rho = \frac{51113}{245800}\rho.$$

The independent exact rational checker enumerated all eight feasible four-dimensional vertices of the one-sided coefficient polytope. Their (A, C) projection has exactly six sharp convex-hull vertices, reproducing the manuscript's source-frame hexagon and the meridional cap. No physical compact positive stress inverse follows from this finite convex calculation.

Formal theorem groups found in the connected GitHub source. The 18 audited Lean modules are grouped as follows:

- *Centre and cones:* `SevenRayCenter`, `ConeBudget`, `ConeOperator`, `FourFieldPhysical`, `ExactCoefficientCone`, `SharpCoefficientDual`, `PositiveCoefficientCone`, `AxialFluxCone`, `JointFluxCone`.
- *Geometric frame and coordinate calculus:* `RotatingFrameDual`, `FrameConnectionDerivative`, `CartesianAxialBridge`, `NonaxisymmetricAxialBridge`, `PhysicalCircleAngular`, `AngularMean`.
- *Source and pressure compatibility:* `MeridionalInverse`, `PressureCompensatedFlux`, `Case006SourceBand`.

These modules cover exact matrix reconstruction, sharp finite coefficient bounds, rotating-frame Frobenius pullback, the frame-connection commutator, Cartesian and cylindrical derivatives, angular mean identities, a pressure-compensated radial primitive retaining its endcap trace, and a *conditional* aligned-frame Case 006 axial interval. The connected source and the full catalogue are available under `docs/unforced-integration/H8_FULL_VOLUME_VERIFICATION_INVENTORY_20261030.md`. The repository also contains 34 associated physical and finite-arithmetic regression modules catalogued there.

Lean evidence status. The repository README reports a historical successful pinned workflow (run 12, ID 35779131542) for the *abstract* nonlinear-closure file. This does not constitute a kernel proof of a source-matched H8 correction. At the time of the connected source check, hosted workflows for the current H8 pull-request head were queued rather than certified passed. Statements available in Lean source must not be described as proved by the hosted kernel for that commit until its actual build and axioms are checked.

Independent finite-arithmetic and Fourier verification. A separately implemented script `verify_h8_independent.py` returned 17 successful assertions. They cover exact fractional identities, complete vertex enumeration, the two scale exponents, the dyadic inward-flux constant, the logarithmic retention boundary, exact pressure polarization on a 16^3 Fourier grid, and the unit Frobenius norm of $k \otimes k/|k|^2$. An additional 26 available local affine-pressure and research-paper Python tests passed. These are finite and symbolic checks, not a proof that the physical pressure-dominance or correction hypotheses hold.

Reviewer notation clarification. The corrected summability law must use the *inverse square* of the relative correction deficit. For clarity the mathematically equivalent statements are

$$\boxed{\sum_{n\geq 0} \tau_n^{1/2-h} \delta_{\tau_n}^{-2} < \infty \quad \Longleftrightarrow \quad \sum_{n\geq 0} \frac{\tau_n^{1/2-h}}{\delta_{\tau_n}^2} < \infty.}$$

At the excluded rate $\delta_{\tau_n}^2 = \tau_n^{1/2-h} \log(1/\tau_n)$, each correct summand is $1/\log(1/\tau_n) \asymp 1/n$, and so the series diverges. The incorrect *product* with δ^2 would converge. The logarithmic retention rate contains the square root $\sqrt{\log(1/\tau)}$.

Full unforced physical release boundary. Define $\mathcal{B}(f,g) = \mathbb{P}[(f\cdot\nabla)g]$, $G = \partial_t U - \Delta U + \mathcal{B}(U,U)$, and $L_U w = \Delta w - \mathcal{B}(U,w) - \mathcal{B}(w,U)$. A complete source-matched nonlinear correction would have to satisfy

$$\boxed{G + w_t - L_U w + \mathcal{B}(w,w) = 0, \quad \nabla\cdot w = 0}$$

everywhere, with the genuine periodic pressure, positive weighted seven-ray stress, and pointwise terminal retention. No authenticated common tuple (U, w, H, ρ, E) achieves those conditions in the present record. The proof ledger remains G0 OPEN, G1–G3 PAPER_CONDITIONAL, and G4–G8 OPEN. No unforced Navier–Stokes singularity or regularity proof is claimed.

Appendix AJ. Pressure-driven retention theorem and two-dimensional findings

Objective. The exact seven-ray coefficient cone and the pressure-driven retention obstruction are analyzed under explicit scaling and flux hypotheses. The two-dimensional diagrams display the sharp feasible coefficient geometry and the nonlocal interface-pressure mechanism. No actual unforced H8 singularity is constructed.

Source-frame one-sided cone. For $g = 15971/122900$, $H_* = 22371/122900$, the sharp one-sided seven-ray (A, C) projection has the six vertices

$$\left(\pm\frac{g\rho}{2}, -g\rho\right), \quad \left(\pm\frac{(5g+H_*)\rho}{4}, \frac{(H_*-g)\rho}{4}\right), \quad \left(\pm\frac{g\rho}{2}, \frac{(g+H_*)\rho}{2}\right).$$

Thus $|A| \le 51113\rho/245800$ in the *source-ray basis*; under a tilted physical frame E the tensor is EHE^{T} and A need not equal a physical cylindrical rz component.

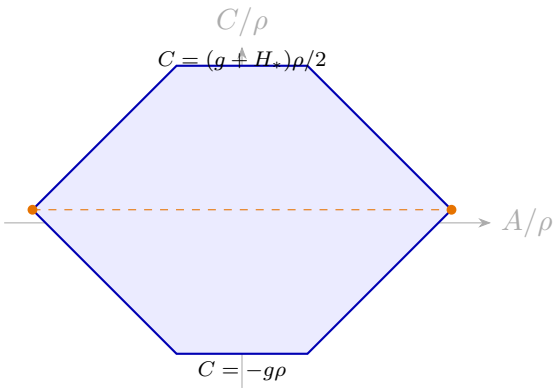


Figure 1: Original two-dimensional sharp six-vertex coefficient slice. Axis geometry is illustrative; the exact vertices are in the preceding display. This is not a source-matched spatial stress.

Conditional pressure-retention obstruction. Let $0 < h < 1/6$, $p = 1/2 - 3h$, $\tau = T - t$ and $I_\tau = [T - \tau, T - \tau/2]$. Assume Appendix AC's nontrivial inward flux, local retention and pressure-dominance hypotheses. Put $\delta_\tau = \tau^{-p/2} \sup_{I_\tau} \|w\|_{L^2(\Omega)} \to 0$ and $\mathsf{S} = U \otimes w + w \otimes U + w \otimes w$. Then

$$\sup_{I_\tau} \|\nabla \mathsf{S}\|_{L^2(\mathbb{T}^3)} \ge c\, \delta_\tau^{-1} \tau^{p/2-1}.$$

Under Appendix AD's additional global velocity bounds, Appendix AE strengthens this to

$$\boxed{\sum_n \frac{\tau_n^{1/2-h}}{\delta_{\tau_n}^2} < \infty, \quad \tau_n = 2^{-n}\tau_0.}$$

Hence $\delta_\tau = O(\tau^{1/4-h/2}\sqrt{\log(1/\tau)})$ is excluded under those additional hypotheses. They have *not* been shown to hold for a complete H8 source and correction.

Boundary of integration. The exact physical equation is $G + w_t - L_U w + \mathcal{B}(w, w) = 0$ with projected $\mathcal{B}$. No verified common tuple (U, w, H, ρ, E) simultaneously solves this equation, the actual mixed/exterior periodic pressure, the positive compact seven-ray stress equation and terminal retention. The full unforced singularity theorem remains **OPEN**.

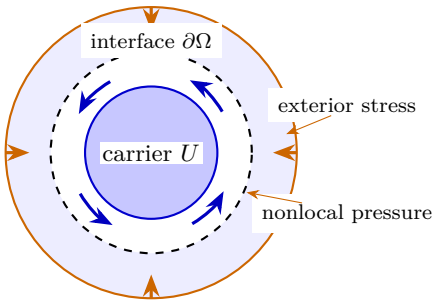


Figure 2: Original transverse slice geometry: the carrier vanishes near the moving interface, while nonlocal pressure can be driven by exterior quadratic velocity stress. These arrows are schematic, not a certified unforced trajectory.

Appendix AK. Exact unforced correction controls and analytic-tail restrictions

Objective. These statements rule out strictly compact total-velocity patching and an exact zero correction on an affine seed plateau. They do not construct the positive H8 stress, source-matched nonlinear correction or unforced blow-up. The analyticity step is established Foias–Temam Gevrey regularity prior art, not a new general Navier–Stokes theorem.

An exact nonzero correction control. On $(\mathbb{R}/2\pi\mathbb{Z})^3$ consider

$$u_{\mathrm{sh}}(t,x,y,z) = (e^{-t}\sin y, 0, 0), \qquad \pi_{\mathrm{sh}} = 0.$$

The field is periodic and solenoidal; $(u_{\mathrm{sh}}\cdot\nabla)u_{\mathrm{sh}} = 0$ and $\partial_t u_{\mathrm{sh}} - \Delta u_{\mathrm{sh}} = 0$. For *any* smooth solenoidal prepared carrier U, set $w = u_{\mathrm{sh}} - U$, $\mathcal{B}(f,g) = \mathbb{P}[(f\cdot\nabla)g]$, $G = U_t - \Delta U + \mathcal{B}(U,U)$, $L_U w = \Delta w - \mathcal{B}(U,w) - \mathcal{B}(w,U)$. The exact identity is

$$\boxed{G + w_t - L_U w + \mathcal{B}(w,w) = 0.}$$

Thus the projected correction equation admits explicit smooth solutions even if the prepared carrier itself does not solve the PDE.

Retention fails for the control. If $\|U(t)\|_2^2 = M_0\tau^p(1+o(1))$ with $M_0 > 0$ and the moving cell has volume $O(\tau^{3/2-h})$, then

$$\|u_{\mathrm{sh}}(t)\|_{L^2(\Omega(t))} \le C\tau^{(3/2-h)/2}, \qquad \frac{\|u_{\mathrm{sh}}\|_{L^2(\Omega)}}{\|U\|_2} = O(\tau^{1/2+h}) \to 0.$$

The two triangle inequalities consequently give $\|w(t)\|_{L^2(\Omega(t))}/\|U(t)\|_2 \to 1$. This explicitly demonstrates that exact projected residual cancellation *alone* gives no local or terminal H8 retention and does not imply a compatible positive seven-ray engineering stress.

Spatial-analyticity obstruction. Classical periodic Navier–Stokes Gevrey smoothing (Foias and Temam, J. Funct. Anal. 87 (1989), 359–369) implies that a smooth unforced solution $u(t_*,\cdot)$ is real analytic at every $t_* > 0$ before its lifespan ends. If it vanishes on an open subset, the identity theorem forces $u(t_*,\cdot) \equiv 0$ everywhere. Thus a nonzero exact solution cannot remain strictly compactly supported in one shrinking torus chart at positive time, even if its auxiliary carrier U is compactly supported.

For the plateau of the localized Case 006 affine seed $U = \tau_*^{-1}(-x,-y,2z)$, an exact correction cannot vanish on an open subregion. Otherwise $\partial_x u_x = -1/\tau_*$ on that subregion, and analyticity would imply the same constant derivative everywhere on the torus. Yet periodicity yields $\int_{\mathbb{T}^3}\partial_x u_x\,dx = 0$, a contradiction.

Constructive compatibility. Analyticity supplies neither a uniform lower bound on the correction's magnitude nor a terminal control of its nonlocal pressure. The source-matched tuple (U, w, H, ρ, E), physical C4 interactions, positivity, all moving-boundary fluxes and pointwise terminal retention are still missing. These elementary tests must not be reported as a new solution of the global unforced regularity question.

Appendix AL. Compact regular-correction obstruction for the affine H8 seed

Objective. An exact prepared affine core is not automatically an exact compact unforced solution. We isolate the failure of corrections with natural amplitude, regular similarity profiles and no nonlocal exterior contribution. This remains a source-seed diagnostic rather than the full four-patch construction.

Theorem AL.1 (Regular compact near-field correction no-go). Let $0 < h < 1/6$, $\tau = T - t$, $A = \frac{1}{2} + h$, $D = \frac{1}{2} - h$, $\varepsilon = \tau^h$ and $S_\tau = \operatorname{diag}(\tau^{1/2}, \tau^{1/2}, \tau^D)$. Take the localized affine source of Appendix AF,

$$U_\tau = \operatorname{curl}\left[\tau^{-1}\chi(r/\sqrt{\tau}, z/\tau^D)(-yz, xz, 0)\right],$$

with fixed smooth compact χ equal to one on a nonempty open reference plateau. Suppose a smooth divergence-free correction is supported in a fixed compact reference set of the same shrinking chart and satisfies

$$w_\tau(x_0 + S_\tau y) = \tau^{-1/2} W^\varepsilon(y), \qquad \sup_{0<\varepsilon<\varepsilon_0} (\|W^\varepsilon\|_{H^s} + \|\varepsilon\partial_\varepsilon W^\varepsilon\|_{H^s}) < \infty, \quad s > 7/2.$$

Then $u = U_\tau + w_\tau$ cannot solve the unforced periodic Navier–Stokes equation at every sufficiently late time.

Proof. On the plateau, $U_\tau = (-x, -y, 2z)/\tau$ and

$$\tau^{3/2+h}[\partial_t U_\tau - \Delta U_\tau + (U_\tau \cdot \nabla)U_\tau + \nabla\pi[U_\tau]]_z = 6\eta + O(\varepsilon^2(1 + |\log\varepsilon|)),$$

by the actual periodic pressure calculation of Appendix AF. With the natural correction scaling, uniform H^s bounds and controlled scaled time derivative, the complete nonpressure axial correction term is $O(\tau^{-3/2})$ on the plateau. In particular the apparently critical $w_\perp \cdot \nabla_\perp U_z$ vanishes there because $\partial_x U_z = \partial_y U_z = 0$. In reference variables the normalized velocity equals $v_U^\varepsilon + \varepsilon W^\varepsilon$ where $v_U^\varepsilon = (\varepsilon F_\perp, F_z)$. Its quadratic stress difference is $O(\varepsilon)$ in H^s. The anisotropic pressure multiplier

$$-\frac{(k, \varepsilon\xi)_i (k, \varepsilon\xi)_j}{|k|^2 + \varepsilon^2\xi^2}$$

has Frobenius norm one at nonzero frequency, so the Sobolev algebra estimate, $H^{s-1} \hookrightarrow L^\infty$, and the smooth periodic-image correction imply

$$\left\|\partial_\eta[\tau^{2A}(\pi[U + w] - \pi[U])]\right\|_\infty = O(\varepsilon).$$

Therefore on each fixed compact reference plateau

$$\tau^{3/2+h}[\partial_t u - \Delta u + (u \cdot \nabla)u + \nabla\pi[u]]_z = 6\eta + O(\varepsilon + \varepsilon^2|\log\varepsilon|).$$

For fixed $\eta > 0$ the right side cannot vanish as $\tau \downarrow 0$. □

Corollary AL.2 (A necessary escape). For this source seed, an exact unforced correction must escape at least one hypothesis: compact confinement to the shrinking chart, uniform rescaled regularity, or natural $\tau^{-1/2}$ amplitude. A global analytic tail or a different H8 carrier is not ruled out.

Constructive compatibility. The full source-matched tuple (U, w, H, ρ, E), nonnegative seven-ray coefficients, all mixed and exterior periodic-pressure contributions, signed inward flux and terminal retention remain unconstructed. The exact unforced singularity theorem remains open.

Appendix AM. Local–far stress and the axial pressure-repair scale

Objective. We apply the standard local–far Poisson pressure decomposition to the affine Case 006 source seed. The splitting acts on the full quadratic velocity correction stress, not on the velocity, so global analytic tails and all mixed products are retained. This is a necessary pressure-localization result, not an H8 singularity construction.

Source and exact pressure decomposition. Let $\tau = T - t$, $0 < h < 1/6$, $A = \frac{1}{2} + h$, $D = \frac{1}{2} - h$, $p = \frac{1}{2} - 3h > 0$, and $\beta = (2A + D)/4 = \frac{3}{8} + \frac{h}{4}$. Let U be Appendix AF's compactified affine seed with a fixed open reference plateau. Fix a compact plateau $P_\tau = x_0 + \operatorname{diag}(\sqrt{\tau}, \sqrt{\tau}, \tau^D)P_0$, with reference axial coordinate $\eta \in [a, b]$, $0 < a < b$. Suppose a hypothetical smooth unforced periodic solution $u = U + w$ has finite global kinetic energy. Define the complete quadratic *velocity* stress

$$\mathsf{S} = u \otimes u - U \otimes U = U \otimes w + w \otimes U + w \otimes w, \qquad \pi_c = \mathcal{Q}(\mathsf{S}),$$

where $\mathcal{Q}(F) = (-\Delta)_0^{-1}\partial_i\partial_j F_{ij}$ is the mean-zero periodic pressure operator. For $0 < d < d_0$ let ζ_d equal one within distance $d/2$ of P_τ and zero beyond distance d. Then

$$\mathsf{S}_{\mathrm{n}} = \zeta_d \mathsf{S}, \quad \mathsf{S}_{\mathrm{f}} = (1 - \zeta_d)\mathsf{S}, \qquad \pi_c = \mathcal{Q}(\mathsf{S}_{\mathrm{n}}) + \mathcal{Q}(\mathsf{S}_{\mathrm{f}}).$$

The cut-off tensor need not be divergence free. This is an exact linear pressure split, including every quadratic mixed stress.

Theorem AM.1 (Critical distance gate). Let

$$Q_z = \big[w_t - \Delta w + (U \cdot \nabla)w + (w \cdot \nabla)U + (w \cdot \nabla)w\big]_z$$

be the complete nonpressure axial correction term. Then on P_0,

$$\boxed{\sup_{P_0}\left|\partial_\eta\left(\tau^{2A}\mathcal{Q}(\mathsf{S}_{\mathrm{n}})\right) + 6\eta\right| \le C\tau^{2h}(1 + |\log\tau|) + \sup_{P_\tau}|\tau^{2A+D}Q_z| + C\tau^{2A+D}d^{-4}.}$$

The constant depends on the fixed source, plateau, period and initial kinetic energy, but not on τ, d.

Proof. The exact global energy equality and $\|U\|_2^2 = O(\tau^p)$ give $\|\mathsf{S}_{\mathrm{f}}\|_1 \le \|u\|_2^2 + \|U\|_2^2 \le C$. The derivative of the periodic double-Riesz pressure kernel is at most $C\,\operatorname{dist}(x, 0)^{-4}$ away from its pole. The far stress is separated from P_τ by $d/2$; hence

$$\|\nabla\mathcal{Q}(\mathsf{S}_{\mathrm{f}})\|_{L^\infty(P_\tau)} \le Cd^{-4}.$$

Its rescaled axial derivative is thus at most $C\tau^{2A+D}d^{-4}$. The genuine periodic pressure calculation of Appendix AF gives the normalized affine-seed axial residual $6\eta + O(\tau^{2h}(1+|\log\tau|))$ on this plateau. The exact axial unforced equation adds $\tau^{2A+D}Q_z$ and the scaled derivative of π_c. Substituting the exact near–far stress split proves the claim. □

Corollary AM.2 (Necessary near stress). If $\sup_{P_\tau}|\tau^{2A+D}Q_z| \to 0$ and $d(\tau)/\tau^\beta \to \infty$, then

$$\partial_\eta(\tau^{2A}\mathcal{Q}(\mathsf{S}_{\mathrm{n}})) = -6\eta + o(1) \quad \text{uniformly on } P_0.$$

If the near pressure term is also subcritical, no such exact unforced solution exists. Thus bounded-energy *far-only* quadratic stress cannot repair the plateau defect at distances much larger than τ^β. At $h = 1/10$, $\boxed{\beta = D = 2/5}$. This matches the axial similarity scale.

Kernel sharpness and claim boundary. For the Euclidean pressure kernel on the axial ray, $K_{zz}(0,0,d) = 1/(2\pi d^3)$ and $\partial_z K_{zz}(0,0,d) = -3/(2\pi d^4)$. Thus d^{-4} cannot be improved for arbitrary L^1 tensor stresses without additional structure. It does not follow that a physical unforced stress attains that rate. The theorem neither constructs a correction w nor authenticates the full four-patch H8 source or positive seven-ray inverse. The common source-matched (U, w, H, ρ, E), signed full energy flux and pointwise terminal blow-up remain open.

Appendix AN. Pressure-Free Carrier Transfer and Critical Correction Amplitude

Objective. Pair the exact unforced equation against a shrinking prepared carrier to obtain a signed nonlinear-injection constraint without pressure-dominance, correction compactness or a global correction L^∞ hypothesis. The result is conditional on the carrier estimates and does not construct an unforced singularity.

Hypotheses. Let $u = U + w$ be a smooth unforced solenoidal solution on $\mathbb{T}^3$, $\tau = T - t$, $0 < h < 1/6$, $p = 1/2 - 3h$, $I_\tau = [T - \tau, T - \tau/2]$, and let U be supported in $\Omega(t)$. Write $M = \|U\|_2^2$, $D = \|\nabla U\|_2^2$ and $\delta_\tau = \tau^{-p/2} \sup_{I_\tau} \|w\|_{L^2(\Omega(t))} \to 0$. Assume uniformly

$$M = M_0 \tau^p (1 + o(1)), \quad D = R_0 M_0 \tau^{p-1} (1 + o(1)), \quad M_0 > 0,\ R_0 > p/2,$$

$$\|U_t + \Delta U\|_2 \le C\tau^{p/2-1}, \quad \|\nabla U\|_\infty \le C\tau^{-1-h}, \quad \|(U \cdot \nabla)U\|_2 \le C\tau^{p/2-1-\kappa}, \quad 0 \le \kappa \le h.$$

The convective bound is on the *unprojected* compactly supported field. Applying the global Leray projector before estimating against local w is not justified without a global L^2 estimate on w.

Lemma AN.1 (Exact pressure-free pairing). For smooth solenoidal $u = U + w$,

$$\boxed{\frac{d}{dt}\langle w, U\rangle = \langle w, U_t + \Delta U + (U \cdot \nabla)U\rangle + \int_{\mathbb{T}^3} w_i w_j \partial_j U_i - \tfrac{1}{2}M' - D.}$$

Proof. Test $u_t - \Delta u + \operatorname{div}(u \otimes u) + \nabla\pi = 0$ against U. Pressure vanishes since $\nabla \cdot U = 0$. The $U \otimes U$ contraction is a divergence; so is $U_i w_j \partial_j U_i = w \cdot \nabla(|U|^2/2)$ because w is solenoidal. The remaining cross term is $w_i U_j \partial_j U_i = \langle w, (U \cdot \nabla)U\rangle$. Now $\langle u, U\rangle = M + \langle w, U\rangle$ and $\langle U, \Delta U + U_t\rangle = -D + M'/2$. Rearrangement proves the identity.

Theorem AN.2 (Signed transfer and correction floor). Set

$$K = \frac{M_0(1 - 2^{-p})}{p}(R_0 - p/2) > 0, \qquad m = \max\{\kappa, h/2\}.$$

Every exact locally retained correction obeys

$$\boxed{\int_{I_\tau} \left[\langle w, U_t + \Delta U + (U \cdot \nabla)U\rangle + \int_{\mathbb{T}^3} w_i w_j \partial_j U_i \, dx\right] dt = K\tau^p + o(\tau^p)}$$

and $\boxed{\delta_\tau \ge c\tau^m}$ for sufficiently small τ. If $|\Omega(t)| \le C\tau^{3/2-h}$, then

$$\boxed{\sup_{I_\tau} \|w\|_{L^\infty(\Omega(t))} \ge c'\tau^{-1/2-h+m}.}$$

Proof. Integrate the exact identity. Local L^2 retention gives $|\langle w, U\rangle| \le C\delta_\tau\tau^p = o(\tau^p)$ at both endpoints. Hence $M'/2 + D$ integrates to $K\tau^p + o(\tau^p)$. The three required interaction integrals are bounded, respectively, by

$$C\delta_\tau\tau^p, \qquad C\delta_\tau\tau^{p-\kappa}, \qquad C\delta_\tau^2\tau^{p-h}.$$

Thus $K/2 \le C(\delta_\tau + \delta_\tau\tau^{-\kappa} + \delta_\tau^2\tau^{-h})$ for small τ, forcing $\delta_\tau \ge c\tau^{\max(\kappa, h/2)}$. The volume bound and Cauchy–Schwarz give the L^∞ floor. No pressure-dominance or global bound on w was used.

Corollary AN.3 (Critical carrier self-advection). If a retained exact correction has the natural local size $\delta_\tau = O(\tau^h)$, its *signed* nonlinear carrier pairing must satisfy

$$\boxed{\int_{I_\tau} \langle w, (U\cdot\nabla)U\rangle\, dt = K\tau^p + o(\tau^p).}$$

Any carrier estimate with $\kappa < h$ contradicts this requirement. The condition involves the unprojected acceleration: a projected norm cannot be substituted unless the nonlocal correction pairing is controlled.

Corollary AN.4 (Compactified no-swirl affine seed). For $U = \operatorname{curl}[\tau^{-1}\chi(r/\sqrt{\tau}, z/\tau^{1/2-h})(-yz, xz, 0)]$ with fixed smooth compact axisymmetric cutoff χ, the explicit $U_r = \tau^{-1/2}F_r$, $U_\theta = 0$ and $U_z = \tau^{-1/2-h}F_z$ show $\|(U\cdot\nabla)U\|_2 \le C\tau^{p/2-1}$, hence $\kappa = 0$. Therefore

$$\boxed{\delta_\tau \ge c\tau^{h/2}, \qquad \sup_{I_\tau} \|w\|_{L^\infty(\Omega)} \ge c'\tau^{-1/2-h/2}.}$$

At $h = 1/10$, the lower L^2 magnitude is $\sup \|w\|_{L^2(\Omega)} \gtrsim \tau^{3/20}$ and the L^∞ amplitude is at least order $\tau^{-11/20}$. This excludes uniformly natural-amplitude local corrections even when w has a global analytic tail. It does not exclude supercritical corrections or the distinct authentic four-patch swirling carrier.

Verification and claim boundary. This is a scaling-specific application of the classical weak–strong carrier-testing/relative-energy method. The companion independent SymPy, unforced-shear and dealiased Fourier suite has 23 local PASS assertions. The proof establishes no authentic U, H, ρ, E source, compact one-sided positive stress inverse, exact nonlinear retained correction, signed periodic-pressure flux or finite-time unforced blowup.

Appendix AO. Axisymmetric Swirl and Signed Carrier Transfer

Objective. Determine whether axisymmetric swirl supplies the signed carrier transfer required by an exact unforced correction. The leading centrifugal force is a radial gradient, which disappears when paired with globally solenoidal w; the remaining axial-gradient coupling is subcritical. No authentic four-patch source or singular unforced solution is asserted.

Assumptions. Let $\tau = T - t$, $0 < h < 1/6$, $p = 1/2 - 3h$, $A = 1/2 + h$, $D = 1/2 - h$, and $I_\tau = [T - \tau, T - \tau/2]$. Let U be a compactly supported smooth axisymmetric solenoidal carrier with uniformly bounded fixed compact similarity profiles

$$U_r = \tau^{-1/2} V_r(R, \eta), \quad U_\theta = \tau^{-A} V_\theta(R, \eta), \quad U_z = \tau^{-A} V_z(R, \eta), \quad R = r/\sqrt{\tau}, \quad \eta = z/\tau^D.$$

The smooth swirl satisfies $V_\theta = O(R)$ at $R = 0$. The retention region Ω_τ contains the *filled* cylinder covering all radial interior to the swirl support, as well as its axial support. This condition is essential, even if the actual carrier is annular.

Lemma AO.1 (Centrifugal cancellation). For $\Psi_\tau(r, z) = \int_r^\infty U_\theta(s, z)^2/s \, ds$,

$$-\frac{U_\theta^2}{r} e_r = \nabla \Psi_\tau - (\partial_z \Psi_\tau) e_z, \qquad \boxed{\int_{\mathbb{T}^3} w \cdot (-U_\theta^2 e_r/r) \, dx = -\int_{\mathbb{T}^3} w_z \partial_z \Psi_\tau \, dx}$$

for every globally divergence-free periodic correction w. The potential is smooth and compactly supported in the filled shrinking cylinder, and $\|\partial_z \Psi_\tau\|_2 \le C\tau^{p/2-1}$. *Proof.* The radial derivative of the integral is $-U_\theta^2/r$; the gradient pairing vanishes by periodic integration by parts. The axial derivative has physical scale τ^{-2A-D} and support volume $O(\tau^{1+D})$, giving exponent $-2A - D + (1 + D)/2 = p/2 - 1$.

Lemma AO.2 (Reduced convective pairing). The exact axisymmetric convective acceleration is

$$\begin{aligned}
[(U \cdot \nabla)U]_r &= U_r U_{r,r} + U_z U_{r,z} - U_\theta^2/r, \\
[(U \cdot \nabla)U]_\theta &= U_r U_{\theta,r} + U_z U_{\theta,z} + U_r U_\theta/r, \\
[(U \cdot \nabla)U]_z &= U_r U_{z,r} + U_z U_{z,z}.
\end{aligned}$$

All noncentrifugal terms have L^2 norm $O(\tau^{p/2-1})$ by direct substitution; the centrifugal pairing has the same bound by Lemma AO.1. Hence

$$\boxed{|\langle w, (U \cdot \nabla)U \rangle| \le C\tau^{p/2-1} \|w\|_{L^2(\Omega_\tau)}}.$$

Theorem AO.3 (Natural-amplitude axisymmetric correction no-go). Suppose $u = U + w$ solves the smooth unforced periodic Navier–Stokes equation, $\operatorname{div} w = 0$, and uniformly on late dyads

$$\|U\|_2^2 = M_0 \tau^p (1 + o(1)), \quad \|\nabla U\|_2^2 = R_0 M_0 \tau^{p-1} (1 + o(1)), \quad M_0 > 0, \ R_0 > p/2,$$

$$\|U_t + \Delta U\|_2 \le C\tau^{p/2-1}, \qquad \|\nabla U\|_\infty \le C\tau^{-1-h}.$$

Let $\delta_\tau = \tau^{-p/2} \sup_{I_\tau} \|w\|_{L^2(\Omega_\tau)} \to 0$ and $|\Omega_\tau| \le C\tau^{1+D}$. Then

$$\boxed{\delta_\tau \ge c\tau^{h/2}, \qquad \sup_{I_\tau} \|w\|_{L^\infty(\Omega_\tau)} \ge c'\tau^{-1/2-h/2}.}$$

Therefore a correction uniformly bounded by $C\tau^{-1/2}$ in the core cannot retain this axisymmetric carrier, regardless of a noncompact exterior analytic tail. *Proof.* The exact pressure-free carrier pairing identity gives

$$\frac{d}{dt}\langle w, U\rangle = \langle w, U_t + \Delta U + (U\cdot\nabla)U\rangle + \int w_i w_j \partial_j U_i - \tfrac{1}{2}(\|U\|_2^2)' - \|\nabla U\|_2^2.$$

Integrating on I_τ and using endpoint retention forces the two interaction terms to total $K\tau^p + o(\tau^p)$, where $K = M_0(1-2^{-p})(R_0 - p/2)/p > 0$. The linear pairings are $O(\delta_\tau \tau^p)$ using Lemma AO.2, and the quadratic term is $O(\delta_\tau^2 \tau^{p-h})$. Thus $K/2 \le C(\delta_\tau + \delta_\tau^2 \tau^{-h})$, so $\delta_\tau \ge c\tau^{h/2}$. The volume bound yields the L^∞ estimate.

Verification and exact claim boundary. For the polynomial swirl $U_\theta = r(1-r^2)^2(1-z^2)^2$ on the unit cylinder, the radial potential equals $(1-r^2)^5(1-z^2)^4/10$ and its derivative is exactly $-U_\theta^2/r$. A cylindrical stream-function test confirms the solenoidal cancellation. Sixteen independent symbolic and exponent checks passed locally. The differential cancellation is classical, not a claim of a new general regularity theorem. The result applies to axisymmetric carriers with filled-core retention and does not authenticate the nonaxisymmetric Case 006 source. The simultaneous (U, w, H, ρ, E) construction, positive stress, full periodic pressure, signed flux, terminal singularity and end-to-end Lean theorem remain OPEN.

Appendix AP. Critical Fourfold Angular Transfer and a Positive Kinematic Control

Objective. Isolate the source component that can deliver the leading signed pressure-free transfer for a natural-amplitude correction to an H8-scale fourfold-equivariant carrier. A positive kinematic example is not an unforced Navier–Stokes solution.

Carrier assumptions. Let $\tau = T - t$, $0 < h < 1/6$, $A = 1/2 + h$, $D = 1/2 - h$, $p = 1/2 - 3h$, $\varepsilon = \tau^h$, $R = r/\sqrt{\tau}$ and $\eta = z/\tau^D$. Suppose U is compactly supported, C_4-equivariant, smooth and divergence free, with

$$U_r = \tau^{-1/2} V_r^\varepsilon, \qquad U_\theta = \tau^{-A} V_\theta^\varepsilon, \qquad U_z = \tau^{-A} V_z^\varepsilon.$$

The profiles have uniform C^2 bounds, fixed compact support, smooth Cartesian axis extension, controlled radial divisions, and $O(\varepsilon)$ convergence in C^1 to V_r^0, V_θ^0, V_z^0. The filled radial potential support lies in a C_4-invariant region $\Omega(t)$ of volume $O(\tau^{1+D})$. Exact divergence reads

$$\frac{1}{R}\partial_\theta V_\theta^\varepsilon = -\varepsilon \left[\frac{1}{R}\partial_R (R V_r^\varepsilon) + \partial_\eta V_z^\varepsilon \right].$$

Consequently $\partial_\theta V_\theta^0 = 0$. The critical candidate is

$$\boxed{F_4(R,\theta,\eta) = \frac{V_\theta^0(R,\eta)}{R}\, \partial_\theta V_z^0(R,\theta,\eta).}$$

It has zero θ mean and only nonzero $m \in 4\mathbb{Z}$ angular harmonics.

Theorem AP.1 (Necessary signed fourfold coupling). Suppose a smooth unforced periodic $u = U + w$, $\nabla \cdot w = 0$, satisfies

$$\|U\|_2^2 = M_0 \tau^p (1 + o(1)), \quad \|\nabla U\|_2^2 = R_0 M_0 \tau^{p-1}(1 + o(1)), \quad R_0 > p/2,\ M_0 > 0,$$
$$\|U_t + \Delta U\|_2 \le C\tau^{p/2-1}, \qquad \|\nabla U\|_\infty \le C\tau^{-1-h},$$
$$\delta_\tau := \tau^{-p/2} \sup_{t \in I_\tau} \|w(t)\|_{L^2(\Omega(t))} = O(\tau^h), \quad I_\tau = [T - \tau, T - \tau/2].$$

On the shrinking cylindrical chart define

$$(\mathcal{P}_* f)(r,\theta,z) = \frac{1}{4}\sum_{j=0}^{3} f(r, \theta + j\pi/2, z) - \frac{1}{2\pi}\int_0^{2\pi} f(r,\phi,z)\, d\phi.$$

Then, with $K = M_0(R_0 - p/2)(1 - 2^{-p})/p > 0$,

$$\boxed{\int_{I_\tau} \tau(t)^{-2A-1/2} \int_{\Omega(t)} (\mathcal{P}_* w_z) F_4 \, dx\, dt = K\tau^p + o(\tau^p).} \tag{145}$$

Necessarily $F_4 \not\equiv 0$ and

$$\boxed{\tau^{-p/2} \sup_{I_\tau} \|\mathcal{P}_* w_z\|_{L^2(\Omega(t))} \ge c\tau^h.}$$

In particular, an axisymmetric correction alone cannot support the critical signed transfer at natural local amplitude.

Proof. The exact cylindrical convective formula, the profile bounds and incompressibility show that the critical acceleration is

$$(U\cdot\nabla)U=\tau^{-2A-1/2}\left[-\frac{(V_\theta^0)^2}{R}e_r+F_4e_z+O_{L^2_{\mathrm{ref}}}(\varepsilon)\right].$$

The large radial centrifugal term is a gradient up to a smaller axial term. Indeed with $\Phi_0(R,\eta)=\int_R^\infty (V_\theta^0(s,\eta))^2/s\,ds$ and $\Phi_\tau=\tau^{-2A}\Phi_0$,

$$-\tau^{-2A-1/2}\frac{(V_\theta^0)^2}{R}e_r=\nabla\Phi_\tau-\tau^{-2A-D}\partial_\eta\Phi_0\,e_z.$$

The smooth compactly extended gradient pairs to zero against globally solenoidal w. Its remaining axial term is smaller by $\tau^{1/2-D}=\varepsilon$. Hence

$$\langle w,(U\cdot\nabla)U\rangle=\tau^{-2A-1/2}\int_\Omega w_zF_4\,dx+O\left(\varepsilon\tau^{p/2-1-h}\|w\|_{L^2(\Omega)}\right).$$

Fourfold periodicity and zero angular mean replace w_z by $\mathcal{P}_*w_z$ exactly. The dyadic error is $O(\tau^{p+h})$, while the $U_t+\Delta U$ and quadratic interaction terms in the exact pressure-free carrier identity are respectively $O(\delta_\tau\tau^p)$ and $O(\delta_\tau^2\tau^{p-h})$, hence $o(\tau^p)$. The carrier-testing identity therefore gives (145). Cauchy–Schwarz and $\|\tau^{-2A-1/2}F_4\|_2=O(\tau^{p/2-1-h})$ prove the lower bound. □

Proposition AP.2 (Explicit positive kinematic channel). Choose nonzero smooth compact flat $f(R^2)$ and $g(\eta)$ supported in $R<1$, $|\eta|<1$. Let

$$\psi_c=R^6f(R^2)g(\eta)\cos4\theta,\qquad V_r=-R^{-1}\partial_\eta\psi_c,\qquad V_z=R^{-1}\partial_R\psi_c.$$

Choose $V_\theta=Ra(R,\eta)$ for a smooth compact $a=1$ on $\operatorname{supp}\psi_c$. Define

$$\psi_w=-4R^6f(R^2)g(\eta)\sin4\theta,\quad W_r=-R^{-1}\partial_\eta\psi_w,\quad W_z=R^{-1}\partial_R\psi_w,$$

and the physical correction $w_r=\tau^{-1/2+h}W_r$, $w_z=\tau^{-1/2}W_z$, $w_\theta=0$. Both the prepared carrier and correction are smooth, compact, exactly solenoidal and C_4 equivariant. On the source support

$$F_4=-4R^{-1}\partial_R(R^6fg)\sin4\theta=W_z,$$

and therefore

$$\boxed{C_0=\int F_4W_z\,R\,dR\,d\theta\,d\eta=\int F_4^2\,R\,dR\,d\theta\,d\eta>0.}$$

Consequently the signed transfer is $C_0(1-2^{-p})\tau^p/p+O(\tau^{p+h})$; scaling w by a constant can match any prescribed positive leading scalar K. This does *not* solve the full unforced equation, positive stress or pressure flux.

For a finite-regularity (not flatly smooth) polynomial diagnostic, $f(X)=(1-X)^4$ on $[0,1]$, $g(\eta)=(1-\eta^2)^4$ on $[-1,1]$,

$$\int_0^1\frac{X^4}{2}(6f+2Xf')^2\,dX=\frac{4}{5005},\quad \int_{-1}^1 g^2\,d\eta=\frac{65536}{109395},\quad \boxed{C_0=\frac{4194304\pi}{547521975}>0.}$$

Smooth flat approximations preserve the sign. The independent exact symbolic checker has 23 passing assertions.

Claim boundary. This is a source-grammar conditional selection theorem and a manufactured kinematic positive control. Group harmonic selection and centrifugal-gradient cancellation are established methods, not broad novelty claims. The original four-patch Case 006 profiles, all ordered overlaps, tilted physical frame, compact positive seven-ray inverse and retained unforced solution are still unauthenticated. All five global PDE and Lean closure gates remain OPEN.

Appendix AQ. Fourfold Solenoidal Residual and Correction Concentration

Objective. A source-specific divergence-free spacetime test detects the fourfold axial convective defect independently of periodic pressure. It establishes a necessary amplitude–concentration tradeoff for a retained nonlinear correction. No genuine Case 006 four-patch source or singular solution is constructed.

Carrier hypotheses. Let $\tau = T - t$, $0 < h < 1/6$, $A = 1/2 + h$, $D = 1/2 - h$, $p = 1/2 - 3h$ and $\varepsilon = \tau^h$. In an injective cylindrical torus chart put $R = r/\sqrt{\tau}$, $\eta = z/\tau^D$. Assume a compact smooth solenoidal C_4-equivariant carrier has

$$U_r = \tau^{-1/2} V_r^\varepsilon, \quad U_\theta = \tau^{-A} V_\theta^\varepsilon, \quad U_z = \tau^{-A} V_z^\varepsilon.$$

The fixed-support similarity profiles have uniformly bounded C^4 derivatives and smooth Cartesian axis extension, converge with $O(\varepsilon)$ error in C^2, and have bounded scaled time derivatives. Exact incompressibility forces $\partial_\theta V_\theta^0 = 0$. Define

$$F(R,\theta,\eta) = \frac{V_\theta^0(R,\eta)}{R} \partial_\theta V_z^0(R,\theta,\eta), \quad J_F = \int |\partial_\theta F|^2 R \, dR \, d\theta \, d\eta > 0.$$

The source F is compact, has smooth Cartesian axis extension, and contains only nonzero fourfold angular modes. Let its support lie in a C_4-invariant shrinking region $\Omega(t)$ of volume $O(\tau^{1+D})$.

Lemma AQ.1 (Compact divergence-free detector). The cylindrical field

$$\boxed{Z_r = 0, \quad Z_\theta = \tau^{-1/2} \varepsilon R \partial_\eta \partial_\theta F, \quad Z_z = -\tau^{-1/2} \partial_\theta^2 F}$$

is exactly solenoidal and compactly supported. Indeed $r^{-1} \partial_\theta Z_\theta = \tau^{-1+h} \partial_\eta \partial_\theta^2 F = -\partial_z Z_z$. Moreover

$$\|Z\|_2 \le C\tau^{D/2}, \quad \|\nabla Z\|_2 \le C\tau^{D/2-1/2}, \quad \|Z_t\|_2 + \|\Delta Z\|_2 \le C\tau^{D/2-1}.$$

The critical axial source is detected with a strictly positive coefficient:

$$\boxed{\left\langle \tau^{-2A-1/2} F e_z, Z \right\rangle = J_F \tau^{p-1} > 0.}$$

The first equality follows from $D - 2A = p - 1$; the second from periodic angular integration by parts.

Theorem AQ.2 (Pressure-free concentration tradeoff). Assume the above carrier additionally satisfies

$$\|U\|_\infty \le C\tau^{-A}, \quad \|\nabla U\|_\infty \le C\tau^{-1-h}, \quad \|U_t - \Delta U\|_2 \le C\tau^{p/2-1}.$$

Suppose a smooth exact unforced periodic Navier–Stokes solution $u = U + w$ with $\nabla \cdot w = 0$ exists for $t < T$. For $I_\tau = [T - \tau, T - \tau/2]$ define

$$\delta_\tau = \tau^{-p/2} \sup_{I_\tau} \|w\|_{L^2(\Omega(t))}, \qquad M_\tau = \sup_{I_\tau} \|w\|_{L^\infty(\Omega(t))}.$$

If $\delta_\tau \to 0$, then for all sufficiently small τ,

$$\boxed{\delta_\tau\, \tau^{1/2+h} M_\tau \geq c > 0.}$$

If the local retention rate is $\delta_\tau = O(\tau^h)$, necessarily

$$\boxed{M_\tau \geq c' \tau^{-1/2-2h}.}$$

At $h = 1/10$ the lower amplitude is $\tau^{-7/10}$, excluding a uniformly natural $\tau^{-1/2}$ near-core correction within these hypotheses. This does not impose any global support condition or derivative estimate on w.

Proof. In cylindrical variables, $(U \cdot \nabla)U_z = U_r \partial_r U_z + (U_\theta/r)\partial_\theta U_z + U_z \partial_z U_z = \tau^{-2A-1/2}(F + O(\varepsilon))$. The radial convective component pairs to zero since $Z_r = 0$. All azimuthal convective contributions tested against the $O(\varepsilon)$-small Z_θ are lower order. Lemma AQ.1 and the carrier bounds therefore give

$$\int_{I_\tau} \langle U_t - \Delta U + (U \cdot \nabla)U, Z \rangle dt = \frac{1 - 2^{-p}}{p} J_F \tau^p + O(\tau^{p+h}).$$

Test the exact unforced equation for $u = U + w$ against Z. Periodic pressure vanishes exactly since $\nabla \cdot Z = 0$. Integrating w_t by parts in time (including both boundary terms), $-\Delta w$ by parts twice in space, and $(U \cdot \nabla)w$ by parts using $\nabla \cdot U = 0$, gives

$$\left| \int_{I_\tau} \langle w_t - \Delta w, Z \rangle dt \right| \leq C \delta_\tau \tau^{p+h},$$

$$\left| \int_{I_\tau} \langle (U \cdot \nabla)w + (w \cdot \nabla)U, Z \rangle dt \right| \leq C \delta_\tau \tau^p,$$

$$\left| \int_{I_\tau} \langle (w \cdot \nabla)w, Z \rangle dt \right| \leq C \delta_\tau M_\tau \tau^{p+1/2+h}.$$

All spatial estimates use w only on $\operatorname{supp} Z \subset \Omega(t)$. The exact zero residual implies

$$\frac{1 - 2^{-p}}{p} J_F \leq C(\tau^h + \delta_\tau + \delta_\tau \tau^h + \delta_\tau \tau^{1/2+h} M_\tau).$$

The first three terms vanish when $\tau \downarrow 0$; $J_F > 0$ forces the fourth to stay positive. This yields both conclusions.

Kinematic calibration and release boundary. For Appendix AP's polynomial fourfold positive control, $\int F^2 R\, dR\, d\theta\, d\eta = 4194304\pi/547521975$ and

$$\boxed{J_F = \frac{67108864\pi}{547521975} > 0.}$$

The polynomial is a finite-regularity diagnostic; smooth flat cutoffs preserve positivity. This theorem shows why a scalar signed-transfer match does *not* by itself solve the exact unforced correction equation: sufficiently concentrated correction self-interaction is necessary. The authentic four-patch source, rotating stress frame, one-sided positive coefficients, actual periodic pressure flux, pointwise terminal blow-up and end-to-end Lean closure remain OPEN.

Appendix AR. Normalized seven-ray identities and sharp scalar coefficient floor

This appendix records two exact algebraic statements underlying the normalized seven-ray tensor representation. Their hypotheses are part of each assertion.

The normalized trigonometric tensor identity. For real r, ℓ, s, c define

$$\begin{aligned} C_4 &= 2 + 16s^2 + 16\ell^2(r^2s^2 - r^4c^2) + 16\ell rcs, \\ C_5 &= 2 + 16s^2 + 16\ell^2(r^2s^2 - r^4c^2) - 16\ell rcs, \\ C_6 &= 1 - 16\ell rcs, \qquad C_7 = 1 + 16\ell rcs, \\ \Pi &= \frac{3}{8} - 2c^2(1 + \ell^2 r^4). \end{aligned}$$

Proposition AR.1. If $s^2 + c^2 = 1$, then

$$\frac{C_4 + C_5 + C_6 + C_7}{16} = 2(1 + \ell^2 r^2 s^2) + \Pi. \tag{146}$$

Proof. Expanding the left-hand side gives

$$\frac{3}{8} + 2s^2 + 2\ell^2 r^2 s^2 - 2\ell^2 r^4 c^2.$$

Subtracting the right-hand side of (146) reduces exactly to $2(s^2 + c^2 - 1)$, which vanishes under the stated hypothesis. □

A sharp scalar floor. Let $\mu, d_1, \ldots, d_7 \in \mathbb{R}$ with $d_i \geq \mu$ for each $1 \leq i \leq 7$. Define the normalized diagonal functional

$$\mathcal{L}(d) = \sum_{i=1}^{7} d_i - \frac{d_2 + d_3 + d_6 + d_7}{16}.$$

Proposition AR.2. For every real μ,

$$\mathcal{L}(d) \geq \frac{27}{4}\mu. \tag{147}$$

The factor $27/4$ is sharp for the stated coordinate floors. If $\mu \geq 0$, the weaker estimate $\mathcal{L}(d) \geq (105/16)\mu$ follows.

Proof. The functional has coefficient 1 at d_1, d_4, d_5 and coefficient $15/16$ at d_2, d_3, d_6, d_7. All seven coefficients are positive, and therefore

$$\mathcal{L}(d) \geq \left(3 + 4\frac{15}{16}\right)\mu = \frac{27}{4}\mu.$$

Equality holds for $d_1 = \cdots = d_7 = \mu$, proving sharpness. For nonnegative μ, the remaining implication uses $\frac{27}{4} - \frac{105}{16} = \frac{3}{16} \geq 0$. □

The two propositions are scalar algebraic statements. They apply to a physical H8 stress field only once the same normalized parameters and one-sided seven-ray coefficient floors have been established for that field.

Appendix AS. Reproducible finite-arithmetic verification

The following self-contained source independently verifies the rational seven-ray witnesses, the six boundary vertices of the axial–azimuthal projection, the exact Case 006 interval, the trigonometric tensor polynomial identity, the sharp scalar floor, and both fourfold polynomial diagnostic integrals. The identities are checked over $\mathbb{Q}$ and by exact symbolic polynomial arithmetic. The positive trigonometric polynomial integral is a kinematic diagnostic, not a solution of the unforced equation.

Python 3 source (requires SymPy):

```
"""Exact␣H8␣finite-arithmetic␣checks␣supporting␣Appendices␣AP,␣AQ,␣AR.
No␣assertion␣here␣certifies␣a␣complete␣unforced␣Navier--Stokes␣solution.
Run:␣python␣-m␣pip␣install␣sympy;␣python␣tests/test_h8_full_paper_20261101.py
"""
from fractions import Fraction as Q
from sympy import symbols, Rational, simplify, integrate, diff, expand, pi

def rays(A, B, C, D):
    return [-Q(3, 2)*D, 2*C-4*D, -2*C-4*D,
            2*A-8*B+4*D, -2*A+8*B+4*D, 8*B, -8*B]

def floors(A, B, C, D):
    base = [Q(172,1229),Q(172,1229),Q(236,1229),
            Q(172,1229),Q(172,1229),Q(172,1229),Q(172,1229)]
    return [x+y for x,y in zip(base,rays(A,B,C,D))]

def test_axial_witness():
    A=Q(51113,245800); B=Q(15971,983200)
    C=Q(16,1229); D=Q(19171,491600)
    assert min(floors(A,B,C,D)) == Q(1,100)

def test_joint_hexagon():
    g=Q(15971,122900); u=Q(19171,122900)
    lo=Q(47913,245800); hi=Q(54313,245800)
    m=Q(16,1229); a=Q(51113,245800)
    vertices=[(-g/2,-g),(g/2,-g),(a,m),
              (g/2,u),(-g/2,u),(-a,m)]
    for A,C in vertices:
        assert -g<=C<=u and abs(A)-C<=lo and abs(A)+C<=hi
        D=(g+Q(22371,122900))/8 if C==m else -g/4
        B=g/8 if A>0 else -g/8
        assert min(floors(A,B,C,D))>=Q(1,100)

def test_case006_band():
    t=Q(-4443,9832); b=Q(15971,122900)
    assert (t-b,t+b)==(Q(-143017,245800),Q(-79133,245800))

def test_trigonometric_tensor():
    r,l,s,c=symbols('r␣l␣s␣c')
    C4=2+16*s*s+16*l*l*(r*r*s*s-r**4*c*c)+16*l*r*c*s
    C5=2+16*s*s+16*l*l*(r*r*s*s-r**4*c*c)-16*l*r*c*s
    C6=1-16*l*r*c*s; C7=1+16*l*r*c*s
    Pi=Rational(3,8)-2*c*c*(1+l*l*r**4)
    difference=simplify((C4+C5+C6+C7)/16-2*(1+l*l*r*r*s*s)-Pi)
    assert simplify(difference-2*(s*s+c*c-1))==0

def test_sharp_scalar_floor():
    assert 3+4*Q(15,16)==Q(27,4)
    assert Q(27,4)-Q(105,16)==Q(3,16)
    # Equality at d_1=...=d_7=mu establishes sharpness.
    for mu in [Q(-2),Q(0),Q(7,3)]:
        assert 7*mu-4*mu/16==Q(27,4)*mu

def test_fourfold_integrals():
    X,z=symbols('X␣z',real=True)
    f=(1-X)**4; g=(1-z*z)**4
    q=X*X*(6*f+2*X*diff(f,X))
    rad=integrate(expand(q*q)/2,(X,0,1))
    axial=integrate(expand(g*g),(z,-1,1))
```

```
    assert rad==Rational(4,5005)
    assert axial==Rational(65536,109395)
    assert simplify(16*rad*axial)==Rational(4194304,547521975)
    assert simplify(256*rad*axial)==Rational(67108864,547521975)

if __name__=="__main__":
    for f in [test_axial_witness,test_joint_hexagon,test_case006_band,
              test_trigonometric_tensor,test_sharp_scalar_floor,
              test_fourfold_integrals]:
        f()
    print("PASS: six independent exact arithmetic groups")
    print("Scope: finite polynomial and cone identities only.")
```

The conditional analytic estimates in Appendices AP and AQ additionally require their stated smoothness, support and localization hypotheses; finite-arithmetic checks alone do not establish those analytic assumptions.

Appendix AT. Exact fourfold mediator energy and carrier-coupling criterion

We record the finite Fourier calculation independently of the unresolved singular-carrier construction. The unit-torus convention is $e_k(x) = \exp(2\pi i k \cdot x)$, with viscosity $\nu > 0$. The finite triadic transfer formula itself is standard [6]; the exact arithmetic below concerns a particular fourfold family, not a general cascade theorem.

The fourfold real seed. Set

$$p = (1,0,0), \quad q = (0,2,1), \quad r = p + q = (1,2,1), \qquad a = (0,1,0), \quad b = (0,1,-2).$$

For real amplitudes A, B, C and $\sigma \in \{-1, +1\}$ prescribe

$$\widehat{h}(p) = iAa, \qquad \widehat{h}(q) = Bb, \qquad \widehat{h}(r) = \sigma Cb.$$

Extend these coefficients by complex conjugation at $-k$ and by the four quarter-turns about the z axis, rotating both wavevectors and polarizations. The resulting real solenoidal trigonometric polynomial h has exactly 20 distinct possible wavevectors: four of type p and eight each of types q, r. The p coefficients counted under opposite quarter-turns coincide, so they are not duplicated in the Fourier series. Let $\mathcal{S} = \operatorname{supp} \widehat{h}$ for generic nonzero amplitudes, and define

$$\mathcal{B}_k(h,h) = \sum_{\ell+m=k} (\widehat{h}(\ell) \cdot m)\, \widehat{h}(m), \qquad \mathbb{P}_k = I - \frac{k \otimes k}{|k|^2} \quad (k \neq 0).$$

Proposition AT.1 (finite mediator enumeration and leakage). Treat A, B, C as independent formal real variables. Among all 170 distinct nonzero pair sums, 124 have a nonzero Leray-projected quadratic coefficient polynomial. Of these, 108 lie outside $\mathcal{S}$. Denote this last set by $\mathcal{N}$. It decomposes into 92 cross-sector-only modes, which admit no contributing pair in one unrotated sector, and 16 modes with contributing pairs entirely within an unrotated sector. Relative to the highest seed wavenumber $|r|^2 = 6$, there are 72 new modes with $|k|^2 > 6$, 8 with $|k|^2 = 6$, and 28 with $|k|^2 < 6$.

The exact off-support quadratic source norm is

$$\begin{aligned} \mathcal{L}(A,B,C) &= \sum_{k \in \mathcal{N}} |\mathbb{P}_k \mathcal{B}_k(h,h)|^2 \\ &= 176A^2B^2 + \frac{8000}{33}A^2C^2 + \frac{1600}{3}B^4 + \frac{1796592}{1105}B^2C^2 + \frac{25376}{35}C^4. \end{aligned} \tag{148}$$

For the 92 cross-sector-only modes, the corresponding sum is

$$\mathcal{L}_\times = 16A^2B^2 + \frac{2720}{33}A^2C^2 + \frac{1600}{3}B^4 + \frac{1796592}{1105}B^2C^2 + \frac{25376}{35}C^4, \tag{149}$$

while the 16 within-sector modes contribute $160A^2(B^2 + C^2)$. The strictly higher-frequency subfamily has

$$\mathcal{L}_{>6} = 8A^2B^2 + \frac{2256}{11}A^2C^2 + \frac{1600}{3}B^4 + \frac{262432}{221}B^2C^2 + \frac{25376}{35}C^4. \tag{150}$$

All three polynomials are independent of the phase sign σ.

Finite verification. For each of the at most 20^2 ordered pairs, form $k = \ell + m$ and accumulate the Gaussian-integer vector $(\widehat{h}(\ell) \cdot m)\widehat{h}(m)$ under the quadratic amplitude monomial. Eliminate $k = 0$ and use

$$|\mathbb{P}_k v|^2 = |v|^2 - \frac{|k \cdot v|^2}{|k|^2}$$

with rational arithmetic, then combine monomials and conjugate terms. The same computation tests sector membership and the $|k|^2$ partition. The stand-alone standard-library verifier `tools/verify_attack24_exact.py` carries out all 20-mode and 108-mode checks with integer Gaussian coefficients and `Fraction` denominators. A separate physical-space pseudospectral implementation reproduces (148) to floating-point round-off for three independent amplitude choices. The finite enumeration is an exact computer-assisted identity, not an analytic theorem concerning infinite spectral evolution.

Initial energy transfer and viscosity. For the isolated unforced Navier–Stokes evolution with initial datum h, local smooth existence gives $\widehat{u}(k,t_0)=0$ on $\mathcal{N}$ and

$$\partial_t\widehat{u}(k,t_0)=-2\pi i\,\mathbb{P}_k\mathcal{B}_k(h,h)\quad(k\in\mathcal{N}).$$

Consequently, for $s\downarrow 0$,

$$\sum_{k\in\mathcal{N}}|\widehat{u}(k,t_0+s)|^2=4\pi^2s^2\mathcal{L}(A,B,C)+o(s^2). \tag{151}$$

This is nonzero initial leakage, not a monotone shell-to-shell cascade or a lower bound at terminal times.

Parseval gives exactly

$$\|h\|_2^2=4A^2+40B^2+40C^2,\qquad \|\nabla h\|_2^2=(2\pi)^2(4A^2+200B^2+240C^2). \tag{152}$$

The instantaneous nonlinear energy derivatives in the three occupied wavevector families $|k|^2=1,5,6$, respectively, are

$$0,\qquad -320\pi\sigma ABC,\qquad +320\pi\sigma ABC.$$

Their sum is zero, as required by incompressibility. In particular, for each of the eight receiver modes of type r,

$$\frac{d}{dt}|\widehat{u}(r)|^2\Big|_{t=t_0}=40\pi\sigma ABC-240\nu\pi^2C^2.$$

For $A,B,C>0$, gain against viscosity at this occupied receiver requires $\sigma=+1$ and $AB>6\pi\nu C$. The choice $\sigma=-1$ used in the original first-mediator calculation has the opposite occupied-receiver energy-transfer direction, although it yields the same off-support polynomial (148). This phase distinction is essential.

Proposition AT.2 (second-generation exit). Let $k_*=(3,2,1)$ and assume the isolated unforced Navier–Stokes evolution begins with the above seed. Then $k_*\notin\mathcal{S}\cup\mathcal{N}$, but

$$\partial_t^2\widehat{u}(k_*,t_0)=16\pi^2\sigma A^2C\,b. \tag{153}$$

In particular, if $AC\neq 0$, the initial 20 modes together with all 108 first-generation modes still fail to form a Fourier-support invariant subsystem.

Proof. Write $s=p+r=(2,2,1)$. The first-derivative identity is $\partial_t\widehat{u}(s,t_0)=4\pi\sigma AC\,b$. At $k_*=p+s$ the only contributing ordered pair involving an initially generated mode is (p,s) and its reverse. The reverse derivative term vanishes because $b\cdot p=0$, whereas

$$(\widehat{h}(p)\cdot s)\,\partial_t\widehat{u}(s,t_0)=(2iA)(4\pi\sigma AC\,b)=8\pi i\sigma A^2C\,b.$$

Since $k_*\cdot b=0$, the Leray projection leaves this vector unchanged. The outer multiplier $-2\pi i$ gives (153). At k_* both the initial amplitude and first derivative vanish, so the viscous term contributes nothing to the second derivative. Consequently $\widehat{u}(k_*,t_0+s)=8\pi^2\sigma A^2C\,b\,s^2+o(s^2)$. This conclusion concerns the isolated initial evolution. An independent physical-space pseudospectral second-derivative calculation reproduces (153) at the tested amplitudes.

A carrier-inclusive necessary test. Let U be any smooth periodic divergence-free carrier at $t_0 < T$ and write $u = U + w$. Define its exact projected residual

$$\mathcal{R}_\nu(U) = \partial_t U - \nu\Delta U + \mathbb{P}\nabla\cdot(U\otimes U)$$

and the carrier-coupling source

$$\mathcal{K}_U(w) = -\mathcal{R}_\nu(U) - \mathbb{P}\nabla\cdot(U\otimes w + w\otimes U).$$

If the full u solves the unforced equation, the correction satisfies

$$\partial_t w - \nu\Delta w + \mathbb{P}\nabla\cdot(w\otimes w) = \mathcal{K}_U(w).$$

Suppose $w(t_0) = h$, so its coefficients vanish on $\mathcal{N}$. Writing $\Pi_{\mathcal{N}}$ for the orthogonal Fourier projector, the triangle inequality and (148) imply

$$\|\Pi_{\mathcal{N}}\partial_t w(t_0)\|_2 \geq \left[2\pi\sqrt{\mathcal{L}(A,B,C)} - \|\Pi_{\mathcal{N}}\mathcal{K}_U(h)(t_0)\|_2\right]_+. \tag{154}$$

Here $[x]_+ = \max(x,0)$. An explicit, generally non-sharp bound is

$$\|\Pi_{\mathcal{N}}\mathcal{K}_U(h)\|_2 \leq \|\mathcal{R}_\nu(U)\|_2 + \|U\|_\infty\|\nabla h\|_2 + \|\nabla U\|_\infty\|h\|_2.$$

Thus first-generation leakage persists in the full correction whenever the actual projected carrier coupling is strictly smaller than $2\pi\sqrt{\mathcal{L}}$; cancellation is not excluded when it is comparable or larger. No value for the actual frozen Version 17 carrier-coupling norm is inferred from this finite computation.

Finally, the exact correction energy relation is

$$\frac{1}{2}\frac{d}{dt}\|w\|_2^2 + \nu\|\nabla w\|_2^2 = -\int_{\mathbb{T}^3} w\cdot(w\cdot\nabla U)\,dx - \langle\mathcal{R}_\nu(U), w\rangle, \tag{155}$$

whereas the full unforced solution must obey $\frac{1}{2}\frac{d}{dt}\|U+w\|_2^2 + \nu\|\nabla(U+w)\|_2^2 = 0$. The finite-mode identities do not control (155) over a shrinking terminal interval.

Scope of the result. The exact data show that distinct fourfold sectors generate many new modes, including both higher and lower frequencies, and they quantify the initial transfer. They do not supply a sustained nonlinear cascade, an energy-compatible terminal-time modulation, a bound on $\Pi_{\mathcal{N}}\mathcal{K}_U$, or non-cancellation of the frozen singular trace. The smooth unrotated invariant shear-advection subsystem from the preceding finite-triad calculation also remains an obstruction to using a single polarization ray as a blow-up mechanism. In particular, no unforced finite-time singularity theorem follows from Appendix AT.

Appendix AU. Spectral population forced by terminal concentration

Let $u(t,\cdot)$ be a smooth, real-valued, periodic divergence-free velocity on the unit three-torus at each time $t < T$, and denote its vector Fourier coefficients by $\widehat{u}(k,t) \in \mathbb{C}^3$. For any fixed $t < T$, smoothness entails absolute convergence of $\sum_{k\in\mathbb{Z}^3} \|\widehat{u}(k,t)\|_{\mathbb{C}^3}$. When $E(t) = \|u(t)\|_2^2 > 0$, define the effective spectral population

$$\mathcal{N}_{\mathrm{eff}}(t) = \frac{\left(\sum_{k\in\mathbb{Z}^3} \|\widehat{u}(k,t)\|_{\mathbb{C}^3}\right)^2}{\sum_{k\in\mathbb{Z}^3} \|\widehat{u}(k,t)\|_{\mathbb{C}^3}^2}.$$

This is an ℓ^1/ℓ^2 concentration quotient, rather than a count of modes above an arbitrary numerical cutoff.

Proposition AU.1 (Necessary terminal spectral population)**.** Suppose that, for a sequence $\tau_j \downarrow 0$, the total velocity $u(T-\tau_j,\cdot)$ satisfies, with constants $c, C > 0$ and $p, \alpha > 0$,

$$\|u(T-\tau_j)\|_2^2 \le C\tau_j^{p}, \qquad \|u(T-\tau_j)\|_\infty \ge c\tau_j^{-\alpha}.$$

Then at every time in this sequence,

$$\mathcal{N}_{\mathrm{eff}}(T-\tau_j) \ge \frac{c^2}{C}\,\tau_j^{-(2\alpha+p)}. \tag{156}$$

If the Fourier series has finite support S_j at these times, its cardinality must satisfy $|S_j| \ge (c^2/C)\tau_j^{-(2\alpha+p)}$.

Proof. The triangle inequality, absolute Fourier convergence, and Parseval give

$$c\tau_j^{-\alpha} \le \|u(T-\tau_j)\|_\infty \le \sum_k \|\widehat{u}(k,T-\tau_j)\|, \qquad \sum_k \|\widehat{u}(k,T-\tau_j)\|^2 = \|u(T-\tau_j)\|_2^2 \le C\tau_j^{p}.$$

Squaring the former and dividing by the latter yields (156). If the support is finite, Cauchy–Schwarz also yields

$$\left(\sum_{k\in S_j} \|\widehat{u}(k)\|\right)^2 \le |S_j| \sum_{k\in S_j} \|\widehat{u}(k)\|^2,$$

so $\mathcal{N}_{\mathrm{eff}} \le |S_j|$. □

For a terminal carrier whose squared energy is comparable to $\tau^{1/5}$, the same upper energy bound on the *full* corrected velocity and a retained lower speed $\|u(T-\tau_j)\|_\infty \ge c\tau_j^{-(1/2+h)}$ force

$$\mathcal{N}_{\mathrm{eff}}(T-\tau_j) \gtrsim \tau_j^{-(6/5+2h)}. \tag{157}$$

The illustrative $h = 1/10$ class requires exponent $7/5$. Neither assumption follows from the 20-mode Fourier calculation in Appendix AT; both refer to the full, unforced correction. Equation (157) is therefore a necessary spectral-concentration test, not a construction.

Proposition AU.2 (Necessary frequency radius under a global band limit)**.** Under the hypotheses of Proposition AU.1, suppose additionally that the Fourier support of the *full* velocity at the terminal times lies in $\{k \in \mathbb{Z}^3 : \|k\|_\infty \le K_0\tau_j^{-\beta}\}$, for constants $K_0, \beta > 0$. Then necessarily

$$\beta \ge \frac{2\alpha + p}{3}.$$

For $p = 1/5$ and $\alpha = 3/5$, this is $\beta \ge 7/15$. A global radius exponent $\beta = 9/20 < 7/15$ is excluded.

Proof. The Fourier support has at most $(2\lfloor K_0\tau_j^{-\beta}\rfloor+1)^3 = O(\tau_j^{-3\beta})$ elements. By Cauchy–Schwarz this bound also controls $\mathcal{N}_{\text{eff}}$. Together with (156), it implies $2\alpha + p \le 3\beta$ as $\tau_j \downarrow 0$. □

In particular, a frequency-migrating exterior field with ceiling $O(\tau^{-9/20})$ cannot by itself serve as the *entire* velocity subject to both the stated energy and speed scalings. The shrinking carrier has higher spatial frequencies, so this conditional band-limit obstruction does *not* rule out a combined carrier–halo construction. The bound is an elementary Fourier/Bernstein consequence, not a dynamical transfer theorem.

The finite Fourier inequality underlying the proof is also encoded in `UnforcedPatching/Attack25SpectralPopulation.lean`. That finite-sum theorem does not by itself formalize the infinite Fourier-series reconstruction or the specific carrier asymptotics; those remain separate physical and analytic verification requirements. The general inequalities in this appendix are standard consequences of Fourier reconstruction and Cauchy–Schwarz and are not presented as new prior-art claims.

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